DNACode™

User Guide for Version 4

Windows and Macintosh

P/N 4000128 RevA

DNACode User Guide

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Preface

1. About This Document

This user guide is designed to be used as a reference in your everyday use of DNACode™. It provides detailed information about the tools and commands of DNACode for the Windows and Macintosh platforms. Any platform differences in procedures and commands are noted in the text.

This guide assumes you have a working knowledge of your computer operating system and its conventions, including how to use a mouse and standard menus and commands, and how to open, save, and close files. For help with any of these techniques, see the documentation that came with your computer.

This guide uses certain text conventions to describe specific commands and functions.

Example	**Indicates**
File > Open	Choosing the Open command under the File menu.
Dragging	Positioning the cursor on an object and holding down the left mouse button while you move the mouse.
CTRL+S	Holding down the Control key while typing the letter *s*.
Right-click / Left-click / Double-click	Clicking the right mouse button / Clicking the left mouse button / Clicking the left mouse button twice.

Some of the illustrations of menus and dialog boxes found in this manual are taken from the Windows version of the software, and some are taken from the Macintosh version. Both versions of a menu or dialog box will be shown only when there is a significant difference between the two.

2. Bio-Rad Listens

The staff at Bio-Rad are receptive to your suggestions. Many of the new features and enhancements in this version of DNACode are a direct result of conversations with our customers. Please let us know what you would like to see in the next version of DNACode by faxing, calling, or e-mailing our Technical Services staff. You can also use Solobug (installed with DNACode) to make software feature requests.

1. Introduction

1.1. A Brief Overview of DNACode

DNACode™ is a software package for analyzing DNA sequencing gels that have been sequenced using the enzymatic method developed by Sanger et al.[1]

The software runs in a Windows or Macintosh environment and has an easy-to-use interface with standard pull-down menus, toolbars, and keyboard commands.

An image of a gel is captured using the controls in the imaging device window and displayed on your computer monitor. Image processing and analysis operations are performed using commands from the menus and toolbars.

Images can be magnified, annotated, rotated, and resized. They can be printed using standard and video printers.

DNACode has robust algorithms that can read the sequence from gels with many lanesets, each containing hundreds of bases. The software can cope with distortions in both lane and band shape. Lanes can be adjusted along their lengths to compensate for any curvature or smiling of gels.

You can share your images among all the Discovery Series™ software, or convert them into TIFF format for compatibility with other Macintosh and Windows applications.

1.2. Digital Data and Signal Intensity

The Bio-Rad imaging devices supported by DNACode are light and/or radiation detectors that convert signals from biological samples into digital data. DNACode then displays the digital data on your computer screen, in the form of gray scale or color images.

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1. Sanger F and Coulson A R. 1977. Proc Natl Acad Sci. USA, 74, 5436.

A data object as displayed on the computer is composed of tiny individual screen pixels. Each pixel has an X and Y coordinate, and a value Z. The X and Y coordinates are the pixel's horizontal and vertical positions on the image, and the Z value is the signal intensity of the pixel.

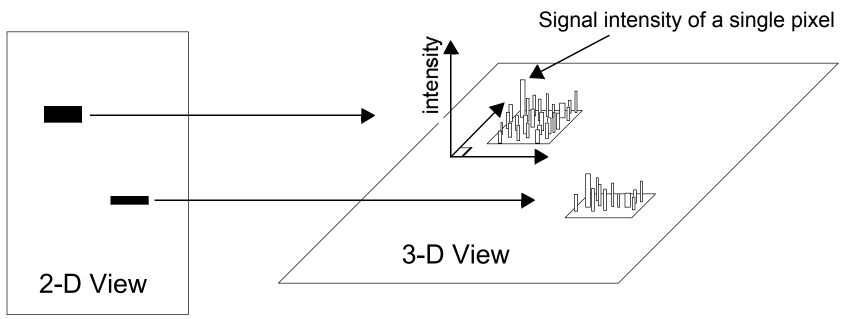


Fig.1-1. Representation of the pixels in two digitally imaged bands in a gel.

For a data object to be visible and quantifiable, the intensity of its clustered pixels must be higher than the intensity of the pixels that make up the background of the image. The total intensity of a data object is the sum of the intensities of all the pixels that make up the object. The mean intensity of a data object is the total intensity divided by the number of pixels in the object.

The units of signal intensity are Optical Density (O.D.) in the case of the GS-700 and 710 densitometers, the Gel Doc with a white light source, or the Fluor-S and Fluor-S MAX MultiImagers with white light illumination. Signal intensity is expressed in counts when using the Personal FX or FX, or in the case of the Gel Doc, Fluor-S, or Fluor-S MAX when using the UV light source.

1.3. About Gel Quality

Optimal strategies for using DNACode will be influenced by the quality of your gels.

If you are using shark's tooth combs, we suggest that you leave a blank lane between lanesets. The automatic sequence reader works most efficiently on gels with well-resolved bands. Sequencing reactions run with ^{35}S-labeled nucleotides typically provide the most discrete bands.

If you cannot read a sequence by eye, the software also will not be able to read it. If you can read the gel by eye but the software cannot, you are probably recognizing visual cues from the gel that the computer does not see. For example, your image exposure may be too long, resulting in overly dark, fat bands, or you may have specific experience with compressions in the A lane using a particular DNA polymerase. You may also know that the first A in a string of As is unusually light in proportion to the others. These are conditions unique to your gels, and the software will not be aware of them.

You may find that you can read higher up into the sequence than the software does after automatic sequencing. This is because the software is purposefully conservative in its base calling. If DNACode is not completely certain of the designation of a particular base, it will label the base ambiguous and let you make the final decision.

If you disagree with some automatic base calls, you can easily overwrite them using DNACode's editing tools and quickly determine the final sequence. It should be noted that the only absolute procedure for resolving remaining ambiguities is to repeat the experiment, sequence the opposite strand, or reclone and resequence the fragment.

1.4. Steps Involved in DNACode

Acquire the Image

↓

Optimize the Image

↓

Read the Sequence

1. Define Data Area (if necessary)
2. Define Lanesets
3. Auto-Sequence

↓

Edit the Sequence

1. Assign/De-assign Incorrect Bases
2. Edit Ambiguous Base Calls
3. Set First and Last Bases

↓

Output the Data

1. Display Sequence Text on Screen
2. Print/Export Data

Fig.1-2. Steps in Sequencing Gel Analysis.

Acquire the Image

Before you can use DNACode to analyze a gel, you need to capture its image into your computer. This may be done with one of the several Bio-Rad instruments supported by this software: the Molecular Imager FX and Personal Molecular Imager FX; the GS-700 and GS-710 Imaging Densitometers; the Gel Doc 1000/2000 gel documentation system; and the Fluor-S and Fluor-S MAX MultiImagers.

The resulting images are stored in files on your hard disk, network file server, or removable storage media.

Optimize the Image

Once you have acquired an image of your gel, you may need to reduce any noise or background density caused by film fogging or the opacity of your carrier medium. A variety of functions exist to minimize background noise while maintaining data integrity.

Read the Sequence

Once the image is ready, you can define the position of the lanesets on your gel. This is done by dragging the cursor from the left edge of the leftmost lane of a laneset across to the right edge of the rightmost lane.

Once your lanesets have been defined, DNACode can automatically read the sequence. Once the sequence has been determined, it will be overlaid on the image of your gel.

Edit the Sequence

After your sequence has been read, you can review the base calls to verify the results of the automatic sequence reader. There are three main functions that you can use to edit the base calls.

- Assign Band adds a missing band to the sequence.
- Make Ambiguous changes the status of a band from certain to uncertain.
- De-assign Band removes an extra band from the sequence.

Output the Data

Once your analysis is complete, the information can be displayed on-screen, sent to a printer, or exported in TIFF or ASCII formats to another system for further analysis.

1.5. Computer Requirements for Running DNACode

DNACode will run on a PC as a Windows 95 / 98 / NT 4.0 32-bit application or on a Macintosh as a Macintosh PowerPC application.

The amount of computer memory required for using DNACode is mainly determined by the size of the images you will scan and analyze. Some images—particularly those scanned at high resolution—can be quite large, and your hard disk may quickly run out of space. For this reason, we recommend that you archive images on a network file server or some kind of removable storage media after your analysis.

PC

The following are the **minimum** system requirements for installing and running DNACode on a PC:

Operating system:	Windows 95 / 98 / NT 4.0.
Processor:	Pentium 166.
RAM:	64 MB for Gel Doc and Fluor-S imaging systems. 128 MB for FX, Personal FX, and densitometers.
Hard disk space:	3 GB. Recommended: Removable storage media (such as an Iomega Jaz drive) or a network file server.
Monitor:	17" monitor, 1024 x 768 resolution, 256 colors.
SCSI:	Required to run all Bio-Rad imaging devices except the Gel Doc. Adaptec SCSI card with EZ-SCSI software.
Printer:	Optional.

Macintosh

Note: The default amount of memory assigned to DNACode on the Macintosh is 64 MB. If the total RAM in your Macintosh is 64 MB or less (the minimum recommended amount is 128 MB), you should reduce the amount of memory assigned to DNACode to 10 MB less than your total RAM. With the DNACode application icon selected, go to File > Get Info in your Finder to reduce the memory requirements for the application. See your Macintosh computer documentation for details.

The following are the **minimum** system requirements for installing and running DNACode on a Macintosh:

Operating system:	System 7.2 or higher.
Processor/Model:	PowerPC Macintosh 9500.
RAM:	64 MB for Gel Doc and Fluor-S imaging systems. 128 MB for FX, Personal FX, and densitometers.
Hard disk space:	3 GB. Recommended: Removable storage media (such as an Iomega Jaz drive) or network file server.
Monitor:	17" monitor, 1024 x 768 resolution, 256 colors.
SCSI:	Required to run all Bio-Rad imaging devices except the Gel Doc. Macintosh has built-in SCSI.
Printer:	Optional.

1.6. Installing DNACode

1.6.a. Installing DNACode on a PC

Before attempting to install and use DNACode on a PC, you should be familiar with your Windows 95/98 or Windows NT 4.0 operating system.

DNACode can be installed from a CD-ROM or you can download the installation program from the Internet. Make sure that Windows is up and running on your computer before attempting to install DNACode. We recommend that you check your hard disk with a disk utility program (such as Norton Disk Doctor) before loading DNACode.

Installing DNACode from a CD-ROM

Insert the Discovery Series™ CD-ROM into the CD-ROM drive on your computer.

Downloading from the Internet

You can download DNACode from the Internet using Bio-Rad's FTP site. Go to ftp://ftp.genetics.bio-rad.com/Public/Discovery Series/Windows/DNACode and click on DNACode.exe. This will download the installer onto your computer. Then you can double-click on the Setup.exe icon on your desktop to begin running the installer.

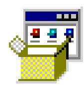


Fig.1-3. Setup.exe icon (Windows).

The Installation Program

The DNACode installation program will guide you through a series of screens. The installer will create a default directory tree under Program Files on your hard disk called Bio-Rad / The Discovery Series / Bin (you can select your own directory if you wish). The main DNACode program will be placed in the Bin directory. Additional directories for storing user profiles and sample images will also be created

The installer will place a DNACode icon on your desktop and create a Discovery Series folder in the Programs folder on your Windows Start menu.

Finally, after installation is complete, the installer will ask you if you want to open DNACode.

1.6.b. Installing DNACode on a Macintosh

DNACode can be installed on a Macintosh from a CD-ROM or you can download the installation program from the Internet.

Installing DNACode from a CD-ROM

Insert the Discovery Series CD-ROM into the CD-ROM drive on your Macintosh.

Fig.1-4. DNACode installation program icon (Macintosh).

Downloading the Installation Program from the Internet

You can download DNACode from the Internet using Bio-Rad's FTP site. Go to ftp://ftp.genetics.bio-rad.com/Public/Discovery Series/Macintosh/DNACode and click on Install DNACode. This will download the installer and place it on your desktop. Then you can double-click on the installer icon to begin running the installer.

The Installation Program

The DNACode installer will guide you through a series of screens. Once installation is complete, the DNACode folder will appear on your desktop.

1.7. Starting DNACode

A hardware protection key (HPK) has been provided with your DNACode software.

Most computers will not need an HPK to run DNACode. However, if you attempt to start DNACode and receive an "Unable to obtain authorization" message, turn off your PC or Macintosh, attach the HPK, and restart the software.

Note: Some parallel port devices such as zip drives may be incompatible with HPKs. Please check with your peripherals vendor.

1.7.a. Attaching the Hardware Protection Key

Before attaching the HPK, you should first attempt to start DNACode without the HPK. If DNACode opens successfully, you can skip this section.

PC

The code for the PC HPK is EYYCY. The code is printed in the lower left corner of the HPK label.

Before proceeding, please turn off your computer. If you have a printer attached to your computer's parallel port, please turn that off as well.

The HPK attaches to the parallel port on the back of your PC. If a printer cable is attached to this port, disconnect it. After you have attached the HPK, you can attach the printer cable to the key itself and restart your computer and printer.

You will also need to install the system driver that allows your DNACode software to "read" the HPK.

Note: Windows NT users must be in the local administrator group to install the HPK driver.

To install the driver, click on your Windows Start menu and select Programs > The Discovery Series. (This directory will not exist until you run the DNACode installer.) Select Install HASP HPK driver to begin installation. After the driver is installed, a message box will appear confirming the installation.

Macintosh

The code for the Macintosh HPK is QCDIY. The code is printed in the lower left corner of the HPK label.

Before proceeding, please turn off your Macintosh.

The Macintosh HPK must be inserted in the Apple Desktop Bus (ADB) path. The ADB is located on the back of your Macintosh.

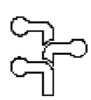


Fig.1-5. Apple Desktop Bus icon on back of Macintosh.

The HPK can be inserted at any point in the ADB path—between the computer and the keyboard, between the keyboard and the mouse, between the keyboard and the monitor, etc. After you have attached the HPK, you can restart your computer.

1.7.b. Opening DNACode

PC

The installation program creates a DNACode application icon on your desktop. To start the program, double-click on this icon.

Fig.1-6. DNACode application icon.

The installer also creates a Discovery Series folder in the Programs folder on your Windows Start menu. You can start DNACode by double-clicking on the icon in this folder.

Macintosh

After installation, the DNACode folder will be open on your desktop. To start the program, double-click on the application icon inside the folder.

1.8. Registering DNACode

When you open DNACode, the Main Registration Screen will pop up welcoming you to DNACode.

Note: If you receive a message saying "Unable to obtain authorization," attach the HPK if you haven't already (refer to the previous section). If you have already attached your HPK, make sure that it is attached correctly and (if you have a PC) your HPK driver is installed. If you continue to have problems, contact Bio-Rad technical service at the numbers listed in section 1.9., Contacting Bio-Rad, below.

The type of screen you see will depend on whether you have an HPK attached or not.

Without HPK

If you do not have an HPK attached, you can register to receive a 30-day free trial of the software.

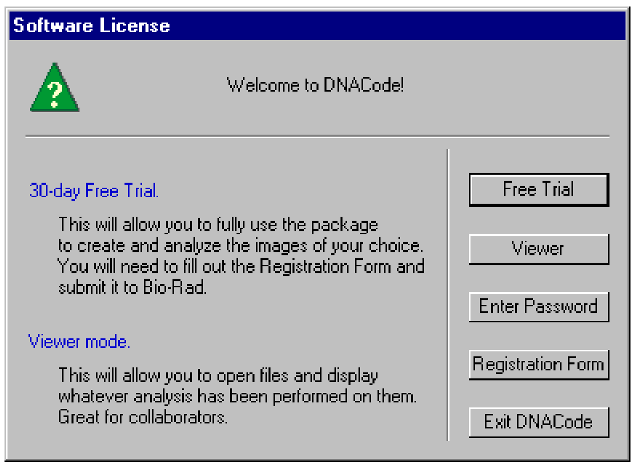


Fig.1-7. Main Registration Screen without HPK

Click on the Free Trial button. This will open the Software License Registration Form.

With HPK

If you have an HPK attached, the Main Registration Screen will reflect the fact that you receive a 30-day temporary license with your HPK (“Your license will expire on _______”).

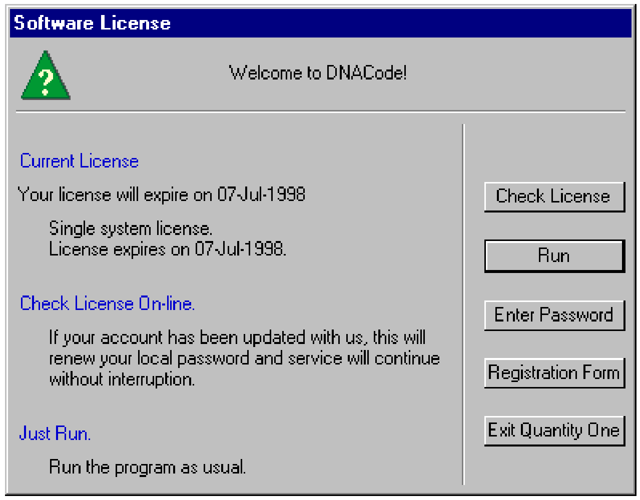


Fig.1-8. Main Registration Screen with HPK.

Click on the Run button to begin using the software.

Some time during the 30-day trial period, click on the Registration Form button to register your software. If your 30-day period has expired, a Free Trial button will appear when you open the software. Click on this button to register your software.

This will open the Software License Registration Form.

1.8.a. Software License Registration Form

To obtain your software license, you will need to fill out in the information in the Software License Registration Form.

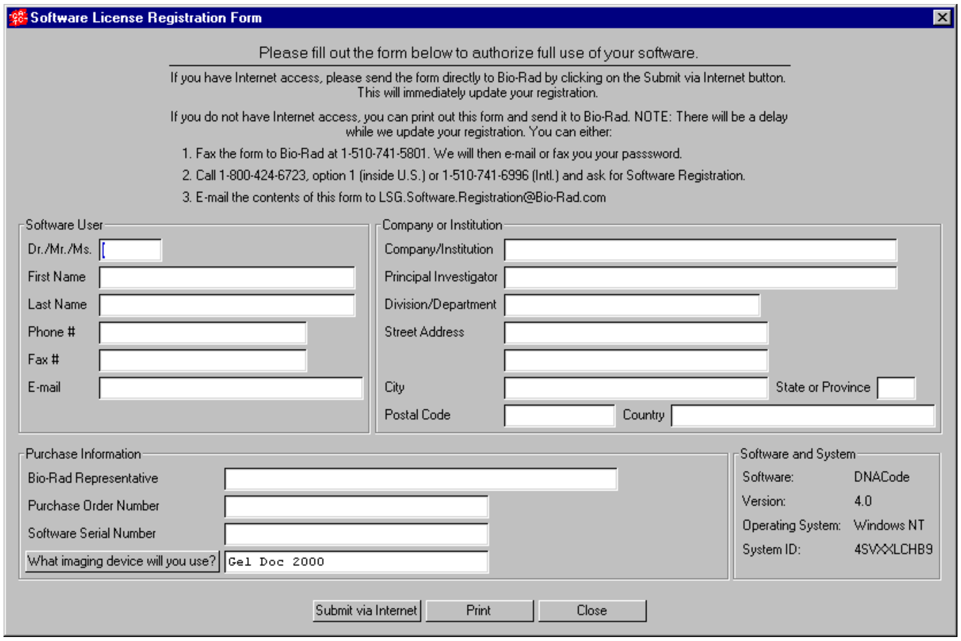


Fig.1-9. Software License Registration Form.

Note: If you do not yet have a Purchase Order Number or Software Serial Number, you may leave these fields blank to receive a trial license.

Submitting the Registration Form by Internet

If you have Internet access on your computer (the same computer on which you loaded DNACode), you can register quickly and easily.

In the Software License Registration Form, click on the Submit via Internet button.

Your information will be submitted automatically over the Internet, and a password will be generated automatically and sent back to your computer. Simply continue to run DNACode as before.

This password will be good for 30 days. During this period, if you have already submitted your Software Serial Number, click on Check License in the Main Registration Screen to update your license. (To access the Main Registration Screen from within DNACode, select Help > Register.)

If you have not yet submitted your Software Serial Number, open the form again, enter the serial number, and resubmit it over the Internet. In 1–2 days, click on the Check License button to update your license.

Submitting the Registration Form by Fax or E-mail

If you do not have Internet access, click on the Print button in the Software License Registration Form and fax the form to Bio-Rad at the number listed on the form. Alternatively, you can enter the contents of the form into an e-mail and send it to Bio-Rad at the address listed in the Registration Form.

If you include your Software Serial Number in the form, Bio-Rad will contact you by fax or e-mail in 2–3 days with a full license.

If you do not include your Software Serial Number in the form, Bio-Rad will contact you in 2–3 days with a temporary 30-day license. Then, some time during the 30-day period, enter the serial number in the form and fax the revised form to Bio-Rad. Bio-Rad will contact you by fax or e-mail in 2–3 days with a full license.

1.8.b. Entering a Password

If you submit your registration information over the Internet, you do not need to enter a password; it is done automatically.

If you fax your registration information, you will receive a password from Bio-Rad. You must enter this password manually.

To enter your password, click on Enter Password in the Main Registration Screen. If you are not currently in the Main Registration Screen, select Register from the Help menu in DNACode.

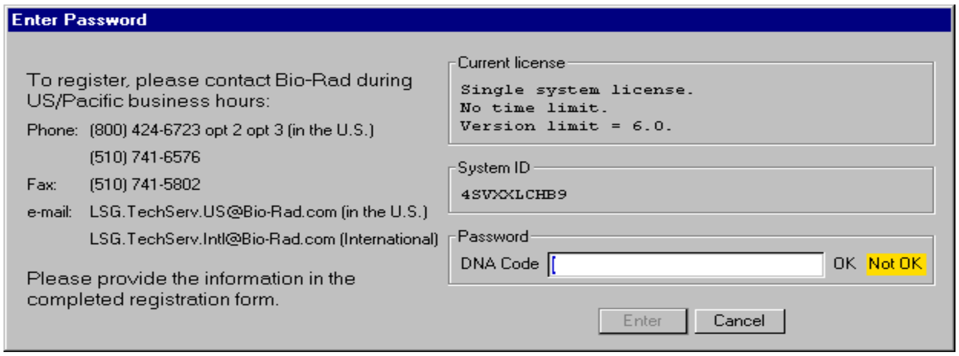


Fig.1-10. Enter Password screen.

In the Enter Password screen, type in your password next to the DNACode prompt.

Once you have typed in the correct password, the OK light next to the password field will change to green and the Enter button will activate. Click on Enter to run DNACode.

1.9. Contacting Bio-Rad

If you have problems with the HPK/password system, please contact Bio-Rad technical service. Technical service hours are from 8:00 a.m. to 4:00 p.m., Pacific Standard Time in the U.S.

Phone: (800) 424-6723, option 2, option 3

(510) 741-6576

Fax: (510) 741-5802

E-mail: LSG.TechServ.US@Bio-Rad.com (in the U.S.)

LSG.TechServ.Intl@Bio-Rad.com (International)

For software registration, phone:

(800) 424-6723, option 1, or (510) 741-6996

2. General Operation

2.1. The Graphical Interface

2.1.a. Menu Bar

DNACode has a standard menu bar with eight pulldown menus that contain all the major features and functions available in the software.

Fig.2-1. Menu bar.

The pulldown menus contain groups of related functions:

- File—General file commands (Open, Save, Print), imaging device acquisition windows (Gel Doc, GS-700, GS-710, Fluor-S, Fluor-S MAX, Personal FX, FX).
- Edit—Annotation tools, commands for selecting and describing lanesets, preferences.
- View—General viewing tools (Zoom Box, Grab), quick analysis tools (Density in Box, Plot Density), Transform function.
- Image—Image processing tools (Crop, Flip, Subtract Background).
- Sequence—Laneset and sequencing commands.
- Search—Commands for searching for bases and strings.
- Window—Commands for tiling windows.
- Help—On-line Help, keyboard commands, software registration.

Below the menu bar is the main toolbar, containing some of the most commonly-used commands. Next to the main toolbar are the status boxes, which provide information about cursor selection and toolbar buttons.

2.1.b. Main Toolbar

The main toolbar in DNACode appears below the menu. It includes buttons for the main file commands (Open, Save, Print), essential viewing tools (Zoom Box, Grab, etc.), and sequencing tools, as well as buttons for opening the secondary toolbars.

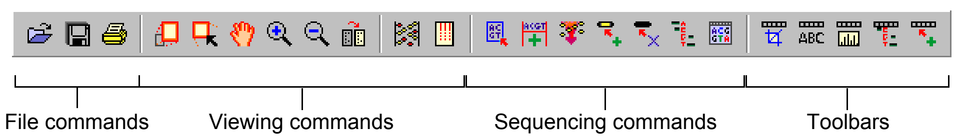


Fig.2-2. Main toolbar.

Tool Help

If you hold the cursor over a toolbar icon, the name of the command will pop up below the icon. This utility is called Tool Help. Tool Help appears on a time delay basis that can be specified in the Preferences dialog box under Edit > Preferences. You can also specify how long the Tool Help will remain displayed.

2.1.c. Status Boxes

There are two status boxes in DNACode. These appear to the right of the main toolbar.

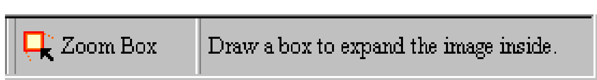


Fig.2-3. Status boxes.

The first box displays any function that is assigned to the mouse (see section 2.2., Mouse-assignable Tools). If you select a command such as Zoom Box, the name and icon of that command will appear in this status box and remain there until another mouse function is selected or the mouse is deassigned.

The second status box is designed to supplement Tool Help (see above). It provides additional information about the toolbar buttons. If you hold your cursor over a button, a short explanation about that command will be displayed in this second status box.

2.1.d. Secondary Toolbars

DNACode has five secondary toolbars, which contain icons for groups of associated functions. You can open these toolbars from the main toolbar or from the View > Toolbars pulldown menu.

The secondary toolbars can be toggled between vertical, horizontal, and expanded formats by clicking on the resize button on the toolbar itself.

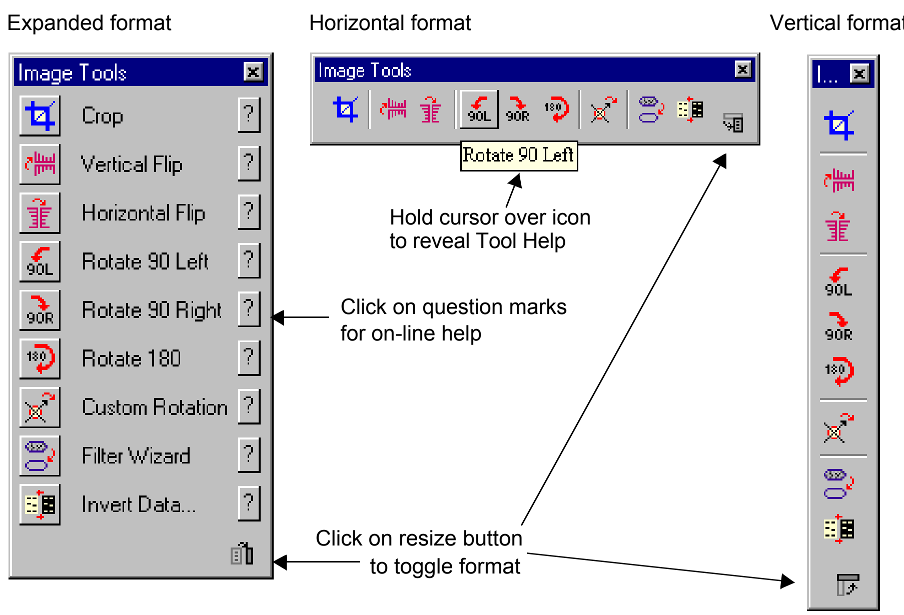


Fig.2-4. Secondary toolbar formats and features.

The expanded toolbar format shows the name of each of the commands. Click on the “?” icon next to the name to display on-line Help for that command.

2.2. Mouse-assignable Tools

DNACode has a number of commands that don't perform actions right away, but instead "assign" a function to your mouse (e.g., Zoom Box, Density at Cursor, Create Laneset). You first select one of these tools from a menu or toolbar, then execute the command by clicking or dragging on the image.

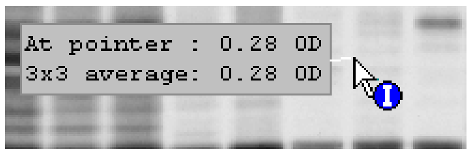


Fig.2-5. Example of a mouse-assignable tool: Density at Cursor

When a "mouse-assignable" function is selected, the cursor appearance changes. The name and icon of the function appear in the status box next to the main toolbar (see section 2.1.c).

Note: To *deassign* a function from the mouse, click on the toolbar button of the assigned function or click in the status box displaying the assigned function. You can also double-click on the Hide Overlays button.

2.3. Keyboard Shortcuts

Many of the functions in DNACode may be executed using keyboard shortcuts. These are listed in the Help menu under Keyboard Layout, as well as in Appendix H of this manual.

The pulldown menus also list the shortcut keys for the menu commands.

2.4. File Commands and Functions

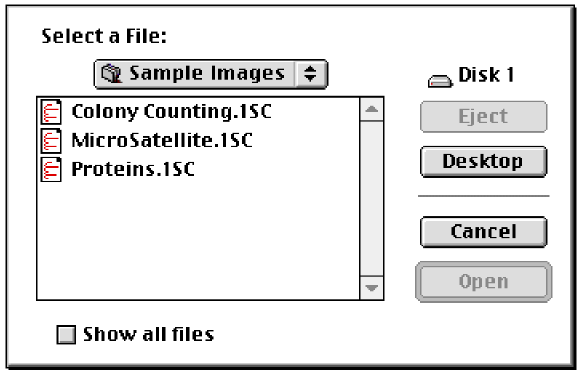
This section describes the basic file commands and functions of DNACode. These are selected from the File menu.

Fig.2-6. File menu.

Open

To open a previously saved image, select Open from the File menu or main toolbar. This opens the Open dialog box.

Macintosh version:



Windows version:

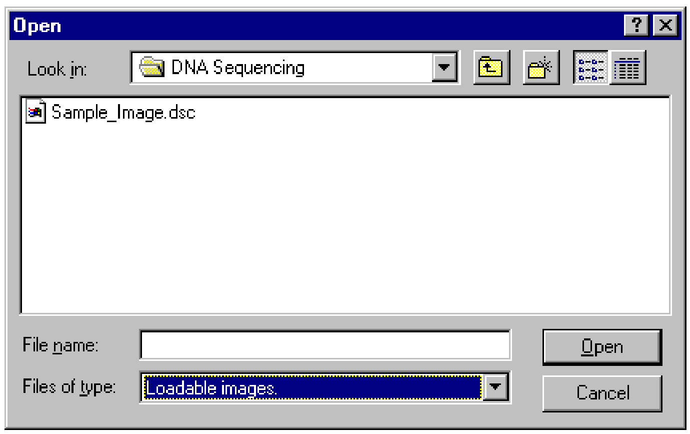


Fig.2-7. Open dialog box.

In the dialog box, open a file by selecting the file name and clicking on the Open button. To look for files in other directories, use the directory pulldown menu at the top of the dialog box.

An image created in the Windows version of DNACode can be opened in the Macintosh version, and visa versa. However, you must add a .1sc extension to your Macintosh files to open them in Windows.

Note: This version of DNACode will open any image created with an earlier version of DNACode.

DNACode comes with a sample image. In Windows, this is located in the Discovery Series / Sample Images / DNA Sequencing directory. On the Macintosh, it is stored in the Sample Images folder in the DNACode folder.

Opening TIFF Images

The Open command can also be used to import TIFF images from other software applications.

There are many types of TIFF formats that exist on the market. Not all are supported by the Discovery Series. There are two broad categories of TIFF files that are supported:

1. 8-bit Grayscale. Most scanners have an option between line art, full color, and grayscale formats. Select grayscale for use with the Discovery Series software. In a grayscale format, each pixel is assigned a value from 0 to 255, with each value corresponding to a particular shade of gray. Hewlett Packard™, Microtek™, and Sharp™ each make scanners that produce compatible 8-bit grayscale images.
2. 16-bit Grayscale. Bio-Rad's Molecular Imager (storage phosphor) systems use 16-bit pixel values to describe intensity of scale. Molecular Dynamics™ imagers also use 16-bit pixel values. The Discovery Series understands these formats and can interpret images from both Bio-Rad and Molecular Dynamics storage phosphor systems.

Note: DNACode can import 8- and 16-bit TIFF images from both Macintosh and PC platforms.

TIFF files that are *not* supported include:

1. 1-bit Line Art. This format is generally used for scanning text for optical character recognition or line drawings. Each pixel in an image is read as either black or white. Because the software needs to read continuous gradations to perform gel analysis, this on-off pixel format is not used.
2. 24-bit Full Color or 256 Indexed Color. These formats are frequently used for retouching photographs and are currently unsupported in the Discovery Series, although most scanners that are capable of producing 24-bit and indexed color images will be able to produce grayscale scans as well.

3. Compressed Files. The software does not read compressed TIFF images. Since most programs offer compression as a selectable option, files intended for compatibility with the Discovery Series should be formatted with the compression option turned off.

Close

To close an image or scan, select File > Close. If the file has changed since it was last saved, a message box will ask you if you want to save the changes before closing.

Close All

File > Close All closes all open images. If you have made changes to any of the images, a message box will open for each unsaved image giving you the option of saving those changes.

Save

File > Save will save a new image or an old image to which you have made changes.

In Windows, new DNACode images will be given a .dsc extension when they are first saved.

Save As

File > Save As can be used to save a new image, rename an old image, or save a copy of a image in a different directory.

In Windows, new DNACode images will be given a .dsc extension when they are first saved.

Save All

Save All on the main toolbar or File menu saves all images that are currently open.

In Windows, new DNACode images will be given a .dsc extension when they are first saved.

Revert to Saved

File > Revert to Saved will reload the last saved version of the image you are working on. This is a quick way to undo any alterations you may have made to an image since you last saved it.

Any changes you have made since last saving the file will be lost. (Also, all open dialog boxes will be closed.) A message box will warn you before completing the command.

File Info

File > File Info displays general information about your image, including the scan date, scan area, number of pixels in the scan, data range, and the size of the scan file. There is also a field where you can type in a file description or comments.

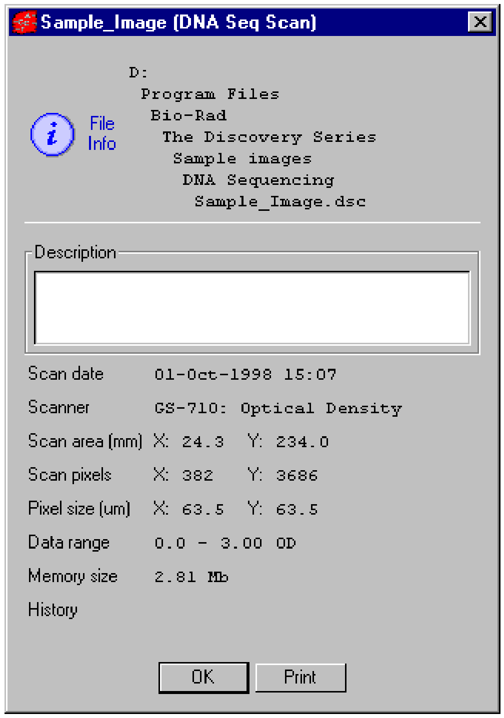


Fig.2-8. File Info box.

Reduce File Size

Image files can be quite large, and computer systems do not have unlimited memory or storage space. If you are having difficulty loading or storing a particular image, you may want to reduce the size of the file by reducing the number of pixels in the image. (You can also trim unneeded parts of an image to reduce its memory size. See section 3.7.a., Cropping Images.)

This function is comparable to scanning at a lower resolution, in that you are increasing the size of the pixels in the image, thereby reducing the total number of pixels and thus memory size.

Note: Reducing the file size of an image will result in some loss of resolution. In most cases this will not affect quantitation. In general, as long as the pixel size remains less than 10 percent of the size of the objects in your image, changing the pixel size will not affect quantitation.

Select File > Reduce File Size to open the Reduce File Size dialog box.

The Reduce File Size dialog box shows you the size of the pixels in the image (Pixel Size: X by Y microns), the number of pixels in the image (Pixel Count: X by Y pixels), and the memory size of the image.

As you increase the size of the pixels, the pixel count will decrease, as will the memory size. You can increase the pixel size in either dimension (see the following figure for an example).

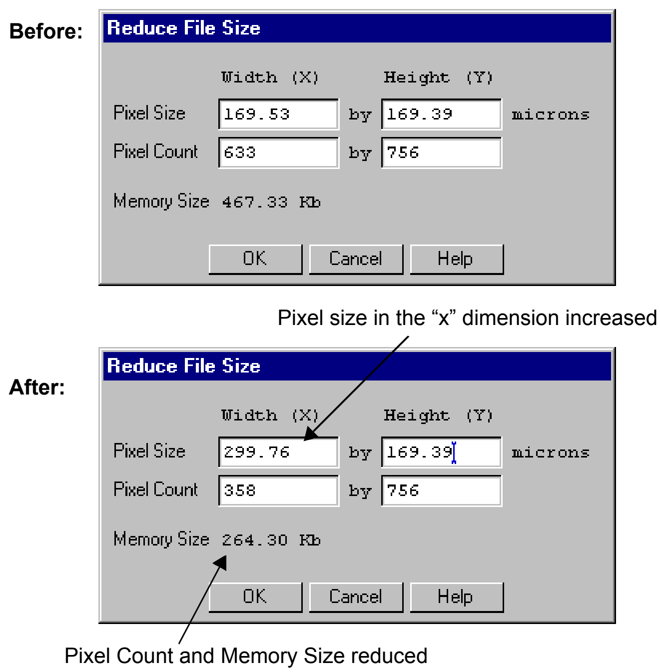


Fig.2-9. Reduce File Size dialog box, before and after pixel size increase.

Note: Since with most 1-D gels you are more concerned with resolving bands in the vertical direction than the horizontal direction, you may want to reduce the file size by making rectangular pixels. That is, keeping the pixel size in the "y" dimension the same, while increasing the size in the "x" dimension. This lets you decrease the total number of pixels and therefore file size without sacrificing detail.

When you are finished, click on the OK button.

A pop-up box will give you the option of reducing the file size of the displayed image or making a copy of the image and then reducing the copy's size.

Reducing the file size is an irreversible process. For that reason, we suggest you make a copy of the image and reduce its file size. That way, if you lose too much resolution, you can simple delete the copy and try again. Once you are happy with the reduced image, don't forget to delete the original. The goal is to save space!

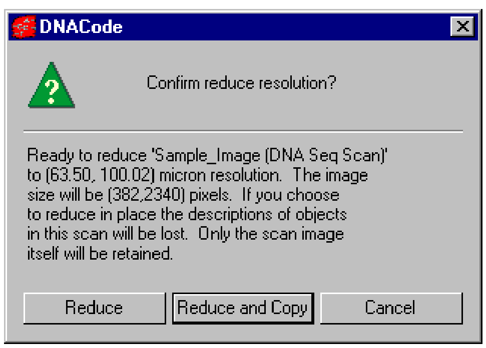


Fig.2-10. Confirm Reduce File Size pop-up box.

If you choose to make a copy of the image, you will be asked to enter a name for the new copy before the operation is performed.

Imaging Device Acquisition Windows

The File menu contains a list of Bio-Rad imaging devices supported by the Discovery Series software. These are:

1. Gel Doc 1000 / 2000
2. GS-700 Imaging Densitometer
3. GS-710 Imaging Densitometer
4. Fluor-S MultiImager
5. Fluor-S MAX MultiImager
6. Personal Molecular Imager FX
7. Molecular Imager FX

Selecting a name in the File menu will open the acquisition window that allows you to scan using that instrument.

See the individual appendices on the imaging devices for more details.

Exit

File > Exit quits the application.

2.5. Preferences

The Preferences dialog box (accessed under the Edit menu) allows you to customize basic features of your system, such as displays and toolbars.

After you have selected your preferences, click on OK to implement any changes you make.

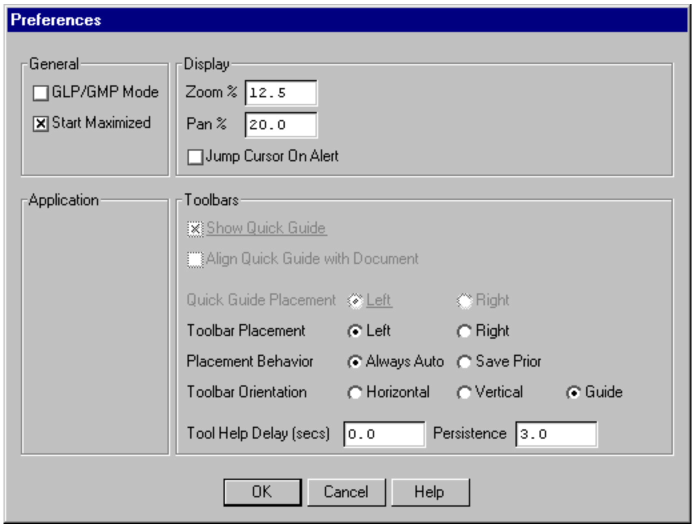


Fig.2-11. Preferences dialog box

2.5.a. General

GLP/GMP Mode

The GLP / GMP Mode checkbox allows you to prevent changes to an image that would change the raw image data. In GLP / GMP mode, the software will not allow the following operations to be performed (these are all located on the Image menu):

- Subtract Background
- Custom Rotation
- Filters
- Invert Data

If a user attempts to use any of these functions in GLP / GMP mode, he will receive a message that that function is not available.

To set GLP / GMP mode, click on the checkbox. A dialog box will pop up asking you to enter a password.

After you enter a password and click on OK, another dialog will ask you to reenter the password for confirmation. Retype the password and click on OK again.

To disable GLP / GMP mode, click on the checkbox to deselect it. The dialog box will pop up asking you to enter the password. When you click on OK, GLP / GMP Mode will be disabled.

Start Maximized

In the Windows version, the Start Maximized checkbox determines whether DNACode occupies the entire computer screen when first opened. If this is unchecked, DNACode's menu and status bars will appear across the top of the screen and any toolbars will appear "floating" on the screen.

2.5.b. Display

The Zoom % field allows you to specify the percentage by which an image moves closer or farther away when you use the Zoom functions. This percentage is based on the size of the image.

Pan % determines the percentage by which the image moves side to side or up and down when you use the arrow keys. This percentage is based on the size of the image.

When an alert box pops up in DNACode, you can set your cursor to automatically go to the OK button in the box by selecting Jump Cursor on Alert.

2.5.c. Toolbars

These settings determine the behavior and positioning of the secondary toolbars.

Toolbar Placement determines on which side of the screen the toolbars will first open.

The Placement Behavior setting determines whether a toolbar will always pop up in the same place and format (Always Auto), or whether it will pop up in the last location they were moved to and the last format selected (Save Prior).

The Toolbar Orientation option buttons specify whether toolbars will first appear in a vertical, horizontal, or expanded format when you open DNACode.

Tool Help Delay allows you to specify the amount of time (in seconds) the cursor must remain over a toolbar icon before the Tool Help appears. First-time users may want to specify a short delay to learn the names of the toolbar functions, while experienced users can specify a longer delay once they are familiar with the icons.

Tool Help Persistence determines how long the Tool Help will linger on the screen after you move the cursor off a toolbar icon.

2.6. User Settings

If DNACode is on a workstation that has multiple users, each user can have his or her own set of DNACode preferences and settings.

In multiple-user situations, DNACode's preferences and settings are associated with individual user names. On a PC with Windows 95/98/NT, your user name is the name you use to log onto the computer. On a Macintosh, your user name is the Owner Name under the File Sharing (OS 8.x) or Sharing Setup (OS 7.x) control panel.

Note: If you do not log onto your PC under Windows 95/98 or do not have a Owner Name on your Macintosh, then you do not have a user name and your preferences and settings will be saved in a generic file.

3. Viewing and Editing Images

This chapter describes DNACode's viewing tools, which you can use to magnify your images and clarify your data. This chapter also describes the tools for cropping, flipping, and rotating images, eliminating and filtering noise, and annotating images.

These tools may be found on the View, Image, and Edit menus.

Note: The following chapters contain instructions for analyzing X-ray films, wet and dry gels, blots, and photographs. For the sake of simplicity, these are all referred to as "gels."

3.1. Magnifying and Positioning Tools

The magnifying and positioning tools in DNACode are located on the View menu and Window menu; some of these functions are also found on the main toolbar.

These commands will only change how the image is displayed on your computer screen. *They will not change the underlying data.*

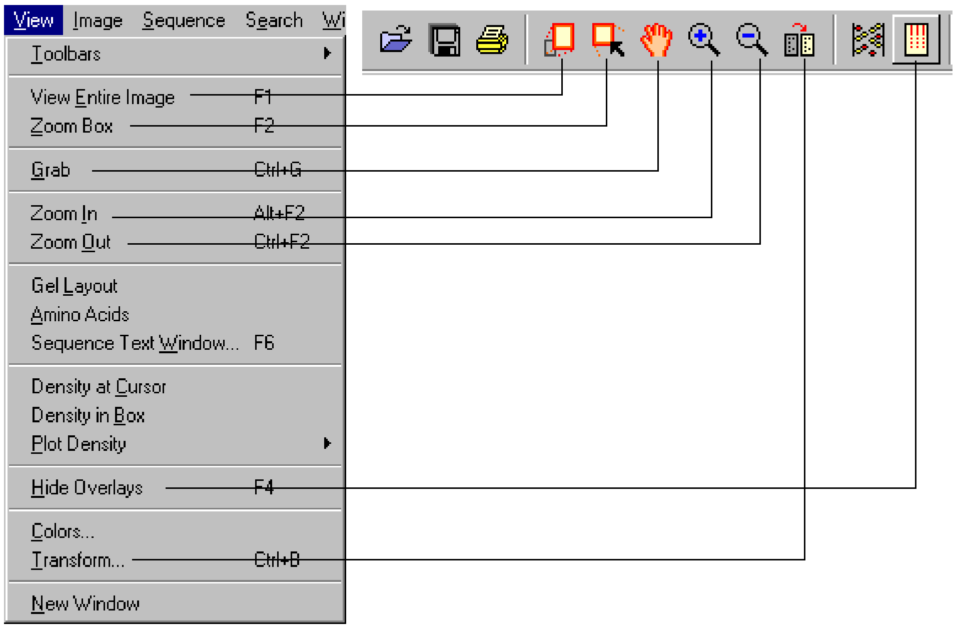


Fig.3-1. Viewing functions on View menu and main toolbar.

Zoom Box

Zoom Box allows you to select a small area of the image to magnify so that it fills the entire image window.

First, click on the Zoom Box button on your main toolbar or select View > Zoom Box. Your cursor arrow will change to a cross. Then drag the cursor across the image to enclose the area you want to magnify. The area of the image you selected will be magnified to fill the entire window.

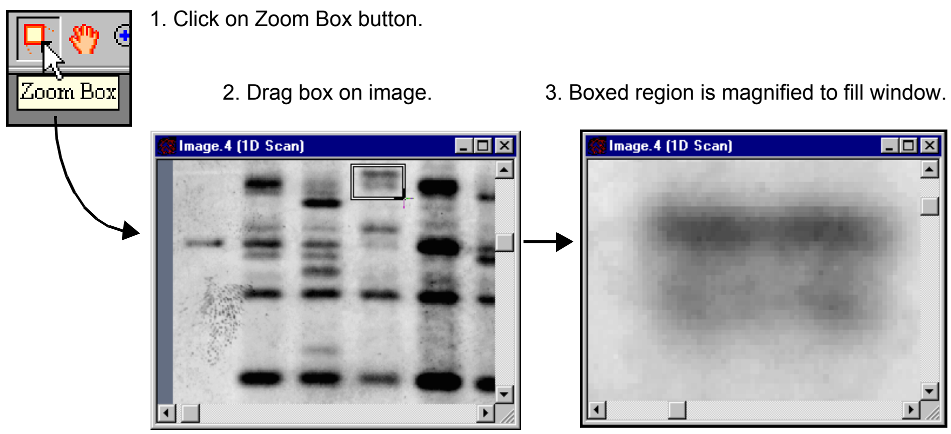


Fig.3-2. Zoom Box tool.

Zoom In/Zoom Out

These tools work like standard magnifying tools in other applications.

Click on the Zoom In or Zoom Out button on your main toolbar (or select from the View menu). Your cursor will change to a magnifying glass. Click on an area of the image to zoom in or zoom out a defined amount, determined by the setting in the Edit > Preferences dialog box.

Grab

This tool allows you to change the position of your image in the image window. Select View > Grab or click on the Grab button. Your cursor will change to a “hand” symbol. Dragging the Grab symbol on the image will move the image in any direction.

You can also move your image inside the image window by using the ARROW keys.

View Entire Image

If you have magnified part of an image or moved part of an image out of view, View Entire Image returns to the original, full view of the image.

Imitate Zoom

If you are comparing two or more images and want to zoom in on the same area on all of them, use the Imitate Zoom command.

First zoom in on one of the images. Then, with that image window still selected (the title bar will be blue), select Imitate Zoom from the Window menu.

The zoom factor and region of the selected image will be applied to all the images.

Note: Imitate zoom will only work on comparable images with similar dimensions.

Tiling Windows

If you have more than one image open, the Tile commands on the Window menu allow you to arrange your images neatly on the screen.

Tile will resize your windows and arrange them on the screen left to right and top to bottom.

Tile Vertical will resize your windows and arrange them side-by-side on the screen.

Tile Horizontal will resize your windows and stack them top-to-bottom on the screen.

3.2. Sequence Viewing Tools

The View menu contains several commands for viewing the results of your sequencing and base calling.

Gel Layout

View > Gel Layout displays any sequencing layout information present in a gel, including the gel's data area, lanesets, and base code order.

Amino Acids

View > Amino Acids divides the DNA sequence into codons and displays their complementary amino acids. As you edit individual bands, the amino acid sequence automatically updates to reflect changes in the base code.

Sequence Text Window

When you have finished editing the base calls for each laneset, you may want to review the results on-screen instead of sending the data to the printer. The Sequence Text Window command opens a window containing the sequence data in a text format.

Select View > Sequence Text Window and click on the laneset whose sequence you want to display. A pop-up window of the sequence will be displayed.

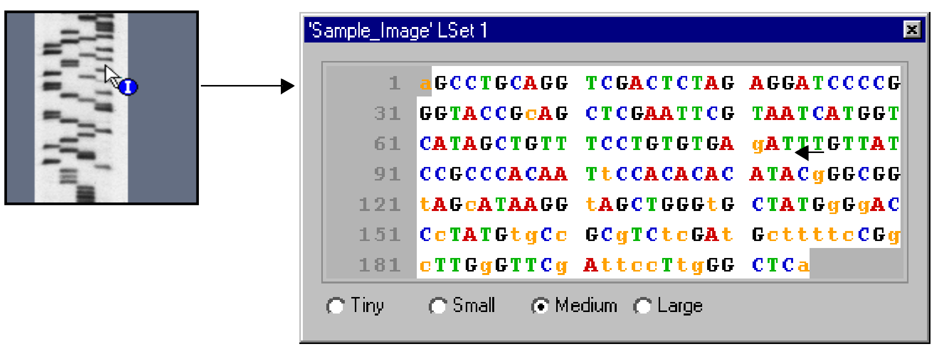


Fig.3-3. Click on a laneset to open the Sequence Text Window.

Unambiguous bases are designated with capital letters (A,C,G,T) while ambiguous bases are designated by the ambiguity code you have specified under Edit > Describe Gel (see section 4.3.).

If you point your cursor at any base in the laneset on the image, that base will appear marked with a box around the letter in the Sequence Text Window.

If only a portion of the laneset is displayed in the image window, only that section will be highlighted in the sequence text window.

If you have more than one laneset in your image, you can open multiple Sequence Text Windows. To conserve screen space, you can resize the text windows using the Tiny, Small, Medium and Large buttons at the bottom of the window.

3.3. Density Tools

The Density Tools (under the View menu and on the Density Tools toolbar) are designed to provide a quick measure of the signal intensity of the data in your image.

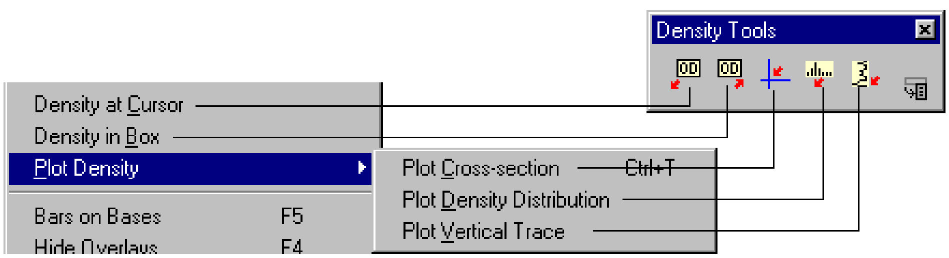


Fig.3-4. Density tools on the menu and toolbar.

Density at Cursor

Density at Cursor displays the signal intensity of the pixel on the image where you click your mouse. It also shows the average intensity for a 3 x 3 pixel box centered on that point.

To use this tool, click on the Density at Cursor button on the Density toolbar or select View > Density at Cursor. Then click on the point of interest on the image.

Density in Box

Density in Box displays the average and total intensity within a boxed region on the image. Select Density in Box from the Density toolbar or View menu, then define the region you want to measure by dragging your cursor across the image.

Plot Density Distribution

Plot Density Distribution displays a histogram of the signal intensity distribution for the part of the image displayed in the image window. The average intensity is marked in yellow on the histogram.

The histogram will appear along the right side of the image. Use the Zoom functions to display the histogram for magnified regions of your image.

Plot Cross-section

Plot Cross-section displays an intensity trace of a cross-section of the image centered on the point where you click your mouse. The horizontal trace is displayed along the top of the image, and the vertical trace is displayed along the side of the image.

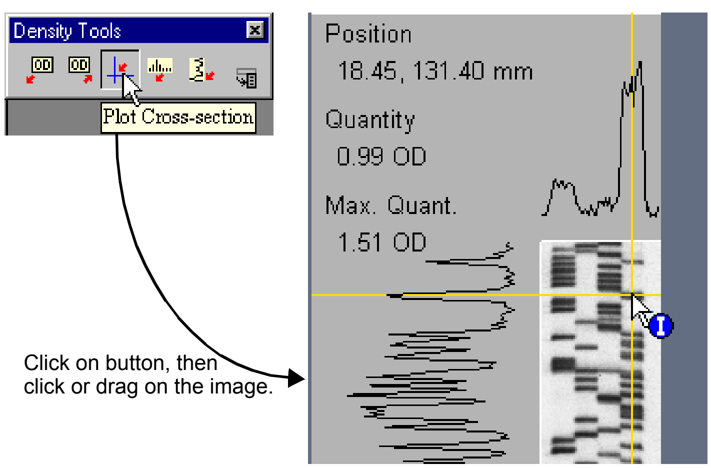


Fig.3-5. Plot Cross-section tool.

The intensity at the point you clicked on is displayed, as is the maximum intensity along the lines of the cross-section. Dragging the mouse with this function selected will continuously update the display.

Plot Vertical Trace

Plot Vertical Trace plots an intensity trace of a vertical cross-section of the image centered on the point where you click your mouse. Dragging the mouse with this function selected will continuously update the display.

Note: The sampling width for the Density traces is one pixel.

3.4. Hiding Overlays

As you create plots and traces and sequence bands on an image, you create a variety of image overlays. Hide Overlays on the main toolbar and View menu is useful for concealing any overlays you have created. It can also be used to erase plots, traces, and information boxes that are created by other functions.

Note: Clicking once on Hide Overlays will conceal the overlays. Clicking twice will deassign any function that has been assigned to the mouse.

3.5. Colors

View > Colors opens a dialog box in which you can adjust the colors of the image, windows, buttons, etc.

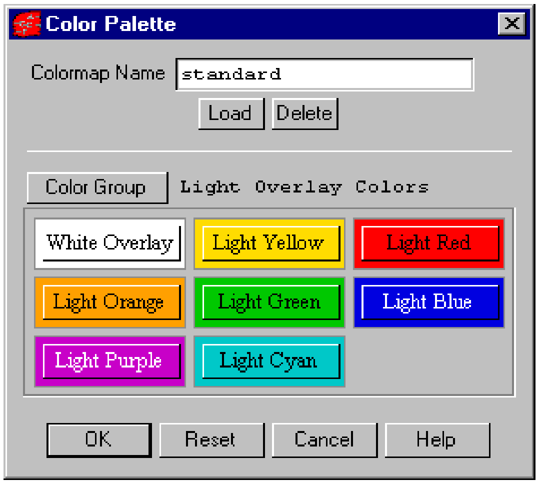


Fig.3-6. Colors dialog box.

Selecting a Color Group

Within the dialog box, the Color Group button allows you to select the colors of a particular group of objects (e.g., pop-up boxes, image colors, etc.). Click on the button to open the list of objects you can change.

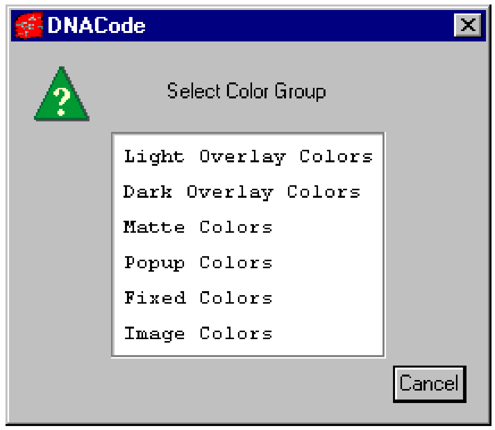


Fig.3-7. List of Color Groups.

Click on a color group in the list to select it.

Changing a Color

After you have selected the color group to change, click on the specific color button to change. The Color Edit dialog box will open, allowing you to adjust the red-green-blue (RGB) values of the color you selected.

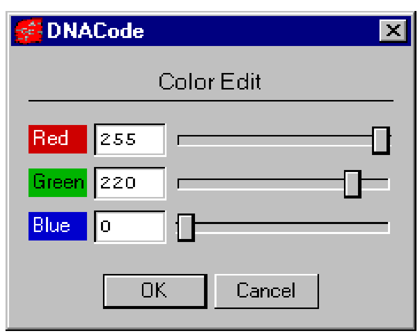


Fig.3-8. Color Edit dialog box.

Drag the sliders or enter a value in the fields. The color of the button will change with your adjustments.

Saving/Selecting a Defined Set of Colors

After you have changed the colors within color groups, you can save these settings for future use on other images. The Colormap Name field displays the name of a defined set of colors and color groups. There are several standard colormaps included with DNACode, or you can create your own.

To select a predefined colormap, click on the Load button.

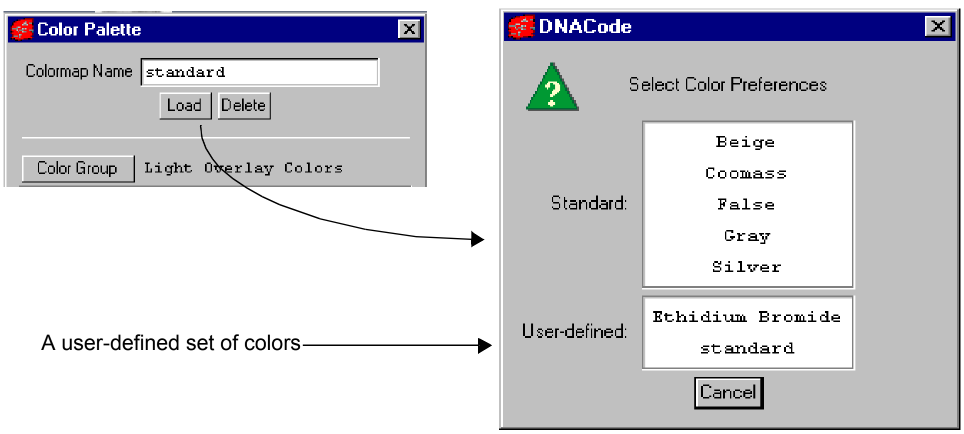


Fig.3-9. Selecting a Colormap.

From the list displayed, click on the set of colors you want to apply.

To create your own colormap, adjust the colors within the color groups as described above and type in a new colormap name. Click on OK to apply your changes.

To remove a colormap, click on the Delete button. Select the colormap to be deleted from the displayed list. A pop-up box will ask you to confirm the deletion.

If you change your mind about applying any changes you make, click on the Cancel button. If you want to return to the Standard colormap, click on the Reset button. All colors will be returned to their default values.

3.6. Transform

If features in your image are indistinct or fine details appear to be lost in background noise, you can use the Transform functions to optimize your displayed image. To open the Transform dialog box, select View > Transform.

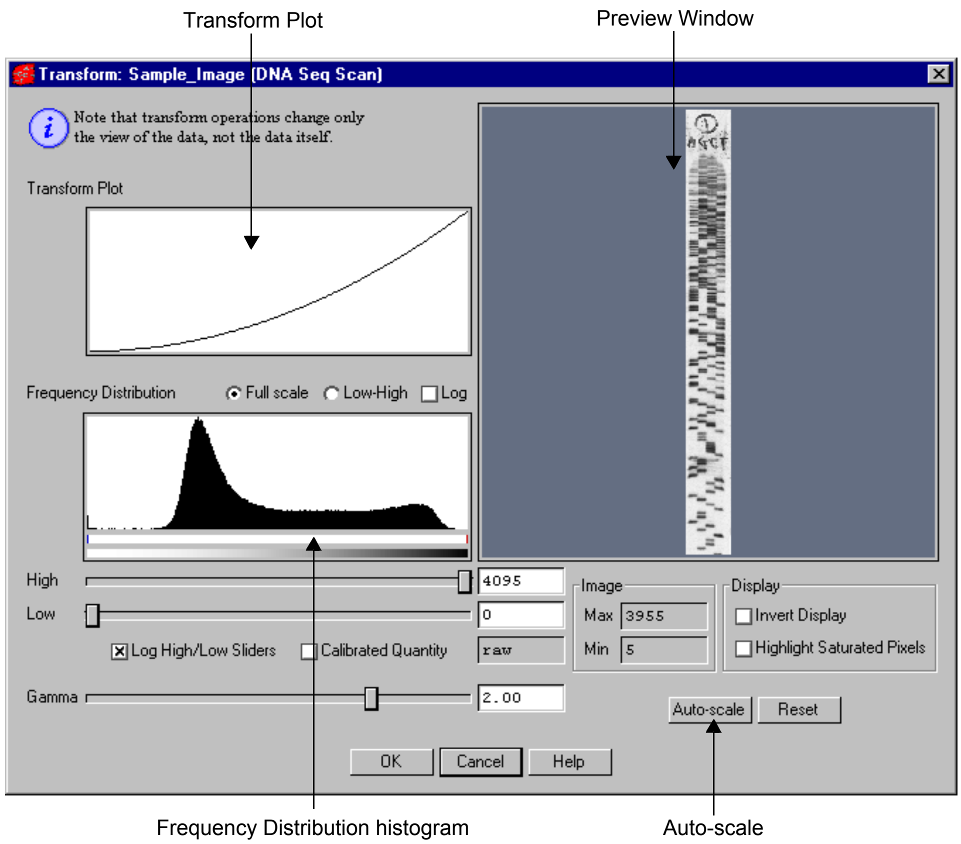


Fig.3-10. Transform dialog box.

The Transform dialog box contains a Preview Window, a Frequency Distribution histogram, a Transform Plot, and three main methods of optimizing your image: Auto-scale, High and Low sliders, and a Gamma slider. You can use these controls to adjust the way the software transforms raw data into visual data.

Note: Any changes made with the Transform controls will only affect how the image is displayed on your screen. *They will not affect the underlying data.*

3.6.a. Subwindows in the Transform Dialog

Preview Window

The Preview Window in the Transform dialog shows a smaller view of the same image that is displayed in the main image window. Changes in the Transform controls are automatically reflected in the Preview Window. They are only applied to the main image when you click on OK.

You can use view tools like Zoom Box and View Entire Image in the Preview Window just as you can in the main image window, to focus on particular regions of interest. You can also use the ARROW keys to move the image within the Preview Window.

Frequency Distribution Histogram

The Frequency Distribution histogram shows the total data range in the image and the amount of data at each point in the range. In a typical scan, there is a signal spike at the left (“gray”) end of the histogram due to background noise.

Transform Plot

The Transform Plot is a logarithmic representation of how the raw pixel data are mapped to the pixels of your computer screen.

3.6.b. Transform Controls

Auto-scale

Clicking on the Auto-scale button will optimize your image automatically. The lightest part of the image will be set to the minimum intensity (e.g., white), and the darkest will be set to the maximum intensity (e.g., black). This enhances minor variations in the image, making fine details easier to see. You can then “fine-tune” the display using the High, Low, and Gamma sliders described below.

High/Low Sliders

If Auto-scale doesn't give you the appearance you want, you can use the High and Low sliders to redraw the image yourself. Dragging the High slider handle to the left will make weak signals appear darker. Dragging the Low slider handle to the right will reduce background noise.

As you drag the sliders, the slider markers on the Frequency Distribution histogram will move. Everything to the left of the Low marker will be remapped to minimum intensity, while everything to the right of the High marker will be remapped to maximum intensity. Using the histogram, you can position the markers at either end of the data range in your image, and use the low slider to cut off the "spike" of background noise.

You can also type specific High and Low values in the text boxes next to the sliders. Clicking anywhere on the slider bars will move the sliders incrementally.

Log High / Low Sliders changes the feedback from the slider handles, so that when you drag them, the slider markers move a shorter distance in the histogram. This allows for finer adjustments when your data is in a narrow range.

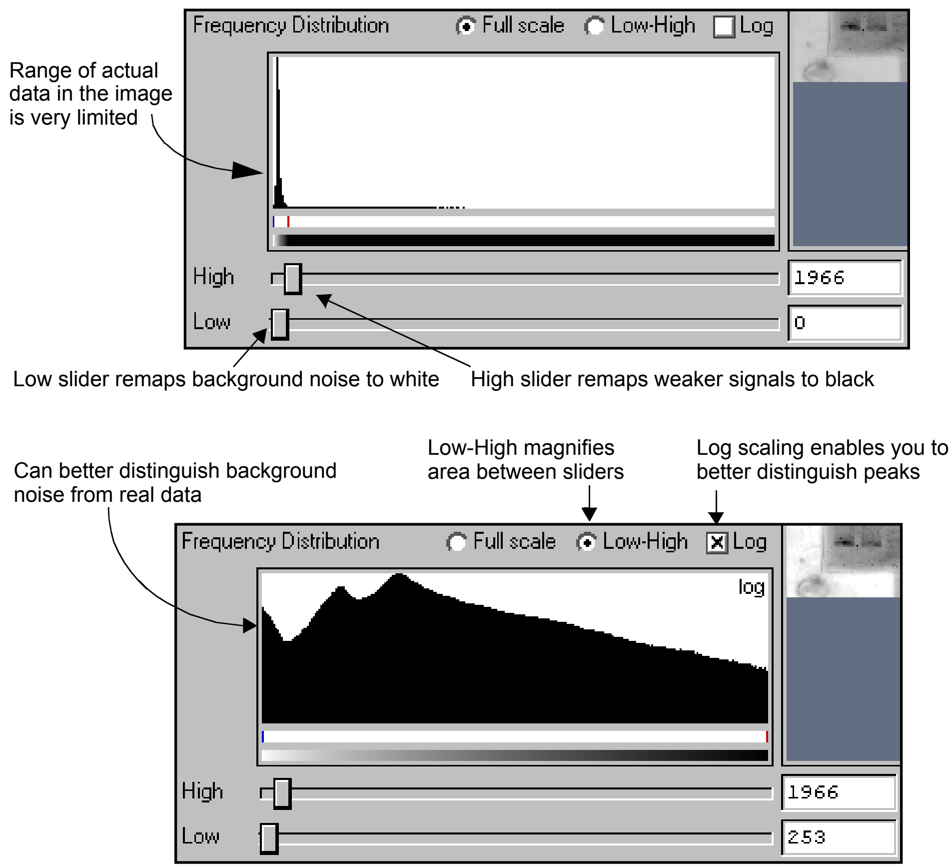


Fig.3-11. Two views of the Frequency Distribution histogram.

Gamma Slider

Some images may be more effectively visualized if their data are mapped to the computer screen in a nonlinear fashion. Adjusting the Gamma slider handle expands or compresses the contrast range at the dark or light end of the range, and this is reflected in the Transform Plot and Preview Window.

3.6.c. Other Features

Full Scale/Low-High

The Full Scale / Low-High option buttons adjust how the range of data in the image is displayed in the Frequency Distribution histogram and Transform Plot. They do not change how the data is displayed in the image window.

Selecting Full Scale adjusts the Frequency Distribution and Transform Plot displays so they show the full intensity range of the image.

Selecting From Low to High magnifies the range between the Low and High sliders. This makes it easier to view your data if it does not occupy the full intensity range of the image.

Log

The Log checkbox changes the way the data is displayed in the histogram so you can better discern subtle changes in signal intensity.

Image Max/Min and Units

Image Max and Min display the range of signal intensity in the image.

The image units are determined by the type of scanner used to create the image. In the case of densitometers, you can select Calibrated Quantity to display your image units in O.D.s.

Invert Display

The Invert Display checkbox flips light bands on a dark background to dark bands on a light background, and visa versa. Once, again, the actual data will not change—only the image.

Highlight Saturated Pixels

When the Highlight Saturated Pixels checkbox is selected, areas of the image with saturated signal intensity are highlighted in red.

Reset

If at any time you want to return to an unmodified view of the scan data, click on Reset.

3.7. Resizing and Reorienting Images

Frequently, your images will require some editing prior to analysis. This section describes the features that allow you to change the size and orientation of your images.

DNACode's image editing tools are located on the Image menu and the Image toolbar.

Note: Many of these image editing commands are irreversible. A pop-up alert box will ask you to confirm each edit that will change your image irreversibly.

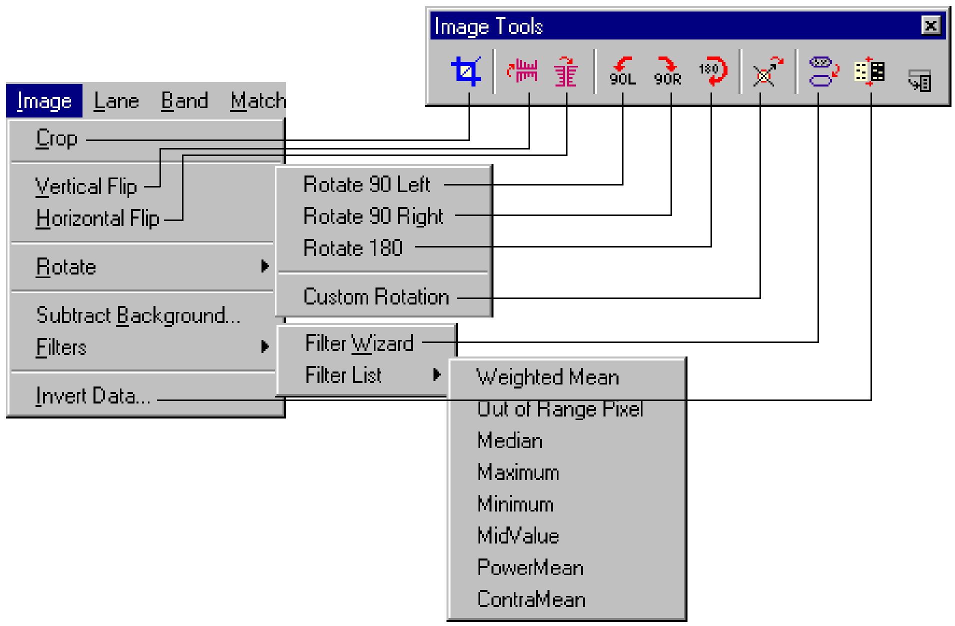


Fig.3-12. Image menu and toolbar.

3.7.a. Cropping Images

To eliminate unwanted parts of an image, you can use the Crop tool. This is also an easy way to reduce the memory requirements of an image.

To crop, click on the Crop button on the Image toolbar or select Image > Crop. Your cursor appearance will change to a Crop symbol.

Define the region to be cropped by dragging the cursor across the image, creating a box. Everything outside the box will be deleted.

Information about the physical dimensions of the crop area (given in millimeters and number of pixels) is listed at the bottom of the crop box, as is information about the memory size of the image inside the crop area.

1. To *reposition* the crop box, position your cursor at the center of the box. The cursor appearance will change to a multidirectional arrow symbol. You can then drag the box to a new position.

2. To *resize* the box, position your cursor on a box border or corner. The cursor appearance will change to a bidirectional arrow. You can then drag that border or corner in or out, resizing the box.

3. To *redraw* the box, position your cursor outside the box. The cursor appearance will change back to the Crop tool, and you can draw another box, replacing the one you just drew.

Once you are satisfied with your crop box, position your cursor within the box slightly off-center. The cursor appearance will change to a scissors symbol. You can then click on the mouse to perform the crop.

A pop-up box will ask you whether you want to: (1) crop the original image, (2) save a copy of the area inside the crop box as a separate image, keeping the original image intact, or (3) cancel out of the cropping operation.

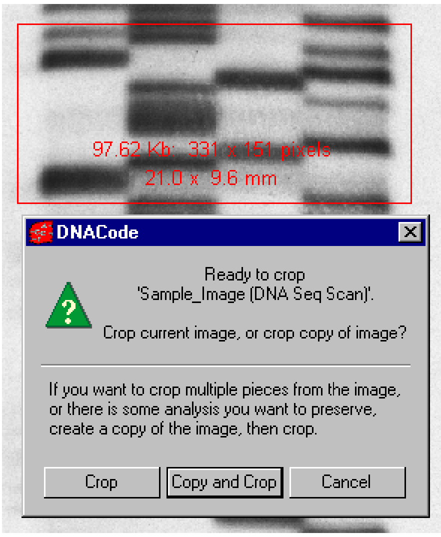


Fig.3-13. Crop box and pop-up Crop dialog.

If you select the Copy and Crop button, a dialog box will be displayed in which you can enter the name of the cropped image and its version number. Click on the OK button once you have finished.

3.7.b. Flipping and Rotating Images

If your image is not properly oriented, you can flip and / or rotate the image.

Note: These actions will erase any overlays you have created or analysis you have performed on an image. You will be asked to confirm the flip / rotation before the command is executed.

Flipping

To flip the image right-to-left, select Horizontal Flip from the Image menu or toolbar. To flip the image top-to-bottom, select Vertical Flip.

90° Rotations

Select Rotate 90 Left, Rotate 90 Right, or Rotate 180 from the Image > Rotate menu or Image toolbar to perform the specified rotation. You will be asked to confirm your choice before the command is executed.

Note: Any analysis performed on the image cannot be rotated and will be lost.

Custom Rotation

If you need to rotate your image in increments other than 90°, you can use the Custom Rotation command.

Select Custom Rotation from the Image > Rotate menu or Image toolbar. A green “plus” sign will appear next to your cursor. Click on the image you want to rotate and a circular overlay with an orange arrow will appear. A small dialog box also will open, indicating the angle of rotation in degrees and radians.

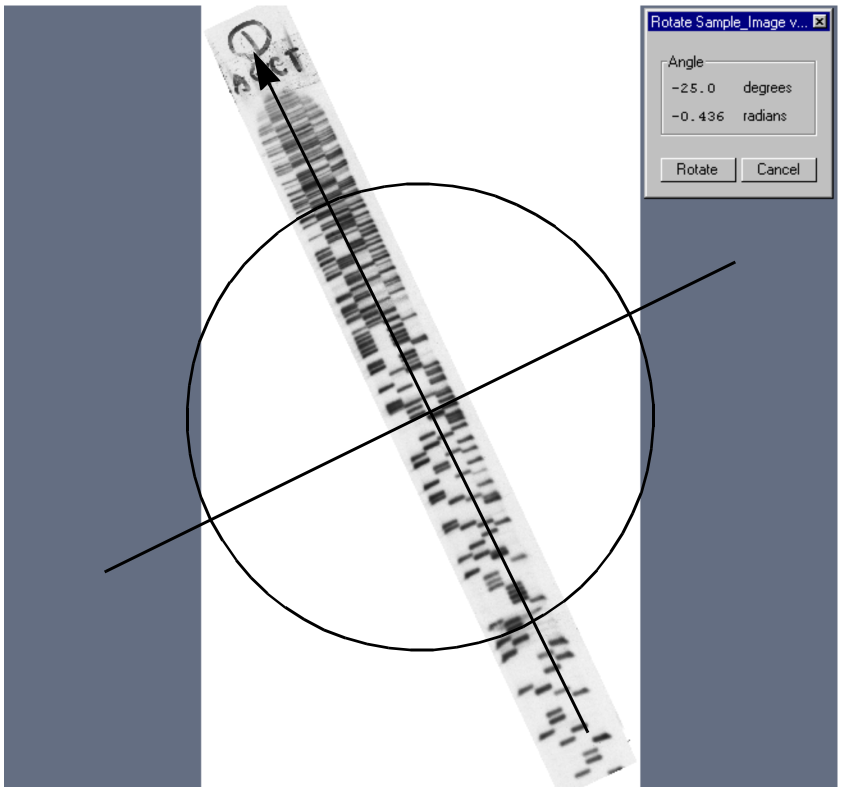


Fig.3-14. Custom rotation; the arrow points in the direction of the new top of image.

To perform the rotation, position your cursor on the arrowhead and drag. As you drag, the arrow will rotate and the angle in the dialog box will change. Position the arrow so that it points in the direction of the new top of the image. You can “fine-tune” your rotation as much as you like.

To complete the rotation, click on the Rotate button in the small dialog box. Another, smaller image window will open containing the rotated image. You will then have the option of renaming your new image and changing the version number.

If you are not satisfied with your rotated image, simply delete it and start over.

Note: Because an image is composed of square or rectangular pixels, Custom Rotation has to perform some minor smoothing on the image to turn it at a non-90^{o} angle. In addition, any analysis performed on the image cannot be rotated and will be lost.

3.8. Background Subtraction

Background subtraction removes noise from your gel to enhance the quality of your image and improve the accuracy of your results.

You may want to experiment with several different subtraction levels to see which works best for your gels. Since background subtraction is a nonreversible process, we suggest that you experiment on copies of the original scan. This way, if you are unhappy with a subtraction, you can simply delete that copy of the image while the original remains unchanged.

To perform a subtraction, select the Subtract Background command from the Image menu. This will open the Subtract Background dialog box.

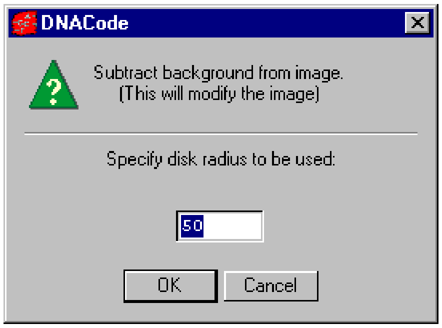


Fig.3-15. Subtract Background box.

Removing background in DNACode uses a process called rolling disk background subtraction. Depending upon the size of a user-defined disk, different amounts of background will be subtracted. A large disk will follow the general trends in background density, while a small disk will be sensitive to smaller fluctuations in density. The smaller the disk, the more density will be detected as background.

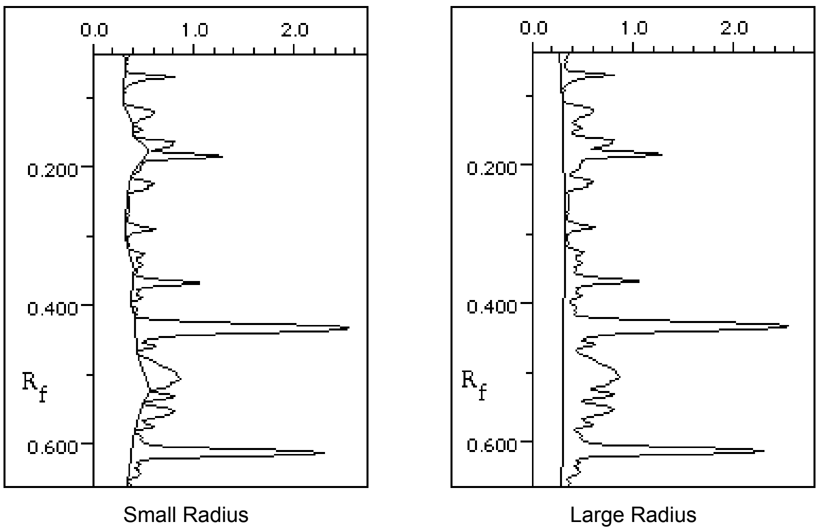


Fig.3-16. Example of the different amounts of density removed by rolling disk subtraction with a small disk radius, as compared to a large disk radius.

You can try different disk sizes to find the one that gives you the desired background subtraction results.

Larger radii result in slower execution and poorer removal of background. Typical values range from 50 to 150.

3.9. Filtering Images

Filtering is a process that removes small noise features on an image while leaving larger features (like data) relatively unaffected. DNACode has a wide range of filters for removing different types of noise from images. Depending on the nature of your data, you will probably need to use only one or two of the available filters. However, you should experiment with several different filters before selecting the ones that work best for your images.

Note: Since filtering is an irreversible process, you will be asked if you want to create a copy of the original image before you filter. If you are experimenting with various filters, you should create copies of your image and compare them side-by-side. If you filter the original image and save it, *you cannot return to the original, unfiltered state.*

3.9.a. Using the Filter Wizard

The Filter Wizard is designed to guide you through the filter selection process. First, you will identify the type of noise in your image. Next, you will select the size of the filter to use on that noise. Finally, you will filter the image.

To open the Wizard, select Image > Filters > Filter Wizard or click on the Filter button on the Image Tools toolbar.

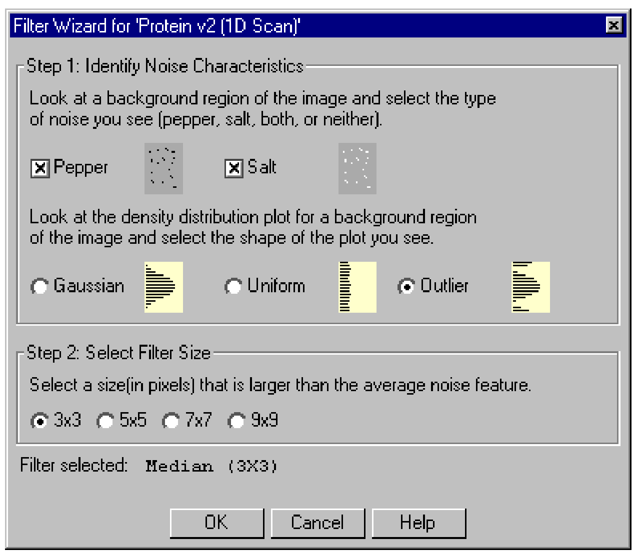


Fig.3-17. Filter Wizard dialog box.

When the Wizard dialog box opens, a vertical plot of the density distribution will pop up next to the image, showing the data range in the image and the amount of signal at each point in the range. (This is the View > Plot Density > Plot Vertical Trace histogram.)

This histogram can be used in conjunction with the Wizard to identify the type of noise in your image.

Step 1: Identify Noise Characteristics

The first step in the Wizard is to identify the type of noise in your image. The Wizard allows you to select from a variety of different noise types and combinations.

To determine the type of noise in your image, it is a good idea to first magnify a background region of the image (i.e., a region where there is no data) using the Zoom Box command on the main toolbar. Note that as you magnify a region, the density distribution histogram changes to reflect only the signal within the region you have magnified.

After examining the magnified region, select one, both, or neither of the check boxes described below.

- **Salt.** This type of noise appears as specks that are lighter than the surrounding background. The density distribution histogram of this type of noise displays noise peaks at the high end of the range. This type of noise is common in electronic cameras with malfunctioning pixels. It can also be caused by dust or lint in the imaging optics or scratches on photographic film. Salt is a type of outlier noise (see below).
- **Pepper.** This type of noise appears as specks that are darker than the surrounding background. The distribution histogram of this type of noise displays noise peaks at the low end of the range. Its causes are similar to those of salt noise. Pepper is a type of outlier noise (see below).

Next, select one of the following option buttons to describe additional features of your noise.

- **Gaussian.** The distribution histogram of this type of noise has a Gaussian profile, usually at the bottom of the data range. This type of noise is usually an electronic artifact created by cameras and sensors, or by a combination of independent unknown noise sources.

- **Uniform noise.** This type of noise appears in the histogram as a uniform layer of noise across the data range of the image.
- **Outlier noise.** This category of noise includes salt and pepper noise (see above). The distribution histogram of this type of noise displays noise peaks at the high and low ends of the range.

After you have identified the type of noise, go to Step 2.

Step 2: Select Filter Size

Image noise is filtered by means of a filtering window (or kernel), which is measured in pixels. This filtering window slides across the image, processing the pixels within it.

The available filter dimensions range from 3 x 3 pixels to 9 x 9 pixels. To select an appropriate size, magnify a background region of your image so that you can see the individual pixels. The filter size you select should be larger than the average noise feature but smaller than your data features.

Note: A smaller filter will alter your image less than a larger filter. Large filters can result in better suppression of noise, but can also blur desirable features in the image.

Step 3: Begin Filtering

After you have completed your selections, the filter name and size will be displayed at the bottom of the Filter Wizard dialog box.

To being filtering, click on the OK button. Because filtering is an irreversible process, a pop-up box will give you the option of filtering the original image, creating a copy of the image to filter, or cancelling out of this operation.

If you choose to Copy and Filter, you will be asked to enter a new name and / or version number for the new copy before the operation is performed. Once this information has been entered, the filtering operation will be performed.

3.9.b. Selecting a Filter Directly

If you already know the type and size of filter you want, you can select it directly by selecting Image > Filters > Filter List. The pulldown menu includes all the filters available in the Discovery Series.

The types of filters in the Discovery Series are:

- **Weighted Mean.** This filter is useful for reducing Gaussian noise. It calculates the weighted mean of the pixels within the filtering window and uses it to replace the value of the pixel being processed.
- **Out of Range Pixel.** This filter is useful for suppressing salt-and-pepper noise; its effect on Gaussian noise is minimal. This filter calculates the mean of the pixel values in the filtering window, including the pixel being processed. If the difference between the mean and the individual pixel value is above a certain threshold, then the individual value is replaced by the mean.
- **Median.** Also useful for suppressing salt-and-pepper noise, this filter calculates the median value of the pixels within the filtering window and uses it to replace the value of the pixel being processed. The median filter produces very little blurring if a small-sized window is selected.
- **Maximum.** This filter is useful for eliminating pepper noise in an image (it worsens the effect of salt noise). It replaces the value of the pixel being processed with the maximum value of the pixels within the filtering window.
- **Minimum.** This filter replaces the value of the pixel being processed with the minimum pixel value within the filtering window. This filter is useful for eliminating salt noise in an image (it worsens the effect of pepper).
- **MidValue.** This filter is useful for suppressing uniform noise within an image; however, it worsens the effect of pepper and salt. This filter replaces the value of the pixel being processed with the mean of the maximum and minimum pixel values within the filtering window.
- **Power Mean.** This filter is useful for suppressing salt and Gaussian noise within an image (it worsens the effect of pepper noise). It replaces the value of the pixel being processed with the power mean of the pixel values within the filtering window.

- **ContraMean.** This filter is useful for suppressing pepper and Gaussian noise within an image (it worsens the effect of salt). It replaces the value of the pixel being processed with the contra-harmonic mean of the pixel values within the filtering window.

To begin filtering, select a filter type from the pull-down list. A pop-up box will ask you to select a filter size.

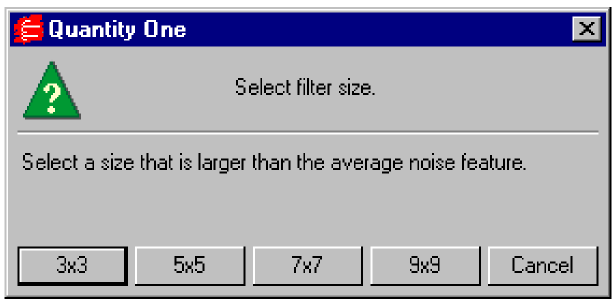


Fig.3-18. Selecting a filter size.

The available filter dimensions range from 3 x 3 pixels to 9 x 9 pixels. (See the previous section for guidance on selecting a size.)

Because filtering is an irreversible process, a pop-up box will give you the option of filtering the original image, creating a copy of the image to filter, or cancelling out of this operation.

If you choose to copy and filter, you will be asked to enter a new name and / or version number for the new copy before the operation is performed. Once this information has been entered, the filtering operation will be performed.

3.10. Invert Data

The Invert checkbox in the Transform dialog box (section 3.6.c.) inverts only the appearance of the image. In some cases, however, you may want to invert the actual image data.

If your image has light bands or spots on a dark background (i.e., the signal intensity of the background is greater than the signal intensity of the sample), you will need to use the Invert Data function before you can analyze the image.

This function is reversible, so if you change your mind, you can always switch it back.

To invert your image data, select Image > Invert Data. You may need to use the Transform function to adjust the appearance of your inverted image.

3.11. Annotating Images

If you want to create and display textual notes directly on your image, select Annotation Tools from the Edit menu or main toolbar.

This will open the Annotation Tools toolbar.

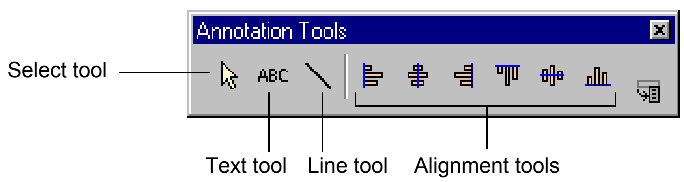


Fig.3-19. Annotation Tools toolbar.

To create an annotation, click on the Text Tool, then click on the image at the spot where you want the annotation to appear. This opens the Text Annotation Properties dialog box.

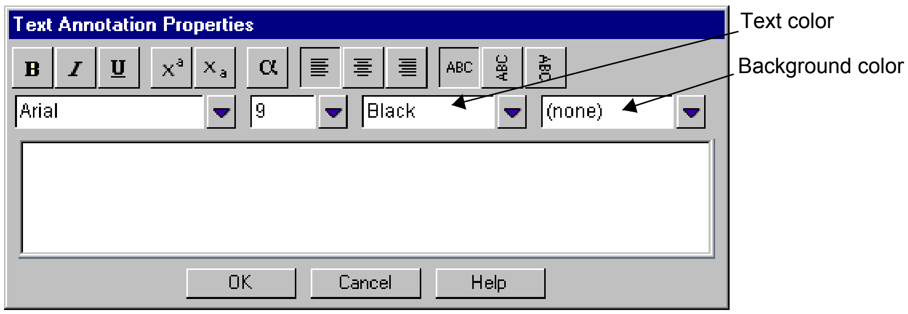


Fig.3-20. Text Annotation Properties dialog box.

To enter your annotation, simply begin typing in the main text field. The buttons in the dialog allow you to select the properties of your text, including format, alignment, and justification.

The pull-down boxes (from left to right) allow you to select the font style, font size, color of the text, and color of the background within the text box.

Once you have typed your annotation text, click on OK.

When you exit the Text Annotation Properties dialog by clicking OK, the annotation you just created will appear on the image at the spot where you originally clicked.

Editing, Positioning, and Copying Individual Annotations

You must have either the Select Tool or the Text Tool assigned to your mouse before performing the following commands. To assign one of these functions, click on the appropriate button on the Annotation Tools toolbar.

- To edit an annotation, double-click on it to open the Text Annotation Properties dialog. The existing text will be displayed and can be edited.
- To move an annotation, position your cursor on it and drag.
- To copy, hold down the CONTROL key while you drag the annotation.
- To delete, click on the annotation once to select it, then press the DELETE key.

Line Tool

You can use the Line Tool on the Annotation Tools toolbar to draw a line between an annotation and an image feature, or between any two points of interest on your image.

Click on the Line Tool button, then drag on your image to create the line.

- To resize or adjust a line, position your cursor on one end of the line (marked by a circle) and drag.
- To move a line, position your cursor on the middle of the line and drag.
- To copy, hold down the CONTROL key while you drag the line.
- To add arrowheads to a line, double-click on the middle of the line. A dialog box will pop up with options to add arrowheads to one or both ends of the line.
- To delete a line, click on it once to select it, then press the DELETE key.

Working with Multiple Annotations/Lines

To edit multiple annotations and / or lines, first click on the Select Tool button on your Annotation Tools toolbar. Then either drag a box around the objects you want to select or hold down the SHIFT key while you click on individual objects.

Note: When dragging a box around objects to be selected, make sure that the box entirely surrounds all the objects.

Each object in the selected group will have a green border.

You can drag the selected objects as a group, or click on the Alignment buttons on the Annotation Tools to align them.

Holding down the CONTROL key while you drag will duplicate all of the objects you have selected. Pressing the DELETE key will delete them.

You can create as many annotations and / or lines as you want.

Viewing Previously Created Annotations

To display previously created annotations after opening an image, select Annotation Tools from the Image menu.

If you have concealed all your overlays using Hide Overlays, clicking on any of the buttons on the Annotation Tools toolbar will display the hidden annotations.

3.12. Erasing All Analysis from an Image

If you want to delete all analysis that has been performed on an image (including lanes, bands, volumes, annotations, etc.), you can use the Clear Analysis command.

Select Clear Analysis from the Edit menu. Since this process is irreversible, DNACode will ask you for confirmation before completing the command.

4. Reading the Sequence

This chapter describes the steps for reading the nucleotide sequence in a laneset of your image.

4.1. Selecting the Data Area

If the image you scanned contains several lanesets and you want to work with a subset of them, you can specify a region to sequence with the Box Data Area command. This is located on the Sequence menu and main toolbar.

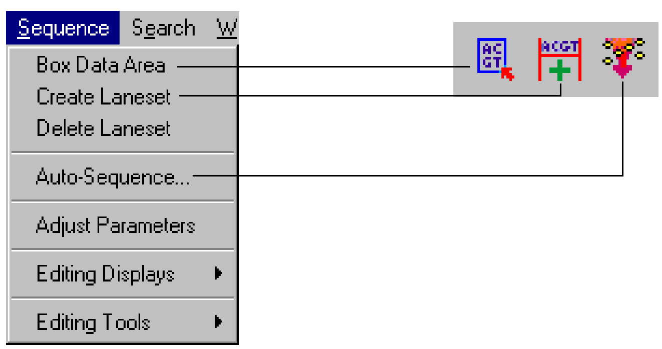


Fig.4-1. Sequence menu and tools.

This function is also useful for excluding regions on the image that are extraneous or unreadable. Such regions are often found at the very top and very bottom of a gel.

The data area should contain at least one laneset (i.e., four adjacent lanes, A,C,G,T—though not necessarily in that order).

The default data area is the entire image.

Select Box Data Area. Note that when you click on the image with this command assigned, the entire image is surrounded by a yellow box, indicating that the entire image is the default data area.

To resize or reposition the data area, place the cursor near one of the sides of the yellow data area box and drag the cursor. Two adjacent sides can be adjusted simultaneously by dragging a corner of the box.

If, after generating the sequence, you want to change the area to include other lanesets or re-sequence only one laneset, set the new data area to include the laneset(s) of interest. Resetting the data area will not erase or change your previous sequences or edits.

4.2. Defining Lanesets

Once you have selected the region of the image to work with, you are ready to specify lanesets. A laneset consists of four adjacent lanes on a gel, each containing one of the sequencing reactions: A, C, G or T.

To group a set of lanes together, select Create Laneset from the Sequence menu or main toolbar.

Magnify the top of the laneset using the Zoom Box tool, and drag a line from the left edge of the leftmost lane across to the right edge of the rightmost lane. It is a good idea to position the ends of the line as close the outside edges of the laneset as possible.

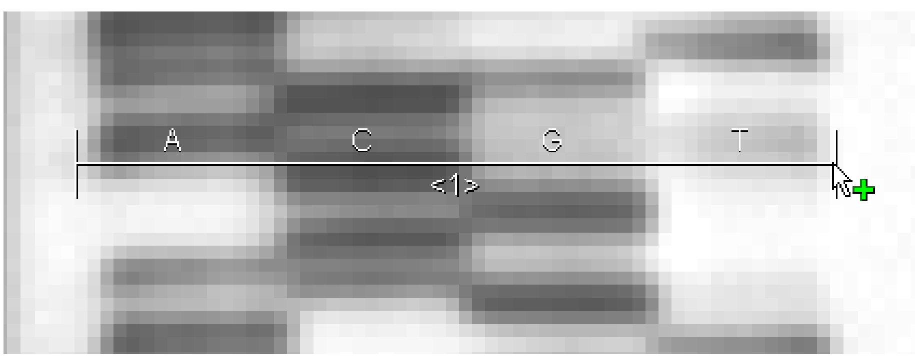


Fig.4-2. To define a laneset, select Create Laneset and drag cursor across lanes.

When you release the mouse button, a laneset anchor bar will appear across the laneset and the default lane assignments will appear across the top of the bar. The laneset number will be displayed below the bar.

Use View Entire Image to return to a full view of the laneset.

Note: If your lane assignments are different than the default, you can change the base code order using the Describe Gel command (see the next section).

Adding Anchor Bars to a Lane Set

If the lanes on your gel are perfectly straight and vertical, you may only need one laneset anchor bar to define a laneset.

In most cases, however, it is a good idea to place at least two anchor bars—one at the top and one at the bottom of the laneset.

If the laneset is distorted (wavy, curved, smiling, slanted, spreading wider towards the lower part of the gel. etc.), you may need to place several anchor bars at different points along its length. This helps DNACode accurately follow each of the lanes in the laneset.

First, magnify the bottom of the laneset using the Zoom Box command. Repeat the steps described above to create another laneset anchor bar.

After you draw a new anchor bar at the bottom of the laneset, border lines will appear down the length of the laneset connecting the ends of the two anchor bars.

Review the length of the laneset to verify that the border lines accurately follow the outer edges of the lanes. If the border lines do not follow the laneset, place other anchor bars between the top and bottom bars. The border lines will be redrawn, taking into account each new anchor bar.

To remove an anchor bar, select Delete Laneset from the Sequence menu and click on the anchor bar. The bar will be removed and the border lines will adjust accordingly.

If the laneset is defined by a single anchor bar, then deleting that anchor bar will delete the entire laneset.

4.3. Describing the Gel

Base code order, ambiguity codes, and other information about your gel and laneset can be specified with the Describe Gel and Describe Laneset functions, both located on the Edit menu.

4.3.a. Describe Gel

To specify the layout of your gel, select Describe Gel from the Edit menu. The Gel Description dialog box will open.

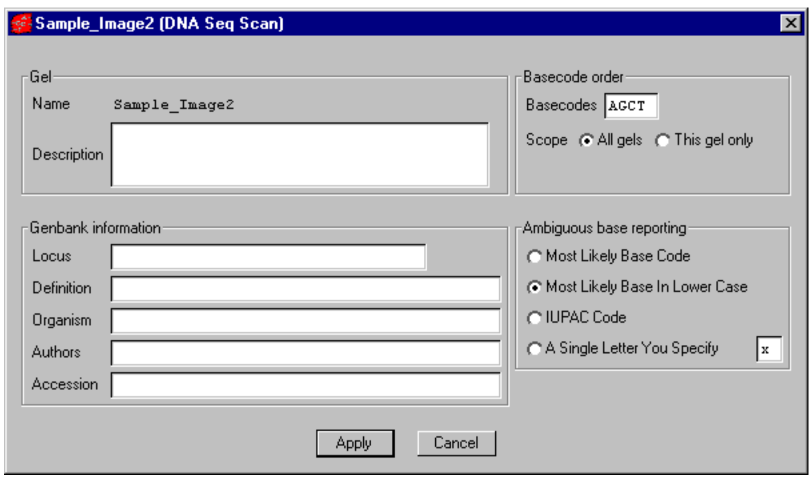


Fig.4-3. Gel Description dialog box.

Enter a description of your gel in the text box next to the Description prompt. Enter Genbank information for your gel in the text boxes next to the appropriate prompts.

Changing the Base Code Order

DNACode uses A, C, G, T as the default laneset order. If the lanes on your gel are not ordered A, C, G, T, you will need to enter the correct order next to the Basecodes prompt.

Note: You can change your basecode order after sequencing; the sequence will remain intact, but will be corrected to show the new, correct base assignments.

If you consistently use this order when running sequencing gels, you can select “All gels” next to the Scope prompt. The order you entered will become the new default order for all gels you open in DNACode.

If this order is unique to one gel, select “This gel only.” The order you entered will only be applied to the current gel, and the default order will not be altered.

Labeling Ambiguous Bases

When reading a sequencing gel, you often will find points where the base calls are ambiguous. For example, you may see bands of equal intensity in both the A and the C lanes at the same position on the gel. Should that base in the sequence be called “A” or “C”?

There are four methods for reporting ambiguous base calls. Indicate your preference by clicking one of the buttons next to the Ambiguous Base Reporting prompt.

Most Likely Base Code will denote ambiguous bases in the same style as unambiguous bases. Both will be displayed as a capital letter—A, C, G or T. On the screen, the ambiguous bases will be displayed in a different color than the unambiguous bases. However, on a printout of the sequence they will be indistinguishable.

Selecting Most Likely Base In Lower Case labels unambiguous bases in capital letters (A, C, G, T) and ambiguous bases in lower-case letters (a, c, g, t).

Selecting IUPAC Code will assign ambiguous bases according to IUPAC conventions.

IUPAC Code	Bases			
	A	C	G	T
a	✓			
c		✓		
m	✓	✓		
g			✓	
r	✓		✓	
s		✓	✓	
v	✓	✓	✓	
t				✓
w	✓			✓
y		✓		✓
h	✓	✓		✓
k			✓	✓
d	✓		✓	✓
b		✓	✓	✓
n	✓	✓	✓	✓

Fig.4-4. IUPAC Codes.

A Single Letter You Specify will report all ambiguous bases with the letter that you designate in the text box next to the button.

After entering the correct order and specifying the gel(s) to which it should be applied, click on the Apply button. Click on the Cancel button to exit the operation.

4.3.b. Describe Laneset

To enter a description for a particular laneset, select Describe Laneset from the Edit menu and click on a laneset. A dialog box will open in which you can enter information for that laneset.

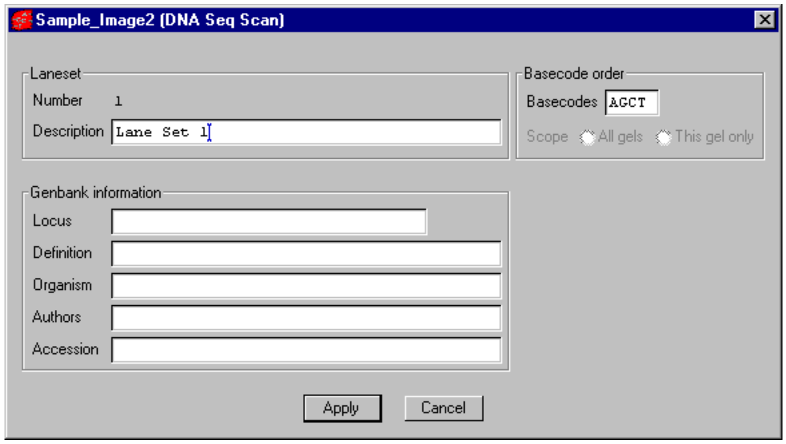


Fig.4-5. Laneset Description dialog box.

Enter a description of your laneset in the text box next to the Description prompt. Enter Genbank information for your laneset in the text boxes next to the appropriate prompts.

You can specify the basecode order of the lanes in a particular laneset. All other lanesets in the data area will continue to use the default order or the one entered in the Describe Gel dialog box.

This can be useful if you accidentally interchanged some lanes when loading your lanesets (e.g., you usually load your sequencing reactions A, T, G, C, but for one set of lanes you reversed the T and the G).

Note: You can change the basecode order after sequencing; the sequence will remain intact, but will be corrected show the new, correct base assignments.

When you are satisfied with the information entered in the dialog box, click on the Apply button.

4.4. Auto-Sequencing Bases

When all your lanesets are defined and their basecode order is set, you are ready to automatically read the sequence. Select Auto-Sequence from the Sequence menu or main toolbar. A pop-box will prompt you to select the auto-sequencing parameters.

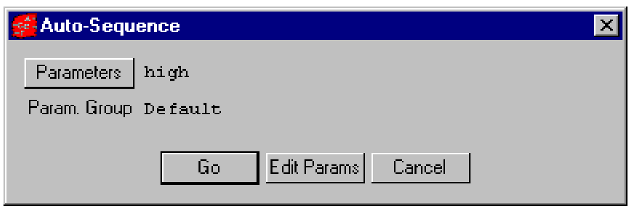


Fig.4-6. Auto-Sequence pop-up box.

The default sequencing parameters are set to “high.” To change these parameters, see below.

To proceed with sequencing, click on the Go button. DNACode will perform a series of steps in the process of reading the sequence. The sequencing progress is indicated with messages in the status bar next to the main toolbar.

Once the sequence has been determined, it is displayed on the image.

Auto-Sequencing Parameters

To select a different set of sequencing parameters, click on the Parameters button. There are three predefined auto-sequence parameter selections: high, standard, and low. These determine the level of sensitivity with which DNACode reads your lanesets.

To examine or edit these parameters, click on the Edit Params button. This will open the Auto-Sequence Parameters dialog box.

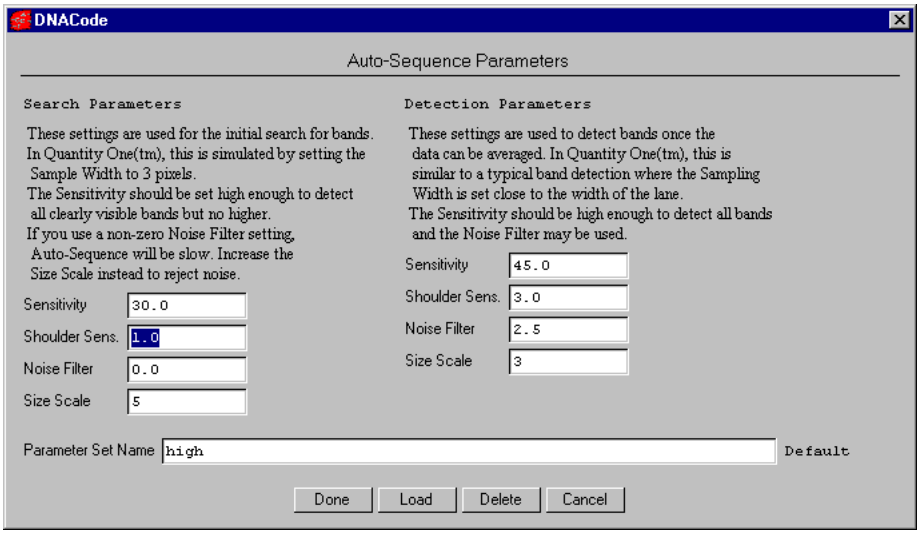


Fig.4-7. Auto-Sequence Parameters dialog box.

The Auto-Sequence function uses two band detection sweeps to determine the sequence of your gel. Parameters for the first sweep can be adjusted in the Search Parameters column on the left side of the dialog box. Parameters for the second sweep can be found in the Detection Parameters column on the right side of the form.

The Search pass uses a 3-pixel sampling width to initially locate bands inside the data area. Because the sampling width is small, using the Noise Filter in the Search sweep will slow the Auto-Sequence function considerably. We suggest that you use the Size Scale parameter instead to reject noise in the first sweep.

The Detection sweep uses a larger sampling width to detect bands once their data can be averaged. In this pass, the sensitivity should be set high enough to detect all the bands and the Noise Filter may be used to eliminate gel background noise.

If you want to experiment with your own sensitivity parameters, you can change the parameters and save your changes by entering a name in the Parameter Set Name text box.

User-defined parameters can be removed by clicking on the Delete button at the bottom of the dialog box. (Predefined parameters cannot be deleted.)

When you are satisfied with the sensitivity parameters for the auto-sequencing function, click on the Done button to return you to the Auto-Sequence box.

4.5. Determining Optimal Auto-Sequence Parameters

If you are having difficulty getting the auto-sequence detector to function optimally on a particular gel, there is a protocol that you can follow to determine the best set of search and detection parameters to enter into the Auto-Sequence Parameters form. Auto-sequencing with parameter values attuned to your particular gel can greatly increase the performance of the detector. Auto-sequencing with incorrect values can result in incorrect base calling and longer detection times.

Since the search and detection sweeps behave differently, they require different steps for determining their optimal detection parameters. Protocols for both types of sequencing are given here.

Note: Both procedures require using either Quantity One or Diversity Database gel analysis software. Neither procedure can be performed using DNACode.

Determining Search Parameters

For more detailed instructions on using the following commands, refer to your Quantity One or Diversity Database User Guide.

1. Start Diversity Database or Quantity One and open the image file of the DNA sequencing gel that you want to analyze.

2. Using the File > Save As command, make a copy of the DNA gel image file with a different name.

3. Working with the copied image, use Frame Lanes and Add / Adjust Anchors (under the Lane menu) to create a lane matrix of approximately 20 lanes over the portion of the gel laneset that you want to auto-sequence. There should be at least three lanes intersecting every band in

the laneset, as well as several lanes that do not intersect any band (see figure below).

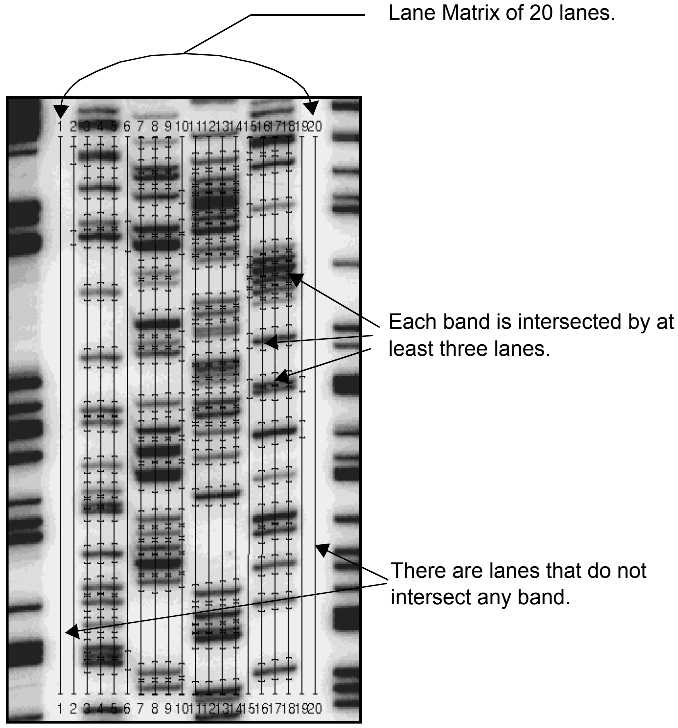


Fig.4-8. Example of the lane matrix.

4. Select Detect Bands (Bands > Detect Bands). The Detect Bands form will appear on-screen next to the lane matrix.

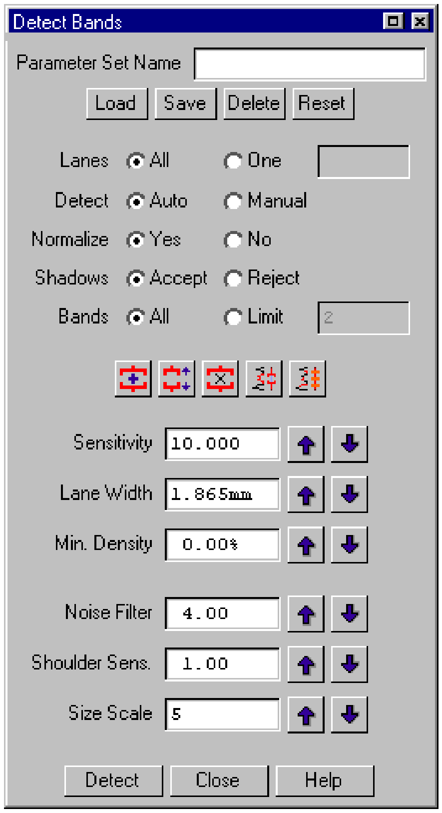


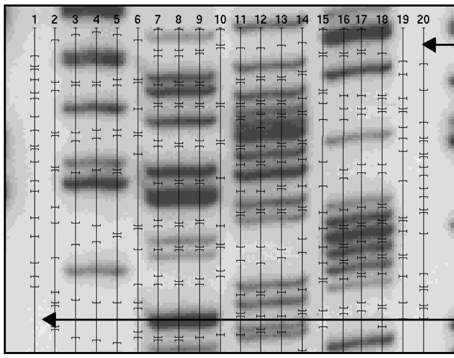
Fig.4-9. The Detect Bands form.

You need to adjust the sampling width of each lane prior to band detection. The sampling width should equal the width of three image pixels (in millimeters). To determine the width of one pixel in your image, select File > File Info. In the File Info dialog box, look at the values next to the Pixel Size prompt. The X value (width) of the pixels in the image is listed in micrometers. Convert this number to millimeters (divide by 1,000) and multiply the value by 3 to determine your sampling width. Enter the new width next to the Lane Width prompt in the Detect Bands form.

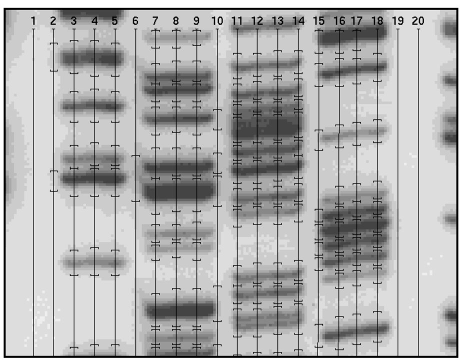
5. You can use the Detect Bands form to experiment with different parameter settings values until you find the numbers that best work for your bands. You then enter these numbers into the Auto-Sequence Parameters form in DNACode. While testing the other parameter settings, be sure that the Normalization, Shadow Rejection, and Band Limit features are turned off and the Minimum Density is set to 0.00%.

Note: Set the Noise Filter to zero when determining Search Parameters, as the use of Noise Filter values greater than zero during the first auto-sequence sweep will excessively increase the time required to sequence your gel. Appropriate Noise Filter values can be determined later, when establishing Detection Parameters.

Sensitivity levels should be set high enough so that all clearly visible bands can be detected, but no higher.



Example 1: The false detection of background noise as bands indicates that the Sensitivity levels are too high.



Example 2: All clearly visible bands have been detected, with no background noise, indicating that the Sensitivity levels are correctly set.

Fig.4-10. How to set Sensitivity levels.

6. Correct values for the Size Scale parameter can be determined by using the Zoom Box function to magnify the image of a single band until the height of the band (in pixels) can be measured by eye.

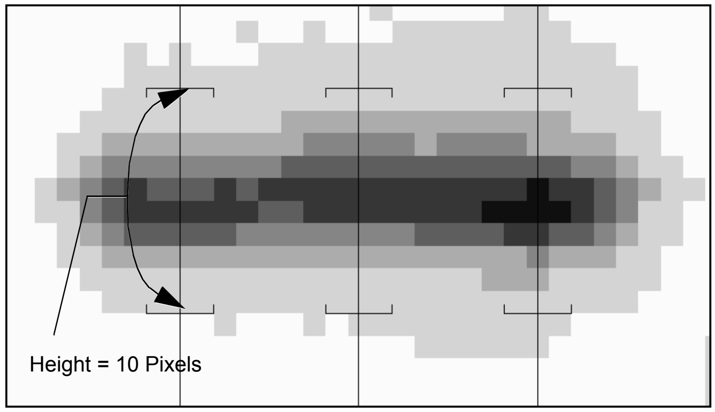


Fig.4-11. How to determine Size Scale.

Size Scale is equal to (height of a band in pixels) / (2 or 3). So for the band in Figure 5-12, with a height of 10 pixels, the appropriate size scale value would be between 5 and 3. Use the largest value that still gives good band resolution; too large a value will reduce the ability to discriminate closely spaced bands.

Determining Detection Parameters

1. Select Frame Lanes and Add / Adjust Anchors from the Lane menu to create a new lane matrix consisting of four lanes, placed over the four lanes of your DNA sequencing gel.
2. Set the sampling width approximately equal to the width of an entire band.
3. Use the Detect Bands feature to find the bands on the lane matrix (see below).

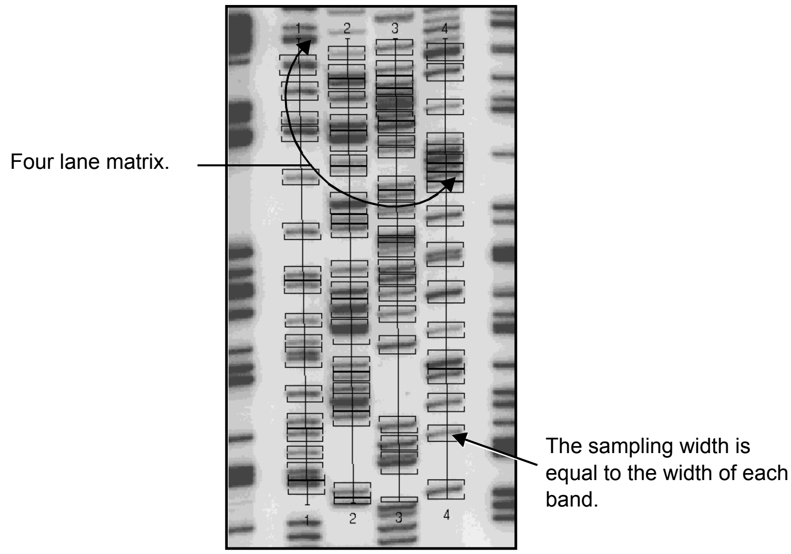


Fig.4-12. The four lane matrix used for determining Detection parameters.

You can use the Detect Bands form to experiment with different parameter settings values until you find the numbers that work best, including values for the Noise Filter. These numbers can then be copied for use on the Detection Parameters part of the Auto-Sequence Parameters form.

Note: On gels where the bands are slanted, the Band Detection features in Quantity One or Diversity Database may not produce good band separation. DNACode, however, can correct for the slant; simply use a smaller Size Scale value in the Detection (second) pass than in the Search (first) pass.

Size Scale values must be odd numbers; 3 is the lowest Size Scale value that typically gives good results. The Noise Filter should be set to a value equal to or less than the Size Scale.

5. Editing the Sequence

This chapter describes the tools for editing the results of auto-sequencing.

5.1. Selecting Lanesets to Display/Edit

Edit > Select All Lanesets will display the sequences of all the lanesets in your gel.

If you want to display the sequences of individual lanesets in your gel, you can use the Select Laneset tool.

Select Edit > Select Laneset, then click on a laneset whose sequence you want to display. Repeat this procedure until you've selected all the lanesets whose sequences you want to display.

Clicking on a selected laneset again with the Select Laneset tool will deselect it.

5.2. Displaying the Sequence

There are several functions for displaying a sequence on-screen. These are found on the Sequence > Editing Displays submenu and on the Display Tools toolbar.

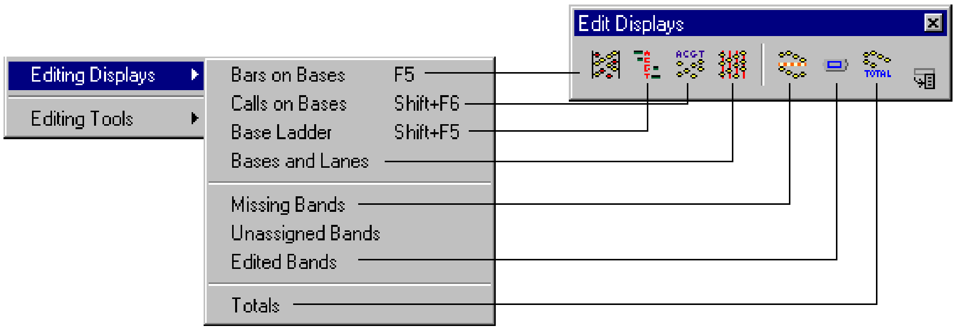


Fig.5-1. Editing Displays submenu and toolbar.

Bars on Bases

Bars on Bases displays horizontal marks on bands that have been detected and included in the sequence. Bands that are included with confidence are marked in yellow, while ambiguous bands are marked in red.

Base Ladder

Base Ladder displays base numbers along the left edge of the lane set and base assignments down the middle. Lines indicate which band is associated with each assignment letter.

Confident assignments are colored yellow and are marked with solid lines. Ambiguous assignments are colored red and are marked with dashed lines.

Green crosses mark both the locations of unassigned bands and the suggested locations of missing bands.

Calls on Bases

Calls on Bases displays a white crosshair at the center of each assigned band, as well as the number of the base in the sequence and its base code letter.

Confident assignments are displayed in yellow; ambiguous assignments are red.

Bases and Lanes

Bases and Lanes creates a display similar to Calls on Bases, but also marks the centers of the lanes with a yellow vertical line.

Missing Bands

Missing Bands displays boxes indicating the locations of possible missing bands.

Bands are labeled “missing” based on predictions made from a model of the band spacing of the assigned bands (i.e., the software decides that the distance between the two nearest bands is too large and no band is detected between them).

Note: You can select Missing Bands once to display the locations of missing bands, then select it again to clear the display.

Edited Bands

Edited Bands displays a dashed box around all the bands that have been edited (assigned, de-assigned, made certain, or made ambiguous) using the tools on the Sequence > Editing Tools submenu (see next section).

Totals

Totals on the Editing Displays submenu or toolbar opens an informational box displaying the statistical information for each laneset in your gel. The information displayed includes the number of bands detected, the number of edited bands, and the type of editing that was performed.

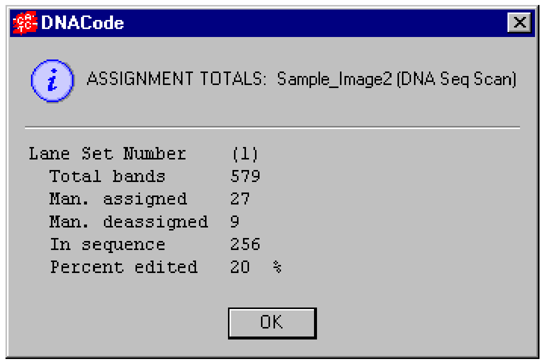


Fig.5-2. Band Totals box.

5.3. Editing the Sequence

The following functions are used to identify and edit ambiguous or incorrectly called bands. They are located on the Sequence > Editing Tools submenu, as well as on the Edit Tools toolbar.

Note: When editing bands, it is usually a good idea to magnify a section of the laneset using the zoom tools to help you place and label bands accurately.

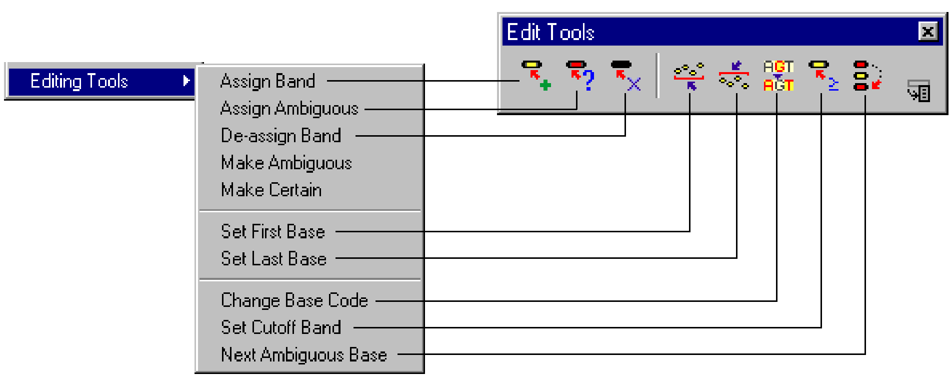


Fig.5-3. Sequence > Editing Tools submenu and toolbar.

Assign Band

Assign Band adds a missing band to the sequence. Select Assign Band from the submenu or toolbar, then click on the undetected band. The sequence will be automatically renumbered to include the new base.

Assign Ambiguous

Assign Ambiguous adds a missing band to the sequence and labels its status as uncertain. Select Assign Ambiguous from the submenu or toolbar, then click on the undetected ambiguous band. The sequence will automatically be renumbered to include the new base.

The newly assigned ambiguous base will be labeled according the preferences you selected under Edit > Describe Gel.

De-assign Band

De-assign Band removes a band from the sequence. Select De-assign Band from the submenu or toolbar, then click on the assigned band. The sequence will be automatically renumbered to reflect the editing.

Set First Base

When reviewing the bottom of a laneset, you may find faint bands or artifacts that were identified as bands by the automatic sequence reader. To specify which band should be considered the first band of the laneset, use the function Set First Base.

Select Set First Base from the submenu or toolbar, then click on the appropriate band. You will be prompted to enter the sequence number of your new first base.

When you complete the command, the new beginning of your laneset will be marked with a blue line. All the bands below that line will not be included in the sequence.

Set Last Base

To specify which band should be considered the last (i.e., topmost) band of the laneset, use the function Set Last Base.

Select Set Last Base from the submenu or toolbar, then click on the appropriate band.

When you complete the command, the new end of your laneset will be marked with a blue line. All the bands above the line will not be included in the sequence.

Make Ambiguous

Make Ambiguous changes the status of a band marked as certain to uncertain. Select Make Ambiguous from the Sequence > Editing Tools submenu, then click on the band.

The band will still be included in the final sequence, but will be marked as ambiguous.

The newly marked ambiguous base will be labeled according the preferences you selected under Edit > Describe Gel.

Make Certain

Make Certain changes the status of a band marked as ambiguous to certain. Select Make Certain from the Sequence > Editing Tools submenu, then click on the ambiguous band to change its status to certain.

Change Base Code

As you are editing, you may come across a base that you consider incorrectly labeled. You can change the code of this base using the Change Base Code command.

Note: This command will only work with bases marked as ambiguous.

Select Change Base Code from the submenu or toolbar and click on an ambiguous band.

A pop-up box will indicate the software's base call next to the "Most likely base" prompt. Indicate the other base(s) that could possibly be correct by clicking the A, C, G, and / or T button(s) next to the "Other possible bases" prompt.

After entering your selections, click on the Done button.

The newly labeled ambiguous base will be marked according the preferences you selected under Edit > Describe Gel.

Set Cutoff Band

If your lanes contain faint or shadow bands that you do not want to include in your sequence, you can use Set Cutoff Band to eliminate erroneous base calls on these bands.

Select Set Cutoff Band from the submenu or toolbar and click on a band that is not dark enough to be considered a real band in the sequence. All bands above it in the lane that are as dark as or lighter than it will be removed from the sequence calls.

This function applies to all bands higher in the laneset because sequencing films are often lighter at the bottom than at the top. Light bands at the bottom of the gel will not be removed if you set the cutoff band above them.

Next Ambiguous Base

Next Ambiguous Base provides a quick way to move from one area of ambiguity to the next. It is useful if you are using the zoom commands to magnify selected regions of the laneset.

Start by magnifying the bottom of the laneset. After editing this section, select Next Ambiguous Base from the submenu or toolbar and click on the image. The magnified area shifts downward, and the next ambiguous base is centered in the window.

5.4. Searching the Sequence

The functions under the Search menu allow you to sequence a laneset for a particular base or string of bases.

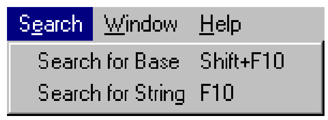


Fig.5-4. Search menu.

Searching for a Specific Base

If you want to find a specific base number, select Search for Base and click on a laneset. This opens a dialog box in which you can enter the number of the base that you want to find.

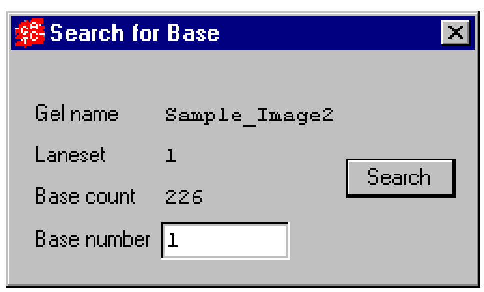


Fig.5-5. Search for Base dialog box.

When you click on the Search button, the dialog box will disappear and the image will be redisplayed with the base you specified highlighted in the center of the window.

To close the dialog box, click on the close button in the upper right corner.

Search for a String of Bases

You can search for a particular sequence of bases using the Search for String function. Select Search for String from the Search menu to open a dialog box in which you can enter your search parameters.

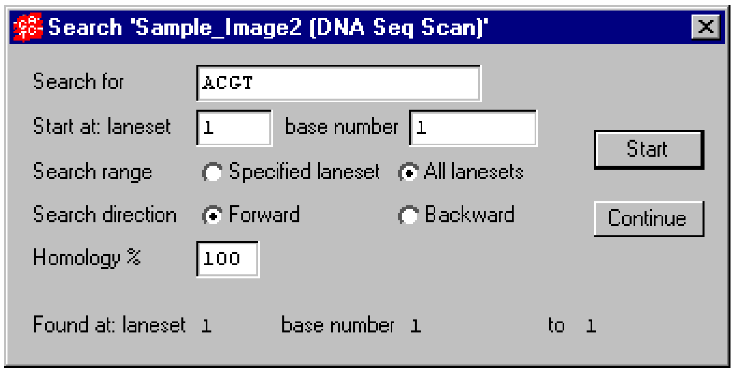


Fig.5-6. Search for String dialog box.

Enter the string of bases that you want to locate (e.g., GGATC) next to the "Search for:" prompt and the number of the base and laneset where the search should begin next to the appropriate prompts.

You can search only in the laneset you specified next to the "Start at: laneset" prompt by clicking on the "Search range: Specified laneset" button, or you can search in all lanesets in the gel by selecting the "All lanesets" button.

Specify the direction of the search in the laneset by selecting the Forward or Backward option button.

Finally, enter a value in the text box next to the Homology % prompt. This number determines how closely the sequence found in the laneset must match the sequence you entered.

When you click on the Start button, the image is redisplayed so the sequence of interest is highlighted and centered in the window. The position (in base numbers) will be indicated in the dialog box.

After the string has been located, you can continue searching by clicking on the Continue button.

To close the dialog box, click on the close button in the upper right corner.

6. Printing and Exporting

When you have finished editing the base calls for each laneset, you can review the results on-screen using the Sequence Text Window command on the View menu or main toolbar. See section 3.2. for more information about this command.

The commands for printing and exporting images are located on the File menu.

6.1. Printing

DNACode offers four different printing options under File > Print:

1. Print Image prints a copy of the image and any overlays that are currently displayed.
2. Print Actual Size prints an actual-size copy of the image.
3. Scan Report prints the image and information about its scan history, number of pixels, data range, etc.
4. Video Print prints images and reports to a video printer.*

*Video printing requires installation of the video board and cable that come with the Gel Doc 1000 / 2000 gel documentation system. The video board and cable can also be ordered separately.

Macintosh Only

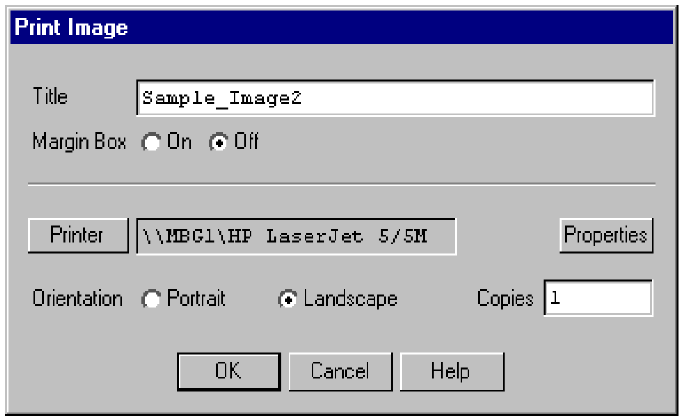
On the Macintosh, specify your printer settings—including paper size, page orientation, etc.—using the File > Print > Page Setup command. This will open the standard Macintosh Page Setup dialog box.

6.1.a. Printing Images

Print Image prints a copy of the active image window and any image overlays that are displayed.

File > Print > Print Image opens the Print Image dialog box.

Windows version:



Macintosh version:

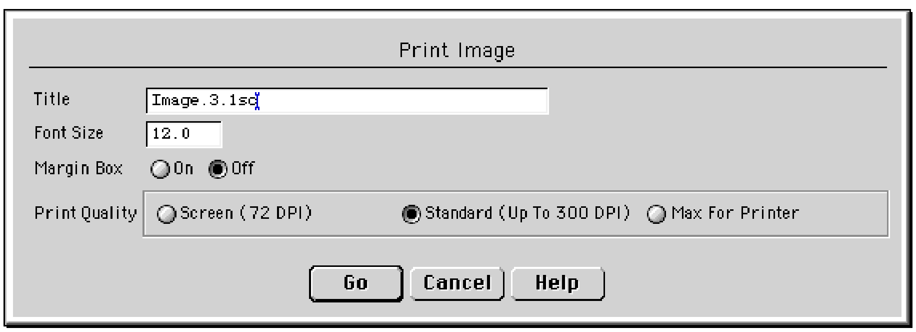


Fig.6-1. Print Image dialog box.

At the top of the dialog box, enter the title that you want to appear above the image when you print.

Choose whether or not a margin border should appear around the edge of the image by clicking either the On or Off button next to the Margin Box prompt.

Print Settings: Windows Only

Click on Printer to display a list of your available printers. After you have selected a printer, you can click on Properties to select the standard Windows print settings for that printer.

Note: If you are using Windows 95 with certain printer drivers, you may experience difficulty printing some overlays and overlay symbols on images. While there is no single solution to this problem, you should try adjusting some settings in the Printer Properties dialog box, under the Graphics and Advanced tabs—specifically, the Dithering, DPI, Rasterizing, and RET settings.

Select the paper orientation by clicking on Portrait or Landscape next to the Layout prompt. (This can also be set in the Properties dialog box.)

Indicate how many copies should be printed in the text box next to the Copies prompt.

When you are satisfied with all the print parameters, click on the OK button to send the image to the printer.

Print Settings: Macintosh Only

Specify a font size for the image title in the text box next to the Font Size prompt. You can use font sizes ranging from 6 to 64, but we recommend you use the default setting of 12.0.

Select the printing resolution by clicking on one of the buttons next to the Print Quality prompt.

- Screen produces a printed image at the resolution of your computer screen (typically 72 dpi).
- Standard produces a printed image at the standard resolution of a typical laser jet printer (300 dpi, unless your printer prints at a lower resolution).
- Max for Printer prints at the maximum resolution of your printer.

When you are satisfied with all the print parameters, click on the Go button to send the image to the printer.

6.1.b. Print Actual Size

You can print your images at their actual size using the Print Actual Size command.

Note: If you are using the Gel Doc or the Fluor-S, you must specify the correct image area size when capturing your images to ensure accurate 1:1 printing. You can specify the image area size in the acquisition window for the instrument. See the appendix on each imaging device for more information.

Select Print Actual Size from the File > Print menu. The standard Print Image dialog box will open (see previous section).

6.1.c. Printing the Scan Report

Scan Report allows you to print out a single-page report of an image and its associated information. The format of the report is designed to provide a concise yet thorough summary of the most relevant features of an image for documentation purposes.

The report includes the following information:

- The image.
- The report date.
- The image title and a brief description.
- The directory location.
- The date the image was scanned.
- The type of imaging device.
- The imaging area and number of pixels.
- Pixel size.
- The intensity range, image color, and memory size.
- Image background information.
- Relevant lane and band information.

To print a scan report of a particular gel, select Scan Report from the File > Print submenu. This will open a smaller version of the standard print dialog box (see Print Image, above).

Note: TIFF images do not contain all the tagged information that would normally be included in an image file (for example, imaging device, scan date, image color, etc.). For this reason, the Scan Report may list this information as "Unknown" for imported TIFF files.

6.1.d. Video Printing

The Video Print command allows you to print images and reports on a video printer. To create a video printout of the active window, select Video Print from the File > Print submenu.

Note: Video printing requires installation of the video board and cable that come with the Gel Doc 1000 / 2000 gel documentation system. The video board and cable can also be ordered separately.

Settings for the Mitsubishi P90W/P91W Video Printer

There are three settings for the Mitsubishi P90W / P91W video printer. Set Contrast to zero, Brightness to zero, and Gamma to 5.

The dip switches should stay in the orientation in which they are shipped: Pin 1 is up (on), and Pins 2–10 are down (off).

6.2. Exporting

6.2.a. Exporting an Image in TIFF Format

To analyze your scanned image using Bio-Rad's Multi-Analyst or Molecular Analyst software, you must first save the image in TIFF format.

File > Export to TIFF Image opens a dialog box in which you can specify your export parameters.

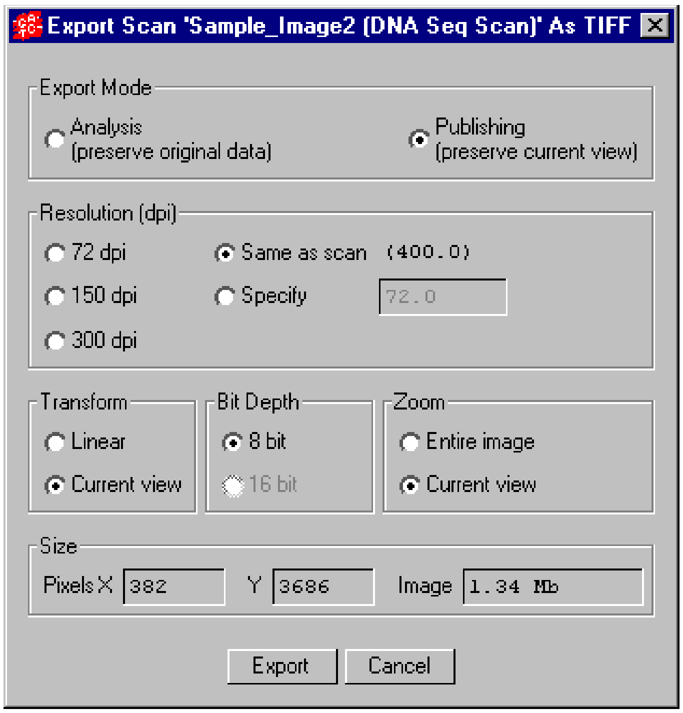


Fig.6-2. Export to TIFF Image dialog box.

Analysis Export Mode

Analysis mode exports the scan data, unmodified by DNACode's viewing controls.

If you select Analysis mode, the other controls in this dialog box will become inactive.

Publishing Export Mode

Publishing mode exports a TIFF image that looks like the image as it is currently displayed in DNACode (with whatever Transform adjustments you may have made).

In Publishing mode, you can specify a resolution for the TIFF image by selecting 72 dpi (typical computer screen resolution), 150 or 300 dpi (standard printing resolutions), the same resolution as the scan file, or any resolution you want to specify (up to the resolution of the scan).

If you have log transformed the image, you can specify a linear transform, or preserve the current view.

If your image is 16 bits, you can compress it to 8 bits for export to TIFF.

Finally, if you are only displaying part of the image due to magnification or repositioning, you can preserve the current view or export the entire image.

Exporting the Image

The size of the pixels in the image and the file size of the image are listed at the bottom of the dialog. When you are ready to export, click on the Export button.

A Save As dialog box will open. The default file name will have a .tif extension, and the file type will indicate that this is a TIFF image. You can change the file name or select a different directory to save in.

DNACode will generate a grayscale TIFF image compatible with Multi-Analyst and Molecular Analyst software.

6.2.b. Exporting Sequences

You can export your DNA sequence to a spreadsheet for analysis using the Export Sequence function.

When you select File > Export > Export Sequence, the Export Sequence dialog box will open.

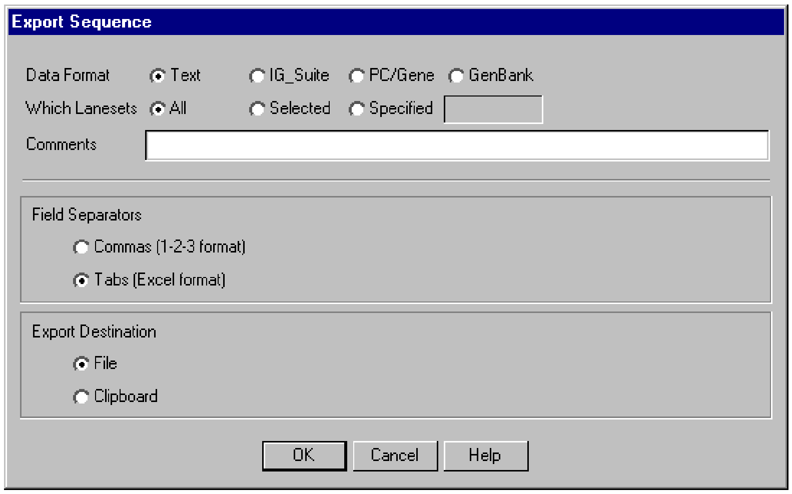


Fig.6-3. Export Sequence dialog box.

At the top of the dialog box, select the format in which the file should be written by clicking on the appropriate button next to the Data Format prompt.

Next to the Which Lanesets prompt, specify whether you want to export data from all the lanesets in the gel or a subset of the lanesets. If you select Specified, enter the laneset number(s) to export in the text field.

Enter any information about the sequence data in the text box next to the Comments prompt.

Under the Field Separators prompt, select Commas or Tabs to use as separators for the sequence. If you are exporting to a Lotus 1-2-3™ spreadsheet, choose Commas. If you are exporting to Excel™, choose Tabs.

Select your Export Destination: to a File or to the Clipboard.

After entering the information in the dialog box, click on the OK button (Go on the Macintosh). If you selected File as your export destination, a dialog box will open in which you can specify the name and location of your export file.

Appendix A. Gel Doc 1000/2000

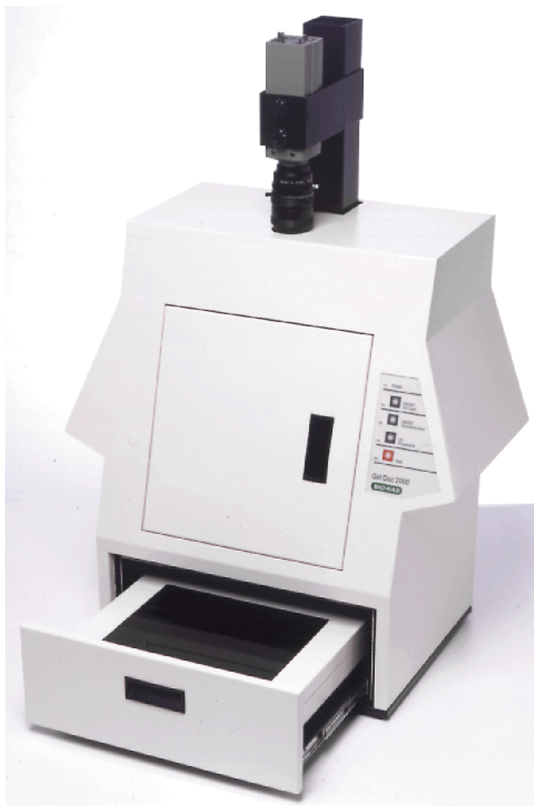


Fig.A-1. Gel Doc 1000/2000.

Before you can begin acquiring images using the Gel Doc® 1000 / 2000 gel documentation system, the system must be properly installed and connected with the host computer. See the Gel Doc hardware manual for installation, startup, and operating instructions.

To use the Gel Doc, you will need to have the Bio-Rad-supplied acquisition board installed in your PC or Macintosh. The drivers for this board will be installed when you install Diversity Database.

Make sure that your Gel Doc camera is turned on. If the camera is not turned on, the Gel Doc acquisition window will open but the screen will be black, and you will be unable to Video Print.

Simulation Mode

Any of the imaging device acquisition windows in Diversity Database can be opened in “simulation mode.” In this mode, an acquisition window will open and the controls will appear active, but instead of capturing real images, the window will create “dummy” images of manufactured data.

You do not need to be connected to an imaging device to open a simulated acquisition window. This is useful for demonstration purposes or practice scans.

To enter simulation mode, make sure the imaging device is disconnected or turned off, then select its name from the File menu. You will be asked if you want to open the simulated window. The title of the window will indicate that it is simulated.

There is no simulated calibration for the GS-700 and GS-710 Imaging Densitometers.

A.1. Opening the Gel Doc Acquisition Window

To acquire images using the Gel Doc, go to the File menu and select Gel Doc.... The acquisition window for the instrument will open, displaying a control panel and a video display window.

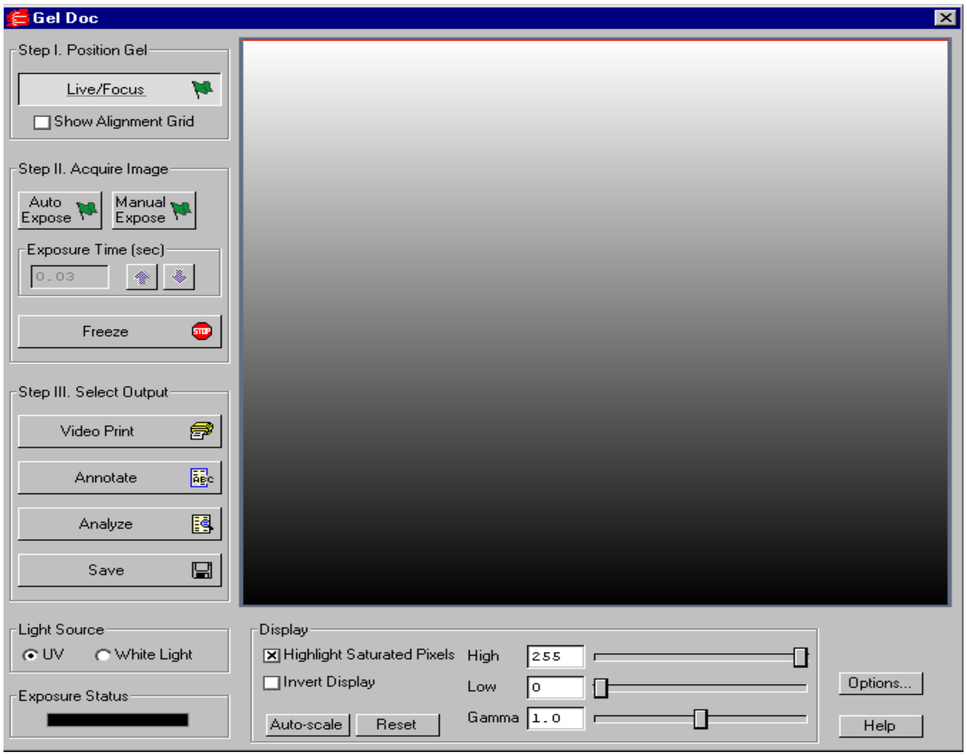


Fig.A-2. Gel Doc acquisition window

The Gel Doc video display window will open in “live” mode, giving you a live video display of your sample. If no image is visible, make sure the camera is on, check the cable connections, and make sure that the protective cap is off the camera lens. Also check to see that the transilluminator is on and working.

The control panel has been arranged from top to bottom to guide you through the acquisition procedure. There are three basic steps to scanning an image using the Gel Doc:

1. Position and focus the gel or other object to be imaged.

2. Acquire the image.

3. Select the output.

A.2. Step 1. Position Gel

The Gel Doc window will open in “live” mode, giving you a live video display of your sample. (Frames will be captured and displayed at about 10 frames per second, depending on the speed of your computer.) In this mode, the Live / Focus button will appear selected.

You can use live mode to zoom and focus the camera on your sample area and position your sample within the area. You can also select the Show Alignment Grid checkbox to facilitate positioning.

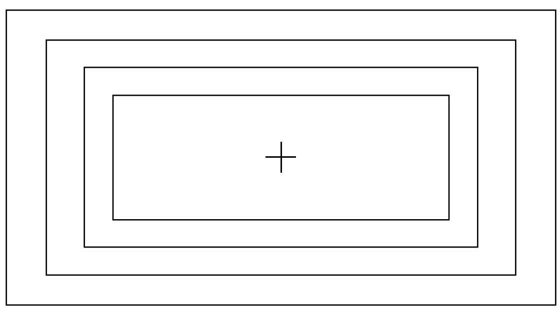


Fig. A-3. Gel Doc alignment grid.

The image should stay in focus while zooming. If focus is not maintained, refer to your Gel Doc hardware manual.

Note: After positioning your sample, you should check the Imaging Area dimensions under Options (see below) to make sure that they conform to the size of the area you are focusing on. To determine the size of the area you are focusing on, you can place a ruler in the Gel Doc box so that it is visible by the camera.

A.3. Step II. Acquire Image

For many white light applications, you can skip this step and save and print images directly from live mode.

For UV light or chemiluminescent applications, or if your aperture is closed down in white light applications, this mode will probably not give you the best possible image. To illuminate faint areas of your sample, you will need to select an exposure time for the image.

“Exposure” refers to the integration of the image on the camera CCD over a set period of time. The effect is analogous to exposing photographic film to light over a period of time. The Gel Doc has two exposure modes: Auto and Manual.

To acquire an image, you will typically perform the following steps:

1. Select Auto Expose to determine an approximate exposure time.
2. After Auto Expose is finished, “fine tune” that exposure time using Manual Expose. You can also change the camera aperture at this time.
3. Click on Freeze or any of the other buttons in the window to stop the exposure process.

Note: If you know the approximate exposure time you want (± 3 seconds), you can skip the Auto Expose step and go directly to Manual Expose.

Auto Expose

Clicking on the Auto Expose button will cancel live mode and begin an automatic exposure. (The Auto Expose button will appear selected throughout the exposure.) If you know the approximate exposure time you want (± 3 seconds), you can skip this step and go directly to Manual Expose.

In Auto Expose mode, the image is continuously integrated on the camera CCD until it has reached a certain percentage of saturated pixels. This percentage is set in the Options dialog box. (Default = 0.75 percent. See Options below.)

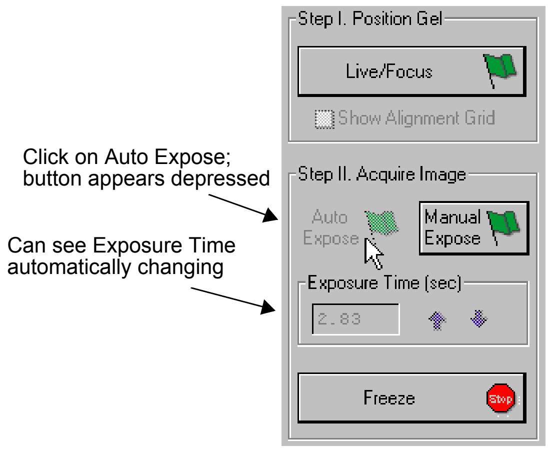


Fig. A-4. Acquiring an image, step 1: Select Auto Expose.

Once an image has reached the specified percent of saturated pixels, it is captured and displayed in the video display window, Auto Expose is automatically deactivated, the exposure time appears active in the Exposure Time field, and Manual Expose is activated.

Note: If you are having difficulty auto-exposing your sample, you can use Manual Expose to adjust your exposure time directly. Most applications only require an exposure time of a few seconds, which can be quickly adjusted using Manual Expose.

Manual Expose

Manual Expose is automatically activated after Auto Expose has deactivated. Or, if you know the approximate exposure time you want, you can simply click on the Manual Expose button.

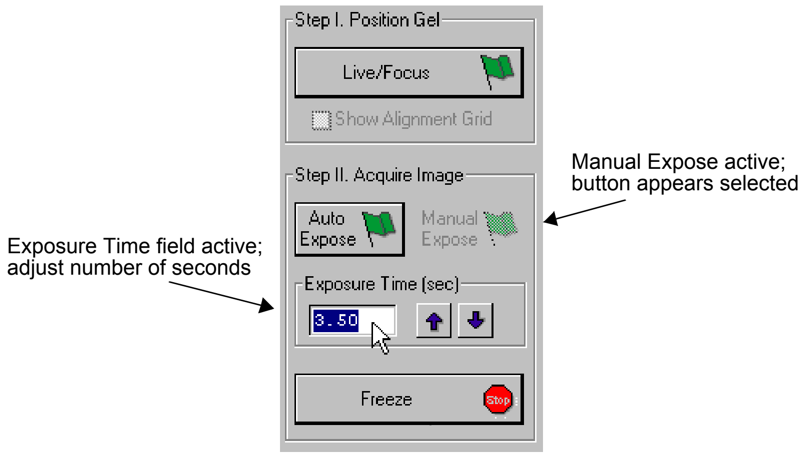


Fig. A-5. Acquiring an image, step 2: Adjust exposure time.

When Manual Expose is active (the button will appear selected), the last captured image will be displayed in the Gel Doc image window. The camera continues to integrate the image on the CCD, updating the display whenever the specified number of seconds is reached.

You can adjust the exposure time by changing the number of seconds in the Exposure Time field. To do so, enter a number directly or use the arrow buttons next to the field.

Freeze

Once you are satisfied with the quality of the displayed image, click on the Freeze button to stop the exposure process. The last full exposure will be displayed in the image window.

Note: Freeze is automatically activated if you adjust any of the subsequent controls (e.g., Video Print, Light Source, Display controls, etc.).

Click on Freeze to stop the exposure process

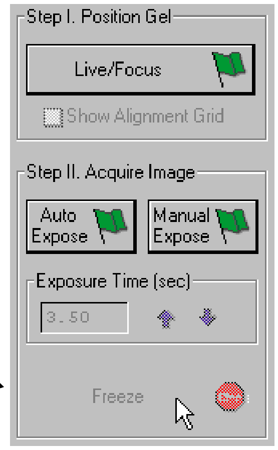


Fig. A-6. Acquiring an image, Step 3: Freeze the exposure.

A.4. Step III. Select Output

The Gel Doc window has several output options.

Video Print

Clicking on Video Print will automatically send the currently displayed frame (either live or integrated) to a video printer. You can add information about your image to the bottom of the printout by selecting the appropriate checkboxes in the Options dialog box. (See Options, below.)

Annotate

Clicking on Annotate will open a separate image window displaying the captured image. The default name for the image will include the date, time, and user (if known).

The Annotation Tools toolbar will also pop up, allowing you to annotate your image.

The image will not be saved until you select Save or Save As from the File menu.

Analyze

Clicking on Analyze will open a separate image window displaying the captured image. The default name for the image will include the date, time, and user (if known).

You can then analyze the image using the other features in Diversity Database.

The image will not be saved until you select Save or Save As from the File menu.

Save

Clicking on Save will open a separate image window displaying the captured image. A Save As dialog box will automatically open displaying the default file name for the image, which will include the date, time, and user (if known). You can then change the file name and storage directory.

You can also save your image as a TIFF image for export to other applications.

A.5. Light Source

The Light Source option buttons allow you to set the type and scale of your Gel Doc data.

UV

Select this light source for fluorescent and chemiluminescent samples. The data will be measured in linear intensity units.

White Light

Select this light source for reflective and transmissive samples. The data will be measured in uncalibrated optical density (uOD) units.

A.6. Exposure Status

The Exposure Status bar shows the progress of your exposure. If your exposure time is greater than 1 second, the status bar display will give you a graphical representation of the remaining time before exposure is complete.

If the exposure time is less than 1 second, the status bar will not refresh itself for each exposure; it will remain at 100 percent.

A.7. Display

The Display controls are useful for quickly adjusting the appearance of your image for output to a video printer. Adjusting these controls will automatically freeze the video display and allow you to alter the image within the Gel Doc window.

These controls are similar to those in the Transform dialog box.

Note: The Display controls will only change the appearance of the image. They will not change the underlying data.

Highlight Saturated Pixels

When this is checked, any saturated pixels in the image will appear marked in red.

Invert Display

This checkbox will switch light spots on a dark background to dark spots on a light background, and visa versa.

Auto-scale

Clicking on Auto-scale will adjust your displayed image automatically. The lightest part of the image will be set to the minimum intensity (e.g., white), and the darkest will be set to the maximum intensity (e.g., black). You can then “fine tune” the display using the High, Low, and Gamma sliders described below.

High/Low Sliders

If Auto-scale doesn't give you the appearance you want, you can use the High and Low sliders to redraw the image yourself. Dragging the High slider handle to the left will make weak signals appear darker. Dragging the Low slider handle to the right will reduce background noise.

You can also type specific High and Low values in the text boxes next to the sliders. Clicking anywhere on the slider bars will move the sliders incrementally.

Gamma Slider

Some images may be more effectively visualized if their data are mapped to the computer screen in a nonlinear fashion. Adjusting the Gamma slider handle changes the light and dark contrast nonlinearly.

Reset

Reset will return the image to its original, unmodified appearance.

A.8. Options

Clicking on the Options button opens the Options dialog box. Here you can specify certain settings for your Gel Doc system.

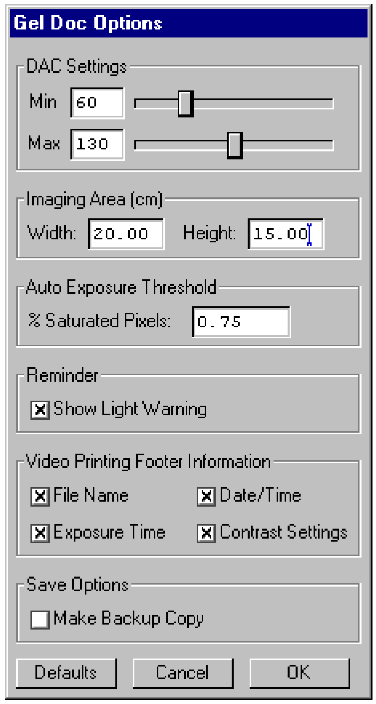


Fig. A-7. Available options in the Gel Doc acquisition window.

Clicking on OK implements any changes you make in this box. Clicking on Defaults restores the settings to the factory defaults.

DAC Settings

Note: The default DAC settings are highly recommended and should be changed with caution.

These sliders may be used to adjust the minimum and maximum voltage settings of your video capture board. The minimum slider defines the pixel value that will appear as white in the image, while the maximum slider defines the pixel value that will appear as black. The slider scale is 0–255, with the defaults set to 60 minimum and 130 maximum.

Imaging Area

These fields are used to specify the size of your imaging area in centimeters, which in turns determines the size of the pixels in your image (i.e., resolution). When you adjust one imaging area dimension, the other dimension will change to maintain the aspect ratio of the camera lens.

Note: Your imaging area settings must be correct if you want to do 1:1 printing. These are also important if you are comparing the size of objects (e.g., using the Volume Tools) between images.

Auto Exposure Threshold

When you click on Auto Expose, the exposure time is determined by the percentage of saturated pixels you want in your image. This field allows you to specify that percentage.

Typically, you will want less than 1 percent of the pixels in your image saturated. Consequently, the default value for this field is 0.75 percent.

Reminder

When this checkbox is selected, the software will warn you to turn off your transilluminator light when you exit the Gel Doc acquisition window or when your system is “idle” for more than 5 minutes.

Note: If you are performing experiments that are longer than 5 minutes (e.g., chemiluminescence), this should be deselected.

Video Printing Footer Information

The checkboxes in this group allow you to specify the information that will appear at the bottom of your video printer printouts.

Save Options

To automatically create a backup copy of any scan you create, select the Make Backup Copy checkbox.

With this checkbox selected, when you save a scan, a backup copy will be placed in the same directory as the scanned image. Windows backup files will have an ".sbk" extension. Macintosh backup files will have the word "backup" after the file name.

This backup copy will be read-only, which means that you cannot make changes to it. You can open it like a normal file, but you must save it under a different file name before editing the image or performing analysis.

Appendix B. GS-700 Imaging Densitometer

Fig. B-1. GS-700 Imaging Densitometer

Before you can begin scanning images with the GS-700 Imaging Densitometer®, your instrument must be properly installed and connected with the host computer. See the hardware manual for installation, startup, and operating instructions.

PC Only: A Note About SCSI Cards

The GS-700 is connected to your computer by a Small Computer System Interface (SCSI) cable. To use the GS-700, you must have a SCSI card installed in your PC. If you have an older PC, you may also need to load the SCSI and WinASPI drivers that came with your card. Newer releases of Windows 95 and Windows NT will already have those drivers installed.

Simulation Mode

Any of the imaging device acquisition windows in Diversity Database can be opened in “simulation mode.” In this mode, an acquisition window will open and the controls will appear active, but instead of capturing real images, the window will create “dummy” images of manufactured data.

You do not need to be connected to an imaging device to open a simulated acquisition window. This is useful for demonstration purposes or practice scans.

To enter simulation mode, make sure the imaging device is disconnected or turned off, then select its name from the File menu. You will be asked if you want to open the simulated window. The title of the window will indicate that it is simulated.

Note: There is no simulated calibration for the GS-700 and GS-710 Imaging Densitometers.

B.1. Opening the GS-700 Acquisition Window

To acquire images using the GS-700, go to the File menu and select GS-700.... The acquisition window for the densitometer will open, displaying a control panel and a scanning window.

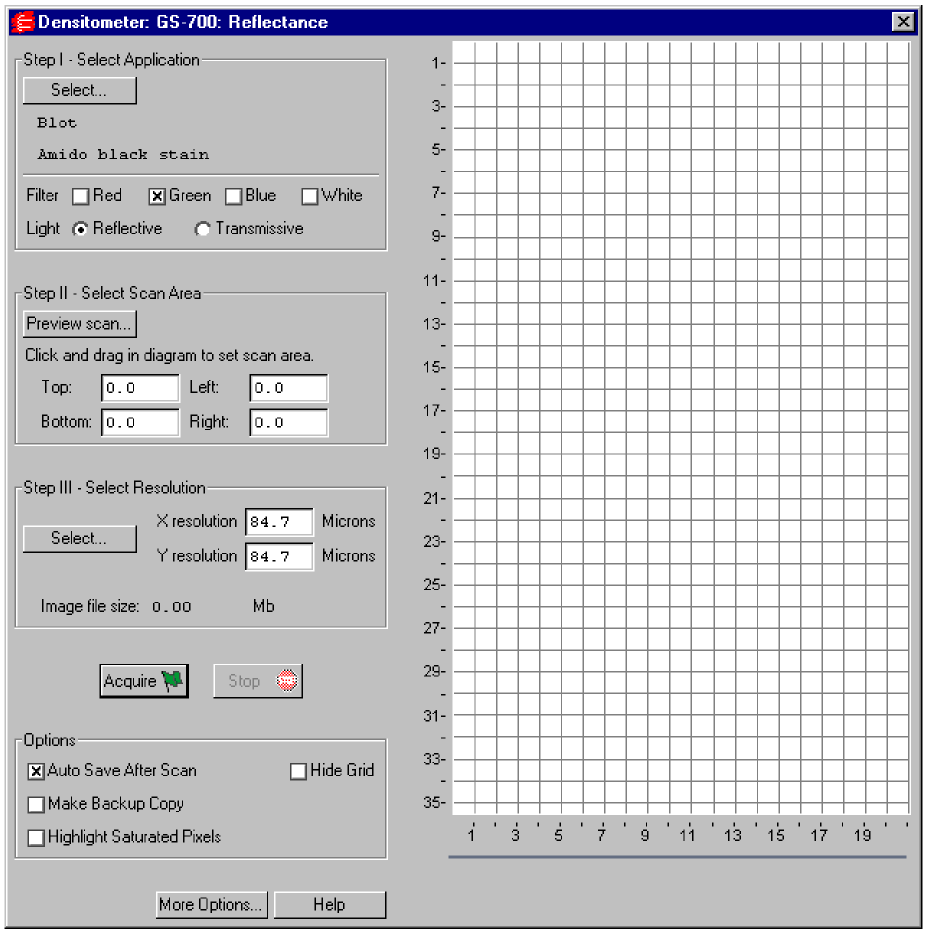


Fig. B-2. GS-700 acquisition window

The scanning window is marked by grid lines that divide the area into square centimeters. These are numbered 1–35 top to bottom and 1–21 left to right if the light source is uncalibrated reflective, and 1–25 top to bottom and 1–20 left to right if the light source is uncalibrated transmissive (see below for details).

To hide the gridlines, click on the Hide Grid checkbox under Options.

The control panel has been arranged from top to bottom to guide you through the acquisition procedure. There are four basic steps to scanning an image using the GS-700:

1. Select the application.
2. Select the scan area.
3. Select the resolution of the scan.
4. Acquire the image.

B.2. Step 1. Select Application

To set the parameters for your particular scan, you can:

1. Select from a list of possible applications, or
2. Choose your own filter and light source settings.

Selecting from the List of Applications

When you select from the list of applications, the software automatically sets the appropriate filter(s) and other parameters for that particular application.

To select from the list of applications, click on the Select button under Select Application.

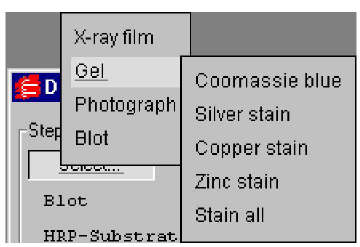


Fig. B-3. Example of the application tree in the GS-700 dialog box.

The applications are listed in a tree that expands from left to right. First you select the category of your application, then you select your particular application.

To exit the tree without selecting, press the ESC key.

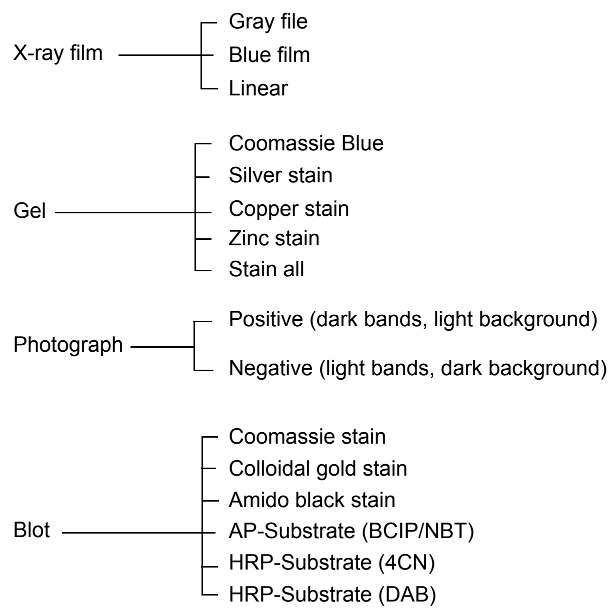


Fig. B-4. Applications available in the GS-700 acquisition window.

Choosing Your Own Settings

If you know the filter and light source settings you want, or want to experiment with different settings, you can choose them yourself.

Next to Filter, click on either the Red, Green, Blue, or White checkbox, or a combination of two of the first three (Red-Green, Green-Blue, Red-Blue).

Filter Color	Wavelength	Application Examples
Red	595–750 nm	Coomassie G-250, BCIP/NBT, Fast Green FCF, Methylene Blue
Green	520–570 nm	Coomassie R-250, Basic Fuchsin
Blue	400–530 nm	Crocein Scarlet
Gray Scale	400–750 nm	Silver Stains, Copper Stains, Film

Fig. B-5. Examples of filter colors and applications.

Next to Light, select the appropriate light source.

- Reflective mode is for scanning opaque mediums such as dried gels on filter paper, arrays, TLC plates, and photographs. The scanning dimensions in this mode are 21 cm x 35 cm (uncalibrated).
- Transmissive mode is for scanning transparent mediums such as films, gels, and slides. The scanning dimensions in this mode are 20 cm x 25 cm (uncalibrated).

B.3. Step II. Select Scan Area

Preview Scan

Before selecting the particular area to scan, you can preview the entire scanning area to determine the exact position of your sample.

Click on Preview Scan. A quick, low-resolution scan of the entire scanning area will appear in the scanning window. Your sample should be visible within a portion of this scan.

Selecting an Area

Using the preview scan as a guide, select your scan area by dragging your mouse within the scanning window. The border of the scan area you are selecting will be marked by a frame.

Note: The scan area you select must be at least 1 cm wide.

When you release the mouse button, the border changes to a dashed blue line, indicating a selected area.

- To *reposition* the scanning area box you have created, position your cursor inside the box and drag. The entire box will move.
- To *resize* the box, position your cursor on a box side and drag. The side you have selected will move.
- To *redo* the box entirely, position your cursor outside the box and drag. The old box will disappear and a new box will be created.

You can also select the scanning area by entering coordinates in the appropriate fields (Top, Bottom, Left, Right). As you enter a coordinate, the position of the scanning area box will change accordingly.

When selecting, be sure to include the entire area of interest, and be generous with borders. You can always crop the image later.

B.4. Step III. Select Resolution

To select from a list of possible scanner resolutions, click on the Select button under Select Resolution. This will open the Select Scan Resolution dialog box.

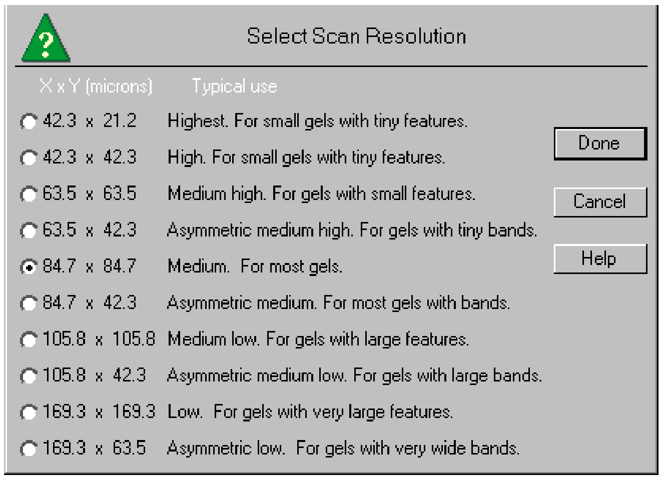


Fig. B-6. Select Scan Resolution dialog box.

Available resolutions are listed from highest to lowest in terms of the dimensions of the resulting pixels (in microns). Smaller pixels equal higher resolution. Each resolution is listed with its typical use.

In general, the size of your pixels should be one-tenth the height of your smallest object.

Some of the resolutions are asymmetrical, meaning that the resolution is higher in the vertical dimension (i.e., the pixels in the resulting image are larger in the horizontal dimension than in the vertical dimension). This is useful for gels with bands, where you are more interested in resolving in the y dimension to determine the size of bands and the spacing between them.

Specifying Your Own Resolution

If you select Oversample under More Options (see section B.7. for details), you can specify your own resolution within the range of 43–169 microns (micrometers). With Oversample selected, enter values directly in the fields next to X resolution and Y resolution in the main acquisition window.

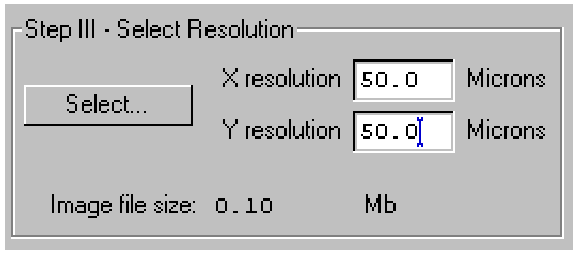


Fig. B-7. Entering a custom resolution (with Oversample selected).

Image File Size

Image File Size (under Select Resolution) shows the size of the scan file you are about to create. If you do not have enough computer memory for the specified file size, an error message will appear when you attempt to scan. If this happens, select a lower resolution or decrease the size of the area to be scanned. (Macintosh users can also increase the application memory partition. See your Macintosh computer documentation for guidance.)

B.5. Acquire the Image

If you want to calibrate your scans, read the following section on calibration before scanning.

To begin to scan, click on the Acquire button. The scanned image will begin to appear in the scanning window, line by line.

To interrupt a scan, click on the Stop button. A message will ask you to confirm the interrupt, and then you will be asked if you want to keep the partial scan. This feature is useful if you overestimated the size of the area you selected.

After the scan is complete, a window will open displaying the scan image, at which point you can save it as a Diversity Database image.

Note: The image will open with a default file name that includes the date, time, and (if applicable) user name. However, unless you have selected Auto

Save After Scan, the file will not be saved until you select Save or Save As from the File menu.

B.6. Calibration

If you have installed a calibration strip on the GS-700 scanner plate, you can automatically calibrate your scans. To order a calibration strip from Bio-Rad, contact your Bio-Rad sales representative.

See the calibration manual for details on installing the calibration strip.

To set the automatic calibration settings, click on the More Options button in the GS-700 acquisition window. This will open the Densitometer Options dialog box.

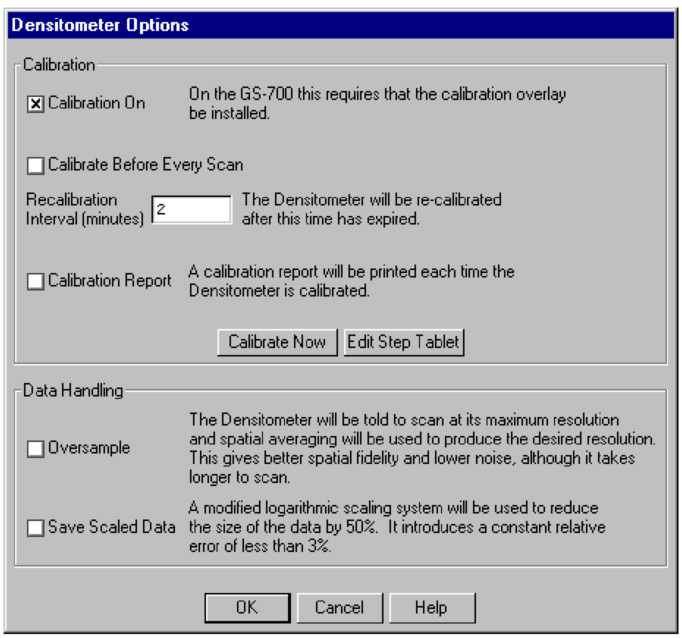


Fig. B-8. More Options in the GS-700 acquisition window.

To enable automatic calibration, click on the Calibration On checkbox.

Note: Your calibration strip must be installed for this feature to work!

With Calibration On selected, the other calibration settings become active.

Calibration Strip Window

When calibration is turned on, a calibration strip window will appear below the main scanning window and the length of the main scanning window will be reduced to 29 cm reflective and 23 cm transmissive.

Every time the densitometer calibrates, an image of the calibration strip will appear in the calibration strip window.

B.6.a. Editing the Step Tablet

Before you can calibrate, you must enter the step tablet values for your calibration strip into the Step Tablet Values form.

In the package with your calibration strip, you will find the manufacturer's printout of the diffuse density values for each segment of your step tablet, as well as a serial number that identifies the step tablet to which those density values belong. Those exact density values must be entered into the computer for the software to associate a correct density value with each step on the step tablet.

Scanning with incorrect step tablet values entered into the form can cause significant errors in the reported densities of your scans.

To enter the density values for your step tablet, click on the Edit Step Tablet button in the Densitometer Options box. The Step Tablet Values dialog box for the GS-700 will open.

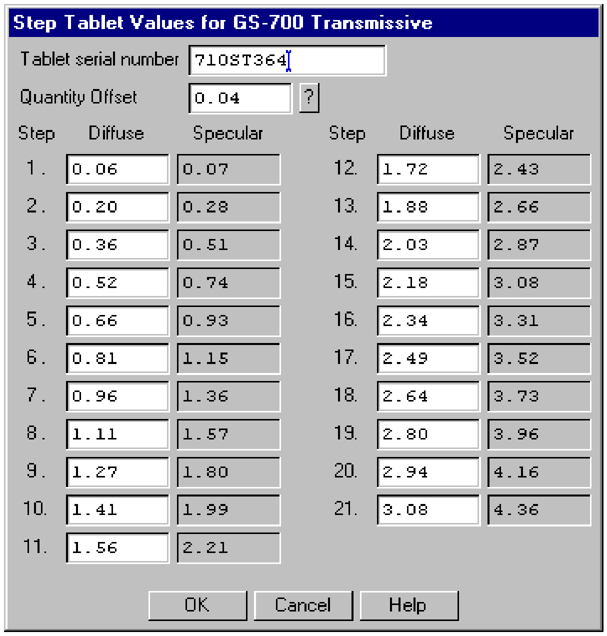


Fig. B-9. Step Tablet dialog box.

The dialog box will indicate whether the displayed form is for the transmissive or reflective step tablet, depending on whether you are in transmissive or reflective mode.

Note: In transmissive scanning, the scanner is calibrated against a transmissive step tablet. In reflective scanning, the scanner is calibrated against a reflective step tablet, unless the reflective calibration strip is uncalibrated. If the reflective calibration strip does not have step tablet values associated with it, you can use the default values in the form.

In the Tablet Serial Number field, you can type in the serial number for the step tablet values you are entering. (The reflective step tablet will probably not have a serial number.)

Your calibration strip will have a clear plastic cover. If the quantity offset value for the clear cover is included in your calibration strip package, you can enter this number in the Quantity Offset field. Otherwise, use the default value.

Finally, enter the values in the appropriate fields under the Diffuse column. After the step tablet is scanned, the software will associate each density value with its corresponding segment on the step tablet.

The step tablet density values do not need to be entered each time you calibrate. Once they have been entered and saved, they will be automatically recalled when the calibration strip is scanned. You only need to enter new values if you use a new step tablet.

When you finished filling out the Step Tablet Values form, click on OK.

Diffuse Versus Specular O.D.

In the step tablet form, you enter O.D. as diffuse density, and then the software automatically calculates the specular density.

Specular density is a measure of the light that passes directly through a medium. Diffuse density includes light that is scattered as it passes through the medium. Step tablet values are given in diffuse density, but are measured by the scanner in specular density, and therefore must be converted according to the specular / diffuse O.D. ratio. This conversion does not affect quantitation.

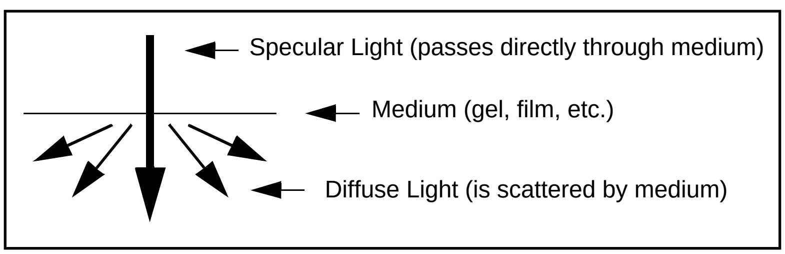


Fig. B-10. Specular and diffuse density

Diffuse density values are converted to specular optical density units according to the following formula:

$$Specular\ \ OD = 1.4\ \ Diffuse\ \ OD$$

B.6.b. Calibration Settings

After you have entered the step tablet values, you can immediately calibrate by clicking on the Calibrate Now button (in the Densitometer Options dialog box).

You can also specify how often you want the densitometer to automatically recalibrate. Either click on the Calibrate Before Every Scan checkbox or enter a Recalibration Interval (in minutes) in the appropriate field.

Note: With calibration turned on, the scanner will automatically recalibrate each time you change your filter or your reflective / transmissive setting. (If you select a different application with the same filter and light settings, it will not auto recalibrate.)

Calibration Report

To print out a calibration report each time the densitometer calibrates, click on the Calibration Report checkbox.

B.7. Other Options

Oversample

This feature allows you to scan at the maximum resolution of the GS-700 (42.3 x 42.3 microns) and then use spatial averaging to create an image with lower resolution. This can result in better images at lower resolution—however, it takes longer to scan.

To turn on oversampling, click on the More Options button in the acquisition window and click on the Oversample checkbox.

With oversampling on, you can specify your own resolution within the range of 43–169 microns by entering values directly in the fields next to X resolution and Y resolution in the main acquisition window.

Auto Save After Scan

To automatically save any scan you create, click on the Auto Save After Scan checkbox.

With this checkbox selected, when you click on Acquire, a Save As dialog box will open asking you to specify a file name and location for the image you are about to create. The scan will begin when you click on the Save button.

Make Backup Copy

If you have checked Auto Save After Scan, you can also automatically create a backup copy of any scan you create.

Click on the Make Backup Copy checkbox. With this checkbox selected, when a scan is created and saved, a backup copy will be placed in the same directory as the scanned image. Windows backup files will have an ".sbk" extension. Macintosh backup files will have the word "backup" after the file name.

This backup copy will be read-only, which means that you cannot make changes to it. You can open it like a normal file, but you must save it under a different file name before editing the image or performing analysis.

Highlight Saturated Pixels

When this is checked, any saturated pixels in the image will appear marked in red.

Hide Grid

To hide the gridlines in the scanning area window, click on the Hide Grid checkbox.

Appendix C. GS-710 Imaging Densitometer

Fig. C-1. GS-710 Imaging Densitometer

Before you can begin scanning images with the GS-710 Imaging Densitometer®, your instrument must be properly installed and connected with the host computer. See the hardware manual for installation, startup, and operating instructions.

PC Only: A Note About SCSI Cards

The GS-710 is connected to your computer by a Small Computer System Interface (SCSI) cable. To use the GS-710, you must have a SCSI card installed in your PC. If you have an older PC, you may also need to load the SCSI and WinASPI drivers that came with your card. Newer releases of Windows 95 and Windows NT will already have those drivers installed.

Simulation Mode

Any of the imaging device acquisition windows in Diversity Database can be opened in "simulation mode." In this mode, an acquisition window will open and the controls will appear active, but instead of capturing real images, the window will create “dummy” images of manufactured data.

You do not need to be connected to an imaging device to open a simulated acquisition window. This is useful for demonstration purposes or practice scans.

To enter simulation mode, make sure the imaging device is disconnected or turned off, then select its name from the File menu. You will be asked if you want to open the simulated window. The title of the window will indicate that it is simulated.

Note: There is no simulated calibration for the GS-700 and GS-710 Imaging Densitometers.

C.1. Opening the GS-710 Acquisition Window

To acquire images using the GS-710, go to the File menu and select GS-710.... The acquisition window for the densitometer will open, displaying a control panel and a scanning window.

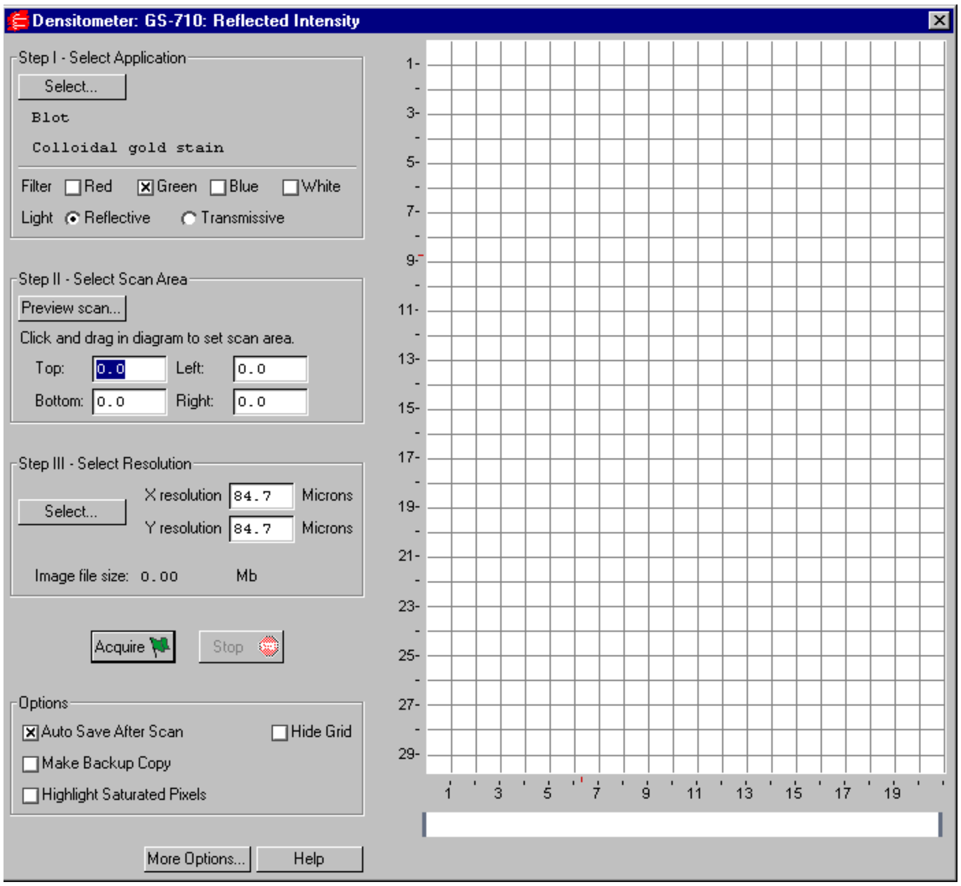


Fig. C-2. GS-710 acquisition window

The scanning window is marked by grid lines that divide the area into square centimeters. These are numbered 1–29 top to bottom and 1–21 left to right if the light source is reflective, and 1–23 top to bottom and 1–20 left to right if the light source is transmissive (see below).

To hide the gridlines, click on the Hide Grid checkbox under Options.

Below the main scanning window is the calibration strip window. Every time the densitometer calibrates, an image of the calibration strip will appear in this window.

The control panel has been arranged from top to bottom to guide you through the acquisition procedure. There are five basic steps to scanning an image using the GS-710:

1. Select the application.
2. Select the scan area.
3. Select the resolution of the scan.
4. Calibrate the instrument. (This is automatic, after you enter the step tablet values before you first scan after installation.)
5. Acquire the image.

C.2. Step 1. Select Application

To set the parameters for your particular scan, you can:

1. Select from a list of possible applications, or
2. Choose your own filter and light source settings.

Selecting from the List of Applications

When you select from the list of applications, the software automatically sets the appropriate filter(s) and other parameters for that particular application.

To select from the list of applications, click on the Select button under Select Application.

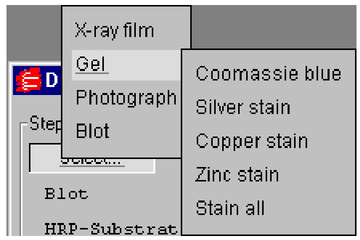


Fig. C-3. Example of the application tree in the GS-710 dialog box.

The applications are listed in a tree that expands from left to right. First you select the category of your application, then you select your particular application.

To exit the tree without selecting, press the ESC key.

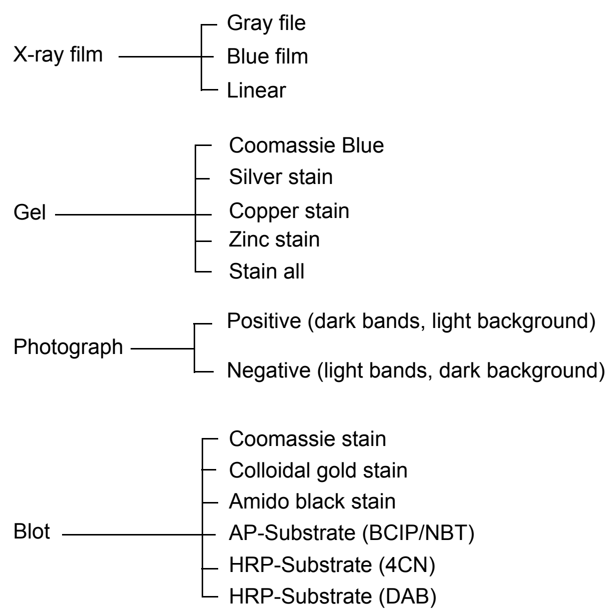


Fig. C-4. Applications available in the GS-710 acquisition window.

Choosing Your Own Settings

If you know the filter and light source settings you want, or want to experiment with different settings, you can choose them yourself.

Next to Filter, click on either the Red, Green, Blue, or White checkbox, or a combination of two of the first three (Red-Green, Green-Blue, Red-Blue).

Filter Color	Wavelength	Application Examples
Red	595–750 nm	Coomassie G-250, BCIP/NBT, Fast Green FCF, Methylene Blue
Green	520–570 nm	Coomassie R-250, Basic Fuchsin
Blue	400–530 nm	Crocein Scarlet
Gray Scale	400–750 nm	Silver Stains, Copper Stains, Film

Fig. C-5. Examples of filter colors and applications.

Next to Light, select the appropriate light source.

- Reflective mode is for scanning opaque mediums such as dried gels on filter paper, arrays, or photographs. The scanning dimensions in this mode are 21 cm x 29 cm.
- Transmissive mode is for scanning transparent mediums such as films, gels, or slides. The scanning dimensions in this mode are 20 cm x 23 cm.

C.3. Step II. Select Scan Area

Preview Scan

Before selecting the particular area to scan, you can preview the entire scanning area to determine the exact position of your sample.

Click on Preview Scan. A quick, low-resolution scan of the entire scanning area will appear in the scanning window. Your sample should be visible within a portion of this scan.

Selecting an Area

Using the preview scan as a guide, select your scan area by dragging your mouse within the scanning window. The border of the scan area you are selecting will be marked by a frame.

Note: The scan area you select must be at least 1 cm wide.

When you release the mouse button, the border changes to a dashed blue line, indicating a selected area.

- To *reposition* the scanning area box you have created, position your cursor inside the box and drag. The entire box will move.
- To *resize* the box, position your cursor on a box side and drag. The side you have selected will move.
- To *redo* the box entirely, position your cursor outside the box and drag. The old box will disappear and a new box will be created.

You can also select the scanning area by entering coordinates in the appropriate fields (Top, Bottom, Left, Right). As you enter a coordinate, the position of the scanning area box will change accordingly.

When selecting, be sure to include the entire area of interest, and be generous with borders. You can always crop the image later.

C.4. Step III. Select Resolution

To select from a list of possible scanner resolutions, click on the Select button under Select Resolution. This will open the Select Scan Resolution dialog box.

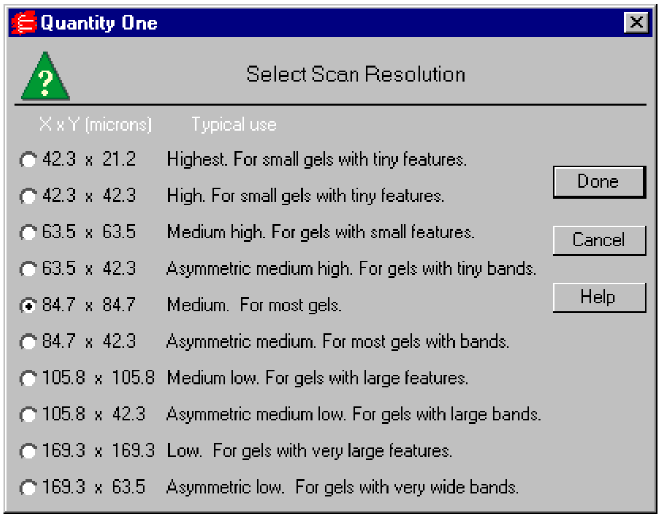


Fig. C-6. Select Scan Resolution dialog box.

Available resolutions are listed from highest to lowest in terms of the dimensions of the resulting pixels (in microns). Smaller pixels equal higher resolution. Each resolution is listed with its typical use.

In general, the size of your pixels should be one-tenth the height of your smallest object.

Some of the resolutions are asymmetrical, meaning that the resolution is higher in the vertical dimension (i.e., the pixels in the resulting image are larger in the horizontal dimension than in the vertical dimension). This is useful for gels with bands, where you are more interested in resolving in the y dimension to determine the size of bands and the spacing between them.

Specifying Your Own Resolution

If you select Oversample under More Options (see section C.7., Other Options, for details), you can specify your own resolution within the range of 43–169 microns (micrometers). With Oversample selected, enter values directly in the fields next to X resolution and Y resolution in the main acquisition window.

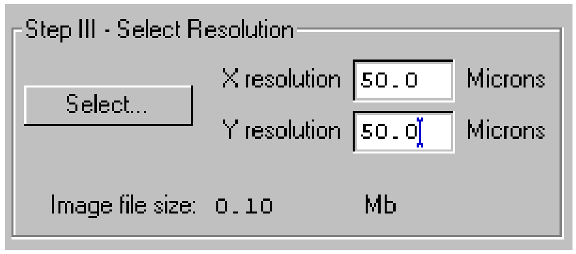


Fig. C-7. Entering a custom resolution (with Oversample selected).

Image File Size

Image File Size (under Select Resolution) shows the size of the scan file you are about to create. If you do not have enough computer memory for the specified file size, an error message will appear when you attempt to scan. If this happens, select a lower resolution or decrease the size of the area to be scanned. (Macintosh users can also increase the application memory partition. See your Macintosh computer documentation for guidance.)

C.5. Calibration

Images scanned using the GS-710 are automatically calibrated. The scanner plate has a built-in calibration strip for both transmissive and reflective scanning.

- The transmissive calibration strip calibrates the densitometer from 0 to 3.0 O.D.
- The reflective calibration strip calibrates the densitometer to 2.0 O.D.

The first time you use the GS-710, you must select some calibration settings to ensure that your calibration is accurate.

To set the automatic calibration settings, click on the More Options button in the GS-710 acquisition window. This will open the Densitometer Options dialog box.

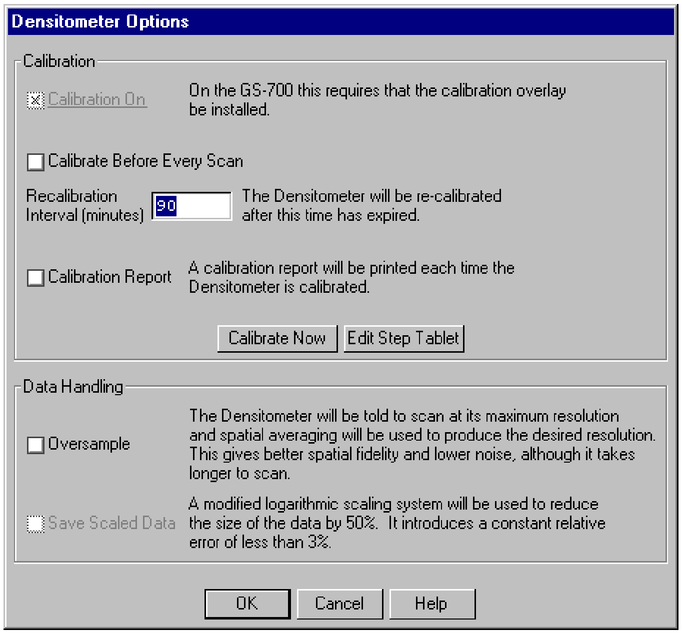


Fig. C-8. Densitometer Options dialog box.

C.5.a. Editing the Step Tablet

Before you scan for the first time, you must enter the step tablet values for your transmissive calibration strip into the Step Tablet Values form.

Note: In reflective scanning, the scanner is calibrated against the reflective step tablet. However, you do not need to enter the values for this step tablet; you can use the default values in the Step Tablet Values form.

Attached to the outside of your GS-710, you will find a copy of the manufacturer's printout of the diffuse density values for each segment of your transmissive step tablet, as well as a serial number that identifies the step tablet to which those density values belong. Those exact density values must be entered into the computer for the software to associate a correct density value with each step on the step tablet.

Note: Scanning in transmissive mode with incorrect step tablet values entered into the computer can cause significant errors in the reported densities of your scans.

To enter the density values for your step tablet, click on the Edit Step Tablet button in the Densitometer Options box. The Step Tablet Values dialog box for the GS-710 will open.

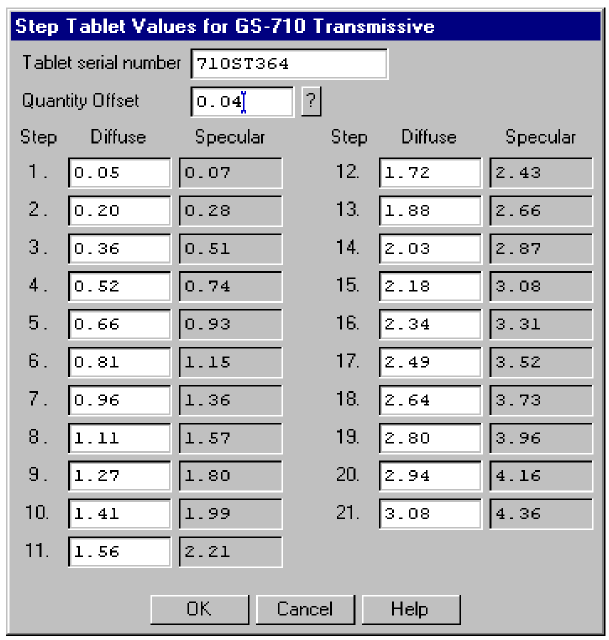


Fig. C-9. Step Tablet dialog box.

The dialog box will indicate whether the values are for the transmissive or reflective step tablet, depending on whether you are in transmissive or reflective mode.

If you are in reflective mode, you <u>do not</u> need to change the default step tablet values.

If you are in transmissive mode, you do need to enter the correct step tablet values that were shipped with your GS-710. You can type the serial number for these values into the Tablet Serial Number field.

The Quantity Offset field does not apply in the GS-710. This value should be set to zero.

Next, enter the values for the transmissive step tablet in the appropriate fields under the Diffuse column. After the step tablet is scanned, the software will associate each density value with its corresponding segment on the step tablet.

The step tablet density values do not need to be entered each time you calibrate. Once they have been entered and saved, they will be automatically recalled when the calibration strip is scanned.

When you are finished filling out the Step Tablet Values form, click on OK.

Diffuse Versus Specular O.D.

In the step tablet form, you enter O.D. as diffuse density, and then the software automatically calculates the specular density.

Specular density is a measure of the light that passes directly through a medium. Diffuse density includes light that is scattered as it passes through the medium. Step tablet values are given in diffuse density, but are measured by the scanner in specular density, and therefore must be converted according to the specular / diffuse O.D. ratio. This conversion does not affect quantitation.

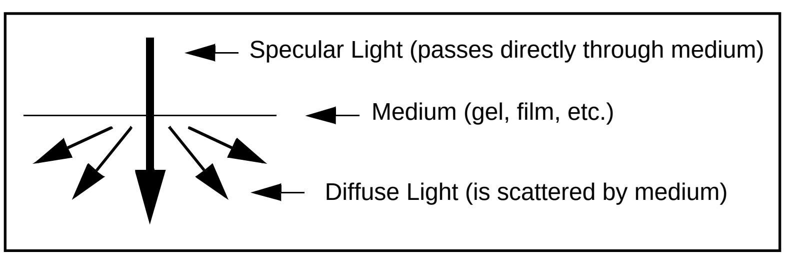


Fig. C-10. Specular and diffuse density

Diffuse density values are converted to specular optical density units according to the following formula:

$$Specular\ OD = 1.4\ \ Diffuse\ OD$$

C.5.b. Calibration Settings

After you have entered the step tablet values, you can immediately calibrate by clicking on the Calibrate Now button (in the Densitometer Options dialog box).

You can also specify how often you want the GS-710 to automatically recalibrate. Either click on the Calibrate Before Every Scan checkbox or enter a Recalibration Interval (in minutes) in the appropriate field.

Note: The scanner will automatically recalibrate each time you change your filter or your reflective / transmissive setting. (If you select a different application with the same filter and light settings, it will not auto recalibrate.)

Calibration Report

To print out a calibration report each time the densitometer calibrates, click on the Calibration Report checkbox.

C.6. Acquire the Image

To begin to scan, click on the Acquire button. The scanned image will begin to appear in the scanning window, line by line.

To interrupt a scan, click on the Stop button. A message will ask you to confirm the interrupt, and then you will be asked if you want to keep the partial scan. This feature is useful if you overestimated the size of the area you selected.

After the scan is complete, a window will open displaying the scan image, at which point you can save it as a Diversity Database image.

Note: The image will open with a default file name that includes the date, time, and (if applicable) user name. However, unless you have selected Auto

Save After Scan, the file will not be saved until you select Save or Save As from the File menu.

C.7. Other Options

Oversample

This feature allows you to scan at the maximum resolution of the GS-710 (42.3 x 42.3 microns) and then use spatial averaging to create an image with lower resolution. This can result in better images at lower resolution—however, it takes longer to scan.

To turn on oversampling, click on the More Options button in the acquisition window and click on the Oversample checkbox.

With oversampling on, you can specify your own resolution within the range of 43–169 microns by entering values directly in the fields next to X resolution and Y resolution in the main acquisition window.

Auto Save After Scan

To automatically save any scan you create, click on the Auto Save After Scan checkbox.

With this checkbox selected, when you click on Acquire, a Save As dialog box will open asking you to specify a file name and location for the image you are about to create. The scan will begin when you click on the Save button.

Make Backup Copy

If you have checked Auto Save After Scan, you can also automatically create a backup copy of any scan you create.

Click on the Make Backup Copy checkbox. With this checkbox selected, when a scan is created and saved, a backup copy will be placed in the same directory as the scanned image. Windows backup files will have an ".sbk" extension. Macintosh backup files will have the word "backup" after the file name.

This backup copy will be read-only, which means that you cannot make changes to it. You can open it like a normal file, but you must save it under a different file name before editing the image or performing analysis.

Highlight Saturated Pixels

When this is checked, any saturated pixels in the image will appear marked in red.

Hide Grid

To hide the gridlines in the scanning area window, click on the Hide Grid checkbox.

Appendix D. Fluor-S MultiImager

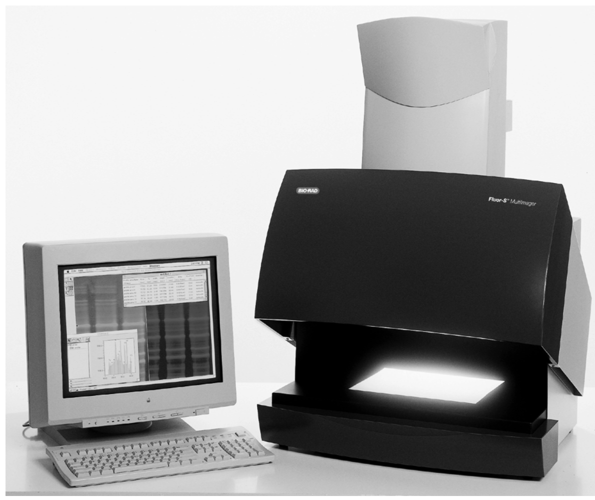


Fig. D-1. Fluor-S MultiImager.

Before you can begin acquiring images using the Fluor-S® MultiImager, the imaging system must be properly installed and connected with the host computer. See the Fluor-S hardware manual for installation, startup, and operating instructions.

Note: Make sure that the temperature light on the Fluor-S is green before attempting to capture an image. If you are using a PC, the Fluor-S should be turned on and the initialization sequence completed *before* the host computer is turned on. See the hardware manual for more details.

PC Only: A Note About SCSI Cards

The Fluor-S is connected to your computer by a Small Computer System Interface (SCSI) cable. To use the Fluor-S, you must have a SCSI card installed in your PC. If you have an older PC, you may also need to load the SCSI and WinASPI drivers that came with your card. Newer releases of Windows 95 and Windows NT will already have those drivers installed.

Simulation Mode

Any of the imaging device acquisition windows in Diversity Database can be opened in “simulation mode.” In this mode, an acquisition window will open and the controls will appear active, but instead of capturing real images, the window will create “dummy” images of manufactured data.

You do not need to be connected to an imaging device to open a simulated acquisition window. This is useful for demonstration purposes or practice scans.

To enter simulation mode, make sure you are not connected to the imaging device, then select its name from the File menu. You will be asked if you want to open the simulated window. The title of the window will indicate that it is simulated.

Note: There is no simulated calibration for the GS-700 and GS-710 Imaging Densitometers.

D.1. Opening the Fluor-S Acquisition Window

To acquire images using the Fluor-S, go to the File menu and select Fluor-S.... The acquisition window for the imager will open, displaying a control panel and a video display window.

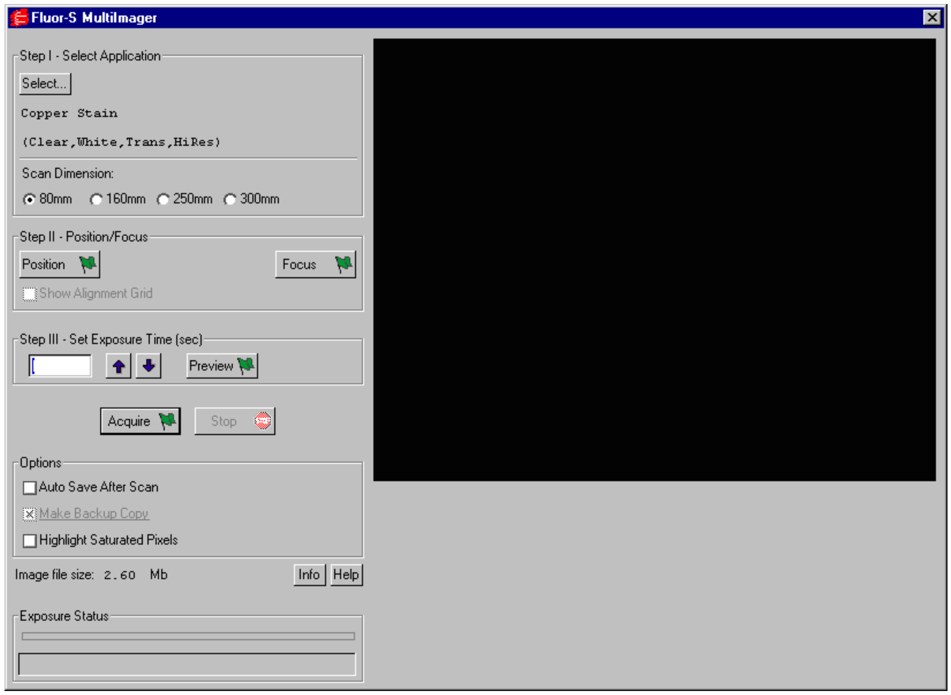


Fig. D-2. Fluor-S acquisition window

When the Fluor-S window first opens, no image will be displayed.

The control panel has been arranged from top to bottom to guide you through the acquisition procedure. There are four basic steps to acquiring an image using the Fluor-S:

1. Select the application.
2. Position and focus the gel or other object to be imaged.
3. Set the exposure time.
4. Acquire the image.

D.2. Step 1. Select Application

To set the appropriate filter and other parameters for the type of object you are imaging, click on the Select button under Select Application. The available applications and their associated settings are listed in a tree that expands from left to right.

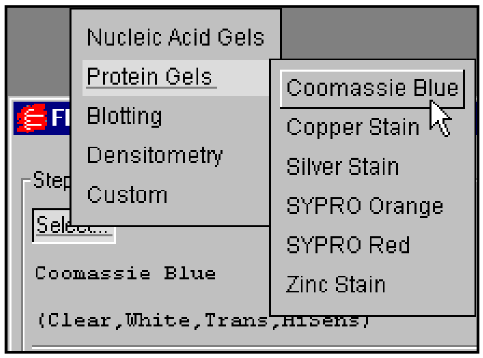


Fig. D-3. The application tree in the Fluor-S acquisition window.

First select your general application, then select the particular stain or medium you are using. If you select one of the chemi applications under “Blotting,” you also have the option of selecting High Resolution or High Sensitivity (see below).

When you select an application, the software automatically sets the appropriate filter (520LP, 530DF60, 610LP, clear, or none) and light type (UV, white, or none) in the Fluor-S for that particular application. With the exception of copper and zinc stains, the software also selects the light source (trans, epi, or neither). With copper and zinc stains, you are prompted to select Epi or Trans as the light source.

Your selection will be displayed below the Select button. To exit the tree without selecting, press the ESC key.

Custom Applications

If your application is not listed and you know the filter and light settings you want to use, you can create and save your own custom application.

From the application tree, select Custom, then Create. This will open a dialog box in which you can name your application and select your settings.

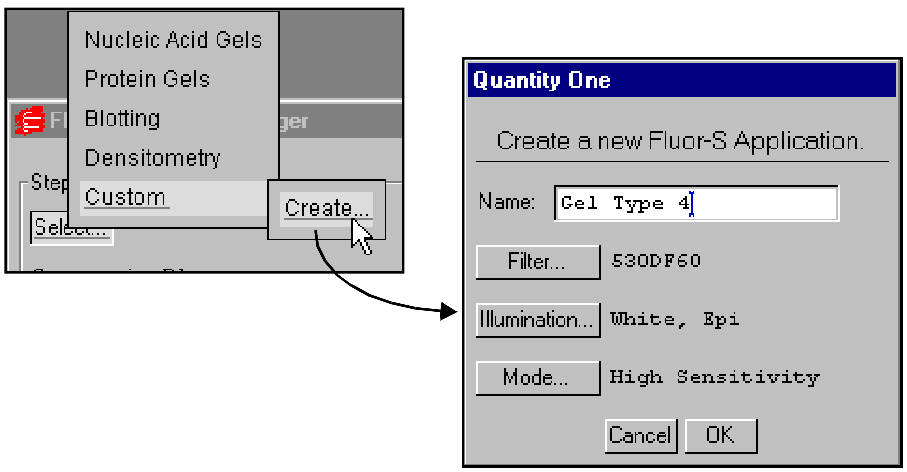


Fig. D-4. Creating a new custom application.

To select the filter (including user-defined), type of illumination, and camera mode, click on the appropriate buttons. Enter a name for your application in the Name field. Click on OK to implement your changes.

After you have created an application, you can select it from the application tree by selecting Custom and the name you created. You can delete the application by selecting Custom, Delete, and the name of the application.

Scan Dimension

If an application uses trans illumination, the Scan Dimension buttons become active. The scan dimension is the distance traveled by the transilluminating light source as it scans horizontally across the platen.

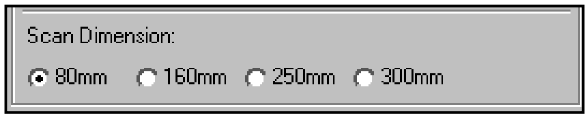


Fig. D-5. Scan dimension settings for trans illumination.

The full scanning range is 300 mm. Select a smaller range if your sample is small and you do not want to wait while the light source travels over the maximum scan width.

High Resolution Versus High Sensitivity

High Resolution and High Sensitivity are mutually exclusive options. High Resolution is the normal operating mode for the Fluor-S. High Sensitivity provides optimal sensitivity for low-light applications such as chemiluminescence.

In High Resolution mode, images are captured at the maximum resolution of the camera.

In High Sensitivity mode, the pixels in the camera are “binned” (i.e., four pixels are combined into one) to increase the amount of signal per pixel. However, combining the pixels results in a reduction in the resolution of the image.

D.3. Step II. Position/Focus

Note: When you click on the Position or Focus button, the light inside the Fluor-S box automatically turns <u>on</u>. To turn the light <u>off</u> while positioning or focusing, hold down the SHIFT key when clicking on the button.

Position

The first step in acquiring an image is centering your gel or other object within the camera frame. To do this, click on the Position button. The Fluor-S will begin capturing a “live” image and updating it every second.

With the Position button selected, look at the image in the acquisition window while you position your object in the center of the platen. If you have a zoom lens on the camera, you can adjust the magnification while you position.

You can select the Show Alignment Grid checkbox to facilitate positioning.

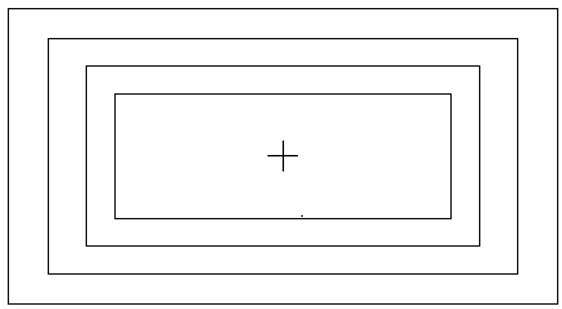


Fig. D-6. Fluor-S alignment grid.

When you are finished positioning, click on the Stop button.

Focus

Note: Before focusing, you should adjust the f-stop on the camera to the lowest setting (i.e., the maximum aperture). This reduces the depth of field, allowing you to more accurately focus the camera. Then, after focusing, increase the f-stop to the desired setting. (See the following table on recommended f-stops.)

After you have positioned your sample, click on the Focus button and look at the image in the acquisition window while aligning the two focusing arrows on the camera lens. While focusing, the camera will limit its focus to a small portion of the sample (this will not affect any zoom lens adjustments you may have made.)

When you are finished focusing, click on the Stop button.

D.4. Step III. Set Exposure Time

When you are ready to capture an image, you will need to select an exposure time. “Exposure” refers to the integration of the image on the CCD over a set period of time. The effect is analogous to exposing photographic film to light.

The exposure time you select should be based on your application. The table provides recommended exposure times for various applications.

Table 1: Recommended f-stops and Exposure Times

Application	f-stop[1] Zoom lens	f-stop[1] 50mm lens	Exposure time[2]
Ethidium bromide	f-4	f-2.8	5–30 sec.
SYBR Green	f-4	f-2.8	5–30 sec.
SYPRO Orange	f-4	f-2.8	5–30 sec.
Radiant Red	f-4	f-2.8	5–30 sec.
Fluorescein	f-4	f-2.8	30 sec.–2 min.
Texas Red	f-4	f-2.8	30 sec.–2 min.
Colorimetric Stains (e.g., Coomassie or Silver)	f-11	f-11	1–5 sec.
X-ray Film	f-11	f-11	1–5 sec.
Photographs	f-11	f-11	1–5 sec.
Chemi[3]		f-1.8	1–10 min.

[1]For sharper focusing, close the f-stop down 1–2 stops from full open while focusing.
[2]Increase exposure time two fold for each step increase in f-stop.
[3]For chemi applications, the 50mm lens is recommended. Always remove the 660 filter.

Note: For most applications, you can select an exposure time, capture an image, study it, then adjust the exposure time accordingly. Repeat this procedure as many times as necessary to obtain a good image. For chemiluminescent samples, which degrade over time and emit low levels of light, select a high exposure time initially.

You can enter an exposure time (in seconds) directly in the field, or use the Arrow buttons to adjust the exposure time in 10 percent increments.

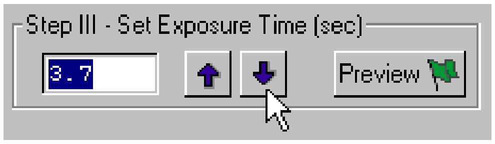


Fig. D-7. Selecting an exposure time.

Preview

For shorter exposures, you can use Preview to test different exposure times. Click on the Preview button create a preview exposure and display it in the acquisition window.

A preview scan takes only half as long to create as a real scan, because the preview scan does not capture a “dark” image (see below). The progress of the exposure will be displayed in the Exposure Status bar at the bottom of the dialog box.

You cannot save preview scans.

If you want to stop a preview scan that is in progress, click on the Stop button.

D.5. Acquire the Image

When you are ready to acquire an image, click on the Acquire button.

Diversity Database will first acquire the image, and then acquire a “dark” image for the purpose of filtering out dark noise. (Dark noise is a type of image noise that is generated by CCD cameras.)

The progress of each exposure will be displayed in the Exposure Status bar at the bottom of the dialog box.

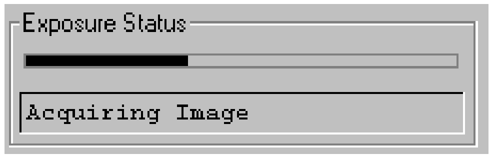


Fig. D-8. Exposure Status bar when acquiring an image.

The type of exposure (image or dark) that is being performed will be noted below the status bar.

Note: A dark image is only acquired the first time you perform a scan with particular application and exposure settings. If you perform subsequent scans with the same settings, no dark image will be acquired.

If you want to stop a scan that is in progress, click on the Stop button.

Saving the Image

After an image has been acquired, a separate window will pop up containing the new image. The image will have a default file name that includes the date, time, and user (if known). To save the image, select Save or Save As from the File menu.

You can then analyze the image using Diversity Database's menu and toolbar functions.

D.6. Options

Auto Save After Scan

To automatically save any image you create, click on the Auto Save After Scan checkbox.

With this checkbox selected, when you click on Acquire, a Save As dialog box will open asking you to specify a file name and location for the image you are about to create. The scan will begin when you click on the Save button.

Make Backup Copy

You can automatically create a backup copy of any scan you create. To do so, first select Auto Save After Scan (see above), then select the Make Backup Copy checkbox.

With this checkbox selected, when you save a scan, a backup copy will be placed in the same directory as the scanned image. Windows backup files will have an ".sbk" extension. Macintosh backup files will have the word "backup" after the file name.

This backup copy will be read-only, which means that you cannot make changes to it. You can open it like a normal file, but you must save it under a different file name before editing the image or performing analysis.

Highlight Saturated Pixels

When this is checked, any saturated pixels in the image will appear marked in red.

Imaging Area Size

To specify the size of your imaging area, click on the Info button and enter each dimension in the appropriate field.

The dimensions of your imaging area are determined by how much you are zooming in or out on your sample. The dimensions determine the size of the pixels in your image (i.e., resolution). When you adjust one imaging area dimension, the other dimension will change to maintain the aspect ratio of the camera lens.

Note: Your imaging area settings must be correct if you want to do 1:1 printing. It is also necessary if you are comparing the quantities of objects (e.g., using the Volume Tools) between images.

File Size of Images

Image File Size (below Options) shows the size of the image file you are about to create. This size is determined by whether you selected High Resolution or High Sensitivity when you selected an application.

If you do not have enough computer memory for the specified file size, an error message will appear when you attempt to acquire an image. (Macintosh users can increase the application memory partition. See your Macintosh computer documentation for guidance.)

Appendix E. Fluor-S MAX MultiImager

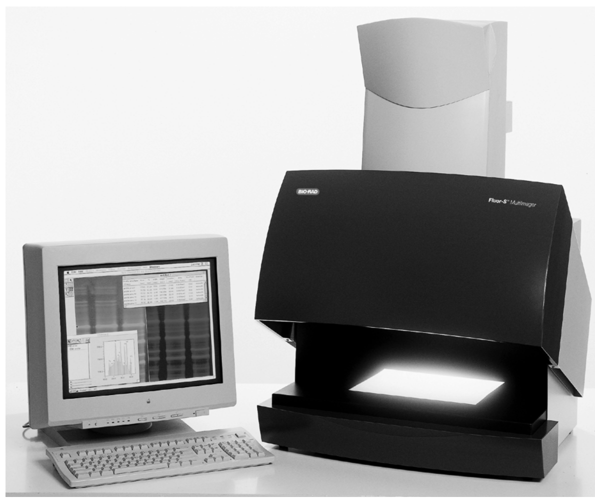


Fig. E-1. Fluor-S MAX MultiImager.

Before you can begin acquiring images using the Fluor-S® MAX MultiImager, the imaging system must be properly installed and connected with the host computer. See the Fluor-S MAX hardware manual for installation, startup, and operating instructions.

Note: Make sure that the temperature light on the Fluor-S MAX is green before attempting to capture an image. If you are using a PC, the Fluor-S MAX should be turned on and the initialization sequence completed *before* the host computer is turned on. See the hardware manual for more details.

PC Only: A Note About SCSI Cards

The Fluor-S MAX is connected to your computer by a Small Computer System Interface (SCSI) cable. To use the Fluor-S MAX, you must have a SCSI card installed in your PC. If you have an older PC, you may also need to load the SCSI and WinASPI drivers that came with your card. Newer releases of Windows 95 and Windows NT will already have those drivers installed.

Simulation Mode

Any of the imaging device acquisition windows in Diversity Database can be opened in “simulation mode.” In this mode, an acquisition window will open and the controls will appear active, but instead of capturing real images, the window will create “dummy” images of manufactured data.

You do not need to be connected to an imaging device to open a simulated acquisition window. This is useful for demonstration purposes or practice scans.

To enter simulation mode, make sure you are not connected to the imaging device, then select its name from the File menu. You will be asked if you want to open the simulated window. The title of the window will indicate that it is simulated.

Note: There is no simulated calibration for the GS-700 and GS-710 Imaging Densitometers.

E.1. Opening the Fluor-S MAX Acquisition Window

To acquire images using the Fluor-S MAX, go to the File menu and select Fluor-S MAX.... The acquisition window for the imager will open, displaying a control panel and a video display window.

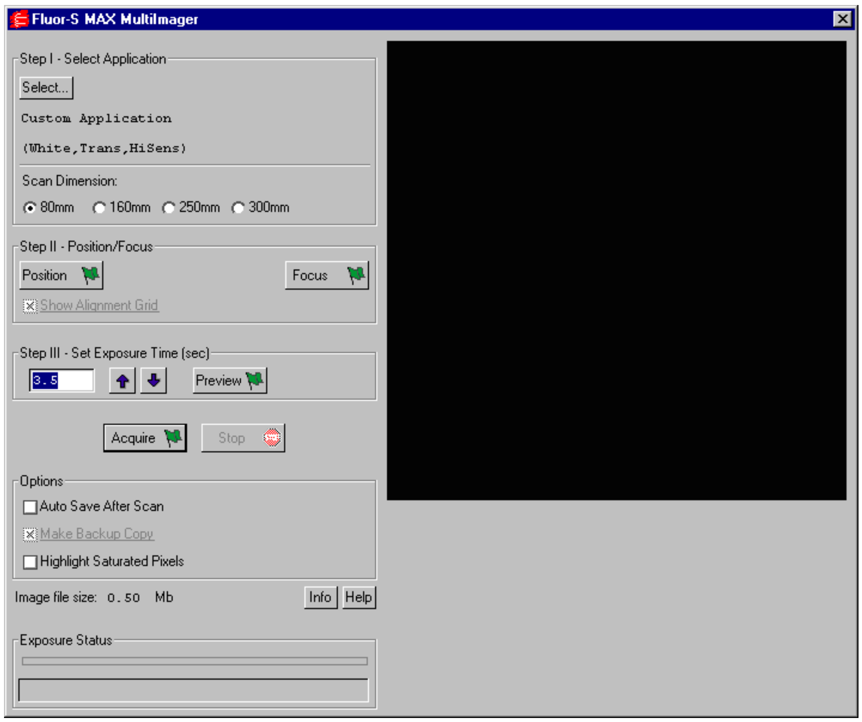


Fig. E-2. Fluor-S MAX acquisition window.

When the Fluor-S MAX window first opens, no image will be displayed.

The control panel has been arranged from top to bottom to guide you through the acquisition procedure. There are four basic steps to acquiring an image using the Fluor-S MAX:

1. Select the application.

2. Position and focus the gel or other object to be imaged.

3. Set the exposure time.

4. Acquire the image.

E.2. Step 1. Select Application

To set the appropriate filter and other parameters for the type of object you are imaging, click on the Select button under Select Application. The available applications and their associated settings are listed in a tree that expands from left to right.

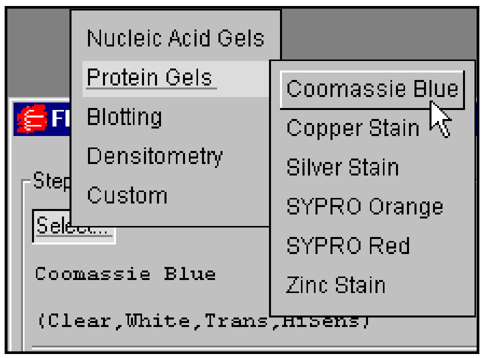


Fig. E-3. The application tree in the Fluor-S MAX acquisition window.

First select your general application, then select the particular stain or medium you are using. If you select one of the chemi applications under “Blotting,” you also have the option of selecting High Sensitivity or Ultra Sensitivity (see below).

When you select an application, the software automatically sets the appropriate filter (520LP, 530DF60, 610LP, clear, or none) and light type (UV, white, or none) in the Fluor-S MAX for that particular application. With the exception of copper and zinc stains, the software also selects the light source (trans, epi, or neither). With copper and zinc stains, you are prompted to select Epi or Trans as the light source.

For applications involving trans illumination, you must also specify a scan dimension (see below).

Your selection will be displayed below the Select button. To exit the tree without selecting, press the ESC key.

Custom Applications

If your application is not listed and you know the filter and light settings you want to use, you can create and save your own custom application.

From the application tree, select Custom, then Create. This will open a dialog box in which you can name your application and select your settings.

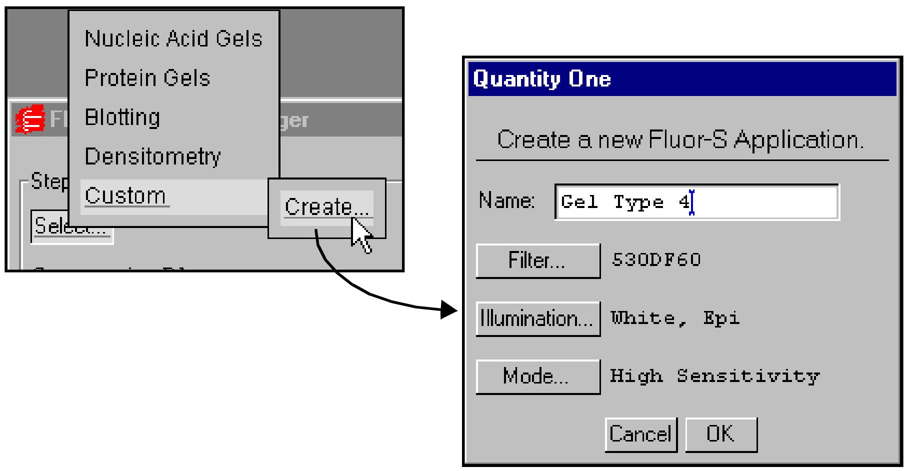


Fig. E-4. Creating a new custom application.

To select the filter, type of illumination, and camera mode, click on the appropriate buttons. Enter a name for your application in the Name field. Click on OK to implement your changes.

After you have created an application, you can select it from the application tree by selecting Custom and the name you created. You can delete the application by selecting Custom, Delete, and the name of the application.

Scan Dimension

If an application uses trans illumination, the Scan Dimension buttons become active. The scan dimension is the distance traveled by the transilluminating light source as it scans horizontally across the platen.

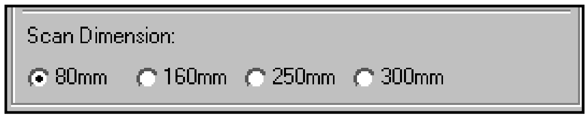


Fig. E-5. Scan dimension settings for trans illumination.

The full scanning range is 300 mm. Select a smaller range if your sample is small and you do not want to wait while the light source travels over the maximum scan width.

High Sensitivity and Ultra Sensitivity

High Sensitivity and Ultra Sensitivity are different camera modes. In Ultra Sensitivity mode, the Fluor-S MAX camera is cooled to –35.0 degrees C. In High Sensitivity, the camera is cooled to –20.0 degrees C.

High Sensitivity is the normal operating mode of the Fluor-S MAX. Ultra Sensitivity provides optimal sensitivity for low-light applications. It may be selected for the chemiluminescence and chemifluorescence applications, or for a custom application.

Note: When you change from High to Ultra Sensitivity or visa versa, there will be a delay of several minutes while the Fluor-S MAX camera cools down or warms up. If you attempt to acquire an image during this period, you will be notified of the changing temperature. If you do not want to wait, you can cancel the mode change.

E.3. Step II. Position/Focus

Note: When you click on the Position or Focus button, the light inside the Fluor-S MAX box automatically turns off. This is because if you focus with the camera at maximum aperture (see note below), leaving the light on would make it difficult to view the image. To turn the light on while positioning or focusing, hold down the SHIFT key when clicking on the button.

Position

The next step in acquiring an image is centering your gel or other object within the camera frame. To do this, click on the Position button. The Fluor-S MAX will begin capturing a “live” image and updating it every second.

With the Position button selected, look at the image in the acquisition window while you position your object in the center of the platen. If you have a zoom lens on the camera, you can adjust the magnification while you position.

You can select the Show Alignment Grid checkbox to facilitate positioning.

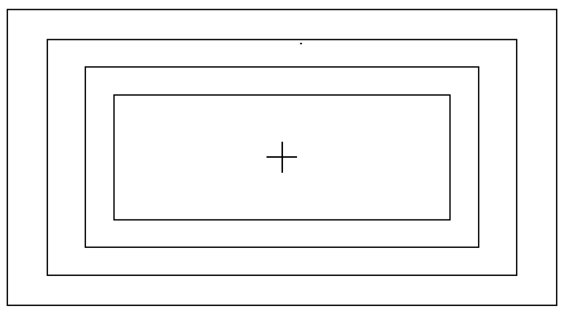


Fig. E-6. Fluor-S MAX alignment grid.

When you are finished positioning, click on the Stop button.

Focus

Note: Before focusing, you should adjust the f-stop on the camera to the lowest setting (i.e., the maximum aperture). This reduces the depth of field, allowing you to more accurately focus the camera. Then, after focusing, increase the f-stop to the desired setting.

After you have positioned your sample, click on the Focus button and look at the image in the acquisition window while aligning the two focusing arrows on the camera lens. While focusing, the camera will limit its focus to a small portion of the sample (this will not affect any zoom lens adjustments you may have made.)

When you are finished focusing, click on the Stop button.

E.4. Step III. Set Exposure Time

When you are ready to capture an image, you will need to select an exposure time. “Exposure” refers to the integration of image captures on the CCD over a set period of time. The effect is analogous to exposing photographic film to light.

The exposure time you select should be based on your application and your “best guess” as to what exposure will give you the best image.

Note: For most applications, you can select an exposure time, capture an image, study it, then adjust the exposure time accordingly. Repeat this procedure as many times as necessary to obtain a good image. For chemi samples, which degrade over time and emit low levels of light, select a high exposure time initially.

You can enter an exposure time (in seconds) directly in the field, or use the Arrow buttons to adjust the exposure time in 10 percent increments.

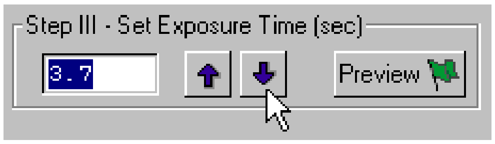


Fig. E-7. Selecting an exposure time.

Preview

For shorter exposures, you can use Preview to test different exposure times. Click on the Preview button create a preview exposure and display it in the acquisition window.

A preview scan takes only half as long to create as a real scan, because the preview scan does not capture a “dark” image (see below). The progress of the exposure will be displayed in the Exposure Status bar at the bottom of the dialog box.

You cannot save preview scans.

If you want to stop a preview scan that is in progress, click on the Stop button.

E.5. Acquire the Image

When you are ready to acquire an image, click on the Acquire button.

Diversity Database will first acquire the image, and then acquire a “dark” image for the purpose of filtering out dark noise. (Dark noise is a type of image noise that is generated by CCD cameras.)

The progress of each exposure will be displayed in the Exposure Status bar at the bottom of the dialog box.

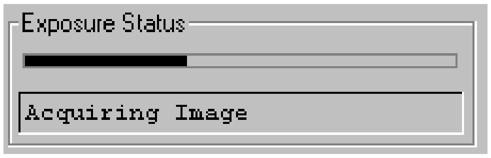


Fig. E-8. Exposure Status bar when acquiring an image.

The type of exposure (image or dark) that is being performed will be noted below the status bar.

Note: A dark image is only acquired the first time you perform a scan with particular application and exposure settings. If you perform subsequent scans with the same settings, no dark image will be acquired.

If you want to stop a scan that is in progress, click on the Stop button.

Saving the Image

After an image has been acquired, a separate window will pop up containing the new image. The image will have a default file name that includes the date, time, and user (if known). To save the image, select Save or Save As from the File menu.

You can then analyze the image using Diversity Database's menu and toolbar functions.

E.6. Options

Auto Save After Scan

To automatically save any image you create, click on the Auto Save After Scan checkbox.

With this checkbox selected, when you click on Acquire, a Save As dialog box will open asking you to specify a file name and location for the image you are about to create. The scan will begin when you click on the Save button.

Make Backup Copy

You can automatically create a backup copy of any scan you create. To do so, first select Auto Save After Scan (see above), then select the Make Backup Copy checkbox.

With this checkbox selected, when you save a scan, a backup copy will be placed in the same directory as the scanned image. Windows backup files will have an ".sbk" extension. Macintosh backup files will have the word "backup" after the file name.

This backup copy will be read-only, which means that you cannot make changes to it. You can open it like a normal file, but you must save it under a different file name before editing the image or performing analysis.

Highlight Saturated Pixels

When this is checked, any saturated pixels in the image will appear marked in red.

Imaging Area Size

The imaging area is the area of a sample that is captured by the camera and displayed in the video display window. To specify the size of your imaging area, click on the Info button and enter each dimension in the appropriate field. When you adjust one imaging area dimension, the other will change to maintain the aspect ratio of the camera lens.

The imaging area will change depending on your zoom factor. For example, if you have zoomed in on a sample that is 4.5 x 4.5 cm, then you would enter 4.5 for the width and height of your imaging area.

Note: Your imaging area settings must be correct if you want to do 1:1 printing. They must also be correct if you want to compare the quantities of objects (e.g., using the Volume Tools) between images.

The imaging area dimensions also determine the size of the pixels in your image (i.e., resolution). A smaller imaging area will result in a higher resolution.

File Size of Images

Image File Size (below Options) shows the size of the image file you are about to create.

If you do not have enough computer memory for the specified file size, an error message will appear when you attempt to acquire an image. (Macintosh users can increase the application memory partition. See your Macintosh computer documentation for guidance.)

Appendix F. Personal Molecular Imager FX

Fig. F-1. Personal Molecular Imager FX

Before you can begin acquiring images using the Personal Molecular Imager® FX, the instrument must be properly installed and connected with the host computer. See the Personal FX hardware manual for installation, startup, and operating instructions.

Note: The Personal FX should be turned on and the initialization sequence completed *before* the host computer is turned on (except in the case of certain Power Macintosh configurations). See the hardware manual for more details.

PC Only: A Note About SCSI Cards

The Personal FX is connected to your computer by a Small Computer System Interface (SCSI) cable. To use the Personal FX, you must have a SCSI card installed in your PC. If you have an older PC, you may also need to load the SCSI and WinASPI drivers that came with your card. Newer releases of Windows 95 and Windows NT will already have those drivers installed.

Simulation Mode

Any of the imaging device acquisition windows in Diversity Database can be opened in "simulation mode." In this mode, an acquisition window will open and the controls will appear active, but instead of capturing real images, the window will create "dummy" images of manufactured data.

You do not need to be connected to an imaging device to open a simulated acquisition window. This is useful for demonstration purposes or practice scans.

To enter simulation mode, make sure you are not connected to the imaging device, then select its name from the File menu. You will be asked if you want to open the simulated window. The title of the window will indicate that it is simulated.

Note: There is no simulated calibration for the GS-700 and GS-710 Imaging Densitometers.

F.1. Opening the Personal FX Acquisition Window

To acquire images using the Personal FX, go to the File menu and select Personal FX.... The acquisition window for the imager will open, displaying a control panel and the scanning area window.

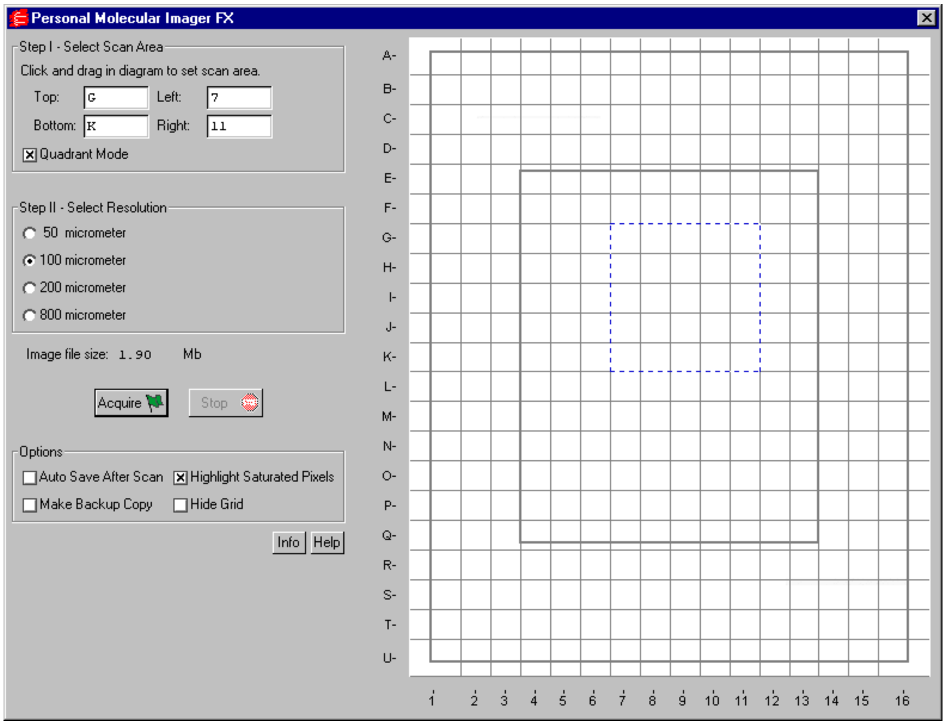


Fig. F-2. Personal FX acquisition window

The default scanning window is marked by grid lines that divide the area into quadrants. There is also an outer box and inner box marked by thicker lines. This conforms to the sample pad for the standard Bio-Rad Exposure Cassette that is supplied with the Personal FX. The quadrants are numbered 1–16 left to right and lettered A–U top to bottom.

If you prefer a scanning window measured in centimeters, deselect the Quadrant Mode checkbox in the control panel by clicking on it. To hide the gridlines, click on the Hide Grid checkbox under Options.

The control panel has been arranged from top to bottom to guide you through the acquisition procedure. There are three basic steps to scanning an image using the Personal FX:

1. Select the scan area
2. Select the resolution
3. Acquire the image

F.2. Step 1. Select Scan Area

To select a scan area, drag your mouse within the scanning window. (In the scanning window, your cursor appearance will change to a cross.) The border of the scan area you are selecting is marked by a frame.

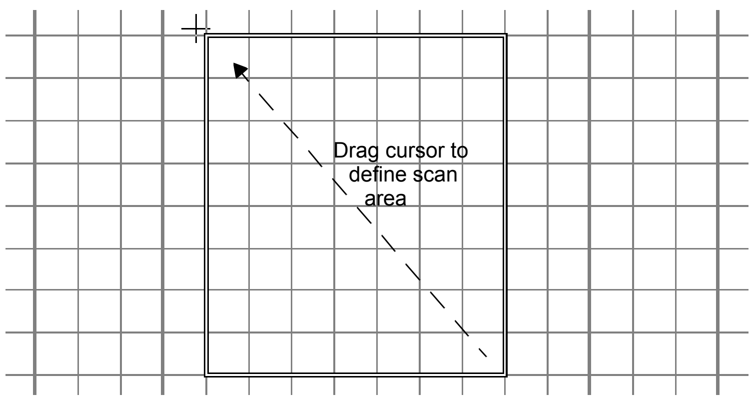


Fig. F-3. Selecting a scan area.

If you are in quadrant mode, note that the frame "locks" onto the next quadrant as you drag. When you release the mouse button, the border changes to a dashed blue line, indicating a selected area.

- To *reposition* the scanning box you have selected, position your cursor inside the box and drag. The entire box will move.
- To *resize* the box, position your cursor on a box side and drag. The side you have selected will move.
- To *redo* the box entirely, position your cursor outside the box and drag. The old box will disappear and a new box will be created.

You can also select the scanning area by entering coordinates in the appropriate fields (Top, Bottom, Left, Right). After you enter a coordinate, the position of the scanning area box will change accordingly.

When selecting, be sure to include the entire area of interest, and be generous with borders. You can always crop the image later.

F.3. Step II. Select Resolution

The Personal FX acquisition window allows you to scan at 50, 100, 200, or 800 micrometers. These resolutions are listed as option buttons in the control panel.

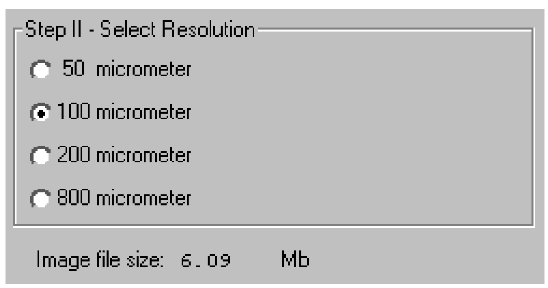


Fig. F-4. Resolution option buttons.

The resolution you select should be based on the size of the objects (e.g., bands, spots) you are interested in. For example:

- 50 micrometer resolution should be reserved for images requiring the highest level of detail, e.g., high density in situ samples, 1,536-well micro-plates, high density arrays, samples with very closely spaced bands. Files scanned at 50 micrometers can be very large.
- 100 micrometer resolution should be used for typical gels and arrays.
- 200 micrometer resolution is useful for gels with large bands and dot blots.
- 800 micrometer resolution should be reserved for very large objects, such as CAT assays.

File Size of Images

Image File Size (below Select Resolution) shows the size of the scan file you are about to create. If you do not have enough computer memory for the specified file size, an error message will appear when you attempt to scan. If this happens, select a lower resolution or decrease the size of the area to be scanned. (Macintosh users can also increase the application memory partition. See your Macintosh computer documentation for guidance.)

F.4. Acquire the Image

Once you have selected your scan area and resolution, you ready to acquire an image.

Click on the Acquire button. There may be a short delay while the image laser warms up; then the scanned image will begin to appear in the scanning window, line by line.

To interrupt a scan, click on the Stop button. A message will ask you to confirm the interrupt, and then you will be asked if you want to keep the partial scan. This feature is useful if you overestimated the size of the area you selected.

Note: If the image you are scanning has more than 10 saturated pixels, you will receive a warning message.

Saving the Image

After the scan is complete, a message will appear asking you if you want to keep the scan. If you select Yes, a separate window will pop up containing the new image. The image will have a default file name that includes the date, time, and user (if known). To save the image, select Save or Save As from the File menu.

You can then analyze the image using Diversity Database’s menu and toolbar functions.

F.5. Options

Auto Save After Scan

To automatically save any scan you create, click on the Auto Save After Scan checkbox.

With this checkbox selected, when you click on Acquire, a Save As dialog box will open asking you to specify a file name and location for the image you are about to create. The scan will begin when you click on the Save button.

Make Backup Copy

You can automatically create a backup copy of any scan you create. To do so, first select Auto Save After Scan (see above), then select the Make Backup Copy checkbox.

With this checkbox selected, when you save a scan, a backup copy will be placed in the same directory as the scanned image. Windows backup files will have an ".sbk" extension. Macintosh backup files will have the word "backup" after the file name.

This backup copy will be read-only, which means that you cannot make changes to it. You can open it like a normal file, but you must save it under a different file name before editing the image or performing analysis.

Highlight Saturated Pixels

When this is checked, any saturated pixels in the image will appear marked in red.

Hide Grid

To hide the gridlines in the scanning area window, click on the Hide Grid checkbox.

Appendix G. Molecular Imager FX

Fig. G-1. Molecular Imager FX

Before you can begin acquiring images using the Molecular Imager® FX, the instrument must be properly installed and connected with the host computer. See the FX hardware manual for installation, startup, and operating instructions.

Note: The FX should be turned on and the initialization sequence completed *before* the host computer is turned on (except in the case of certain Power Macintosh configurations). See the hardware manual for more details.

PC Only: A Note About SCSI Cards

The FX is connected to your computer by a Small Computer System Interface (SCSI) cable. To use the FX, you must have a SCSI card installed in your PC. If you have an older PC, you may also need to load the SCSI and WinASPI drivers that came with your card. Newer releases of Windows 95 and Windows NT will already have those drivers installed.

Simulation Mode

Any of the imaging device acquisition windows in Diversity Database can be opened in “simulation mode.” In this mode, an acquisition window will open and the controls will appear active, but instead of capturing real images, the window will create “dummy” images of manufactured data.

You do not need to be connected to an imaging device to open a simulated acquisition window. This is useful for demonstration purposes or practice scans.

To enter simulation mode, make sure you are not connected to the imaging device, then select its name from the File menu. You will be asked if you want to open the simulated window. The title of the window will indicate that it is simulated.

Note: There is no simulated calibration for the GS-700 and GS-710 Imaging Densitometers.

G.1. Opening the FX Control Panel

To acquire images using the FX, go to the File menu and select FX.... The acquisition window for the imager will open, displaying the control panel for the imager and the scanning area window.

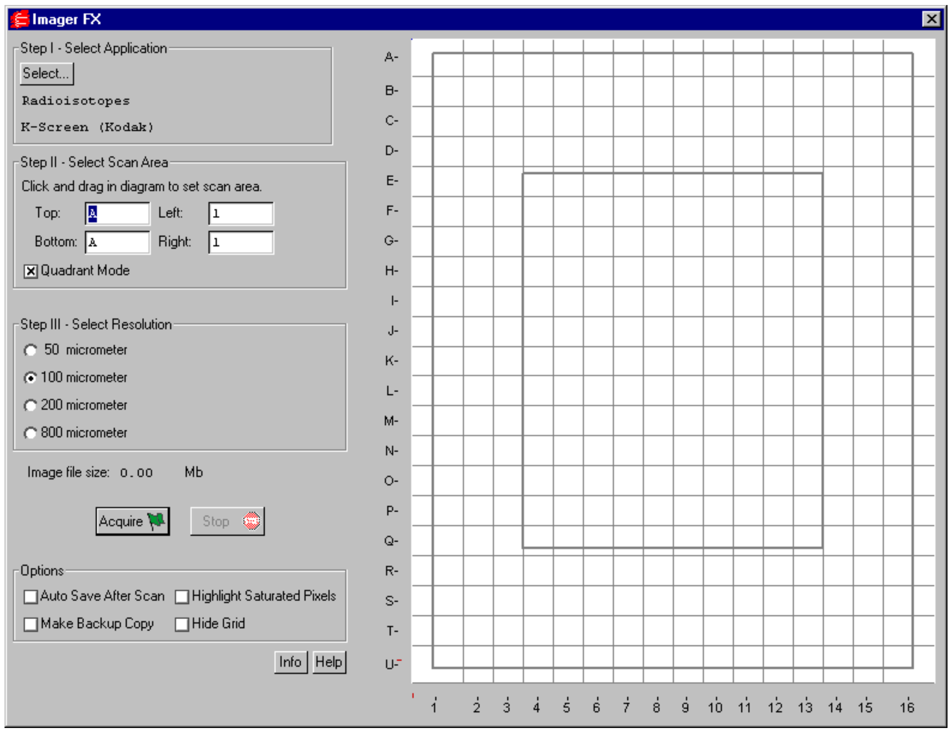


Fig. G-2. FX acquisition window

The default scanning window is marked by grid lines that divide the area into quadrants. There is also an outer box and inner box marked by thicker lines. This conforms to the sample pad for the standard Bio-Rad Exposure Cassette that is supplied with the FX. The quadrants are numbered 1–16 left to right and lettered A–U top to bottom.

If you prefer a scanning window measured in centimeters, deselect the Quadrant Mode checkbox in the control panel by clicking on it. To hide the gridlines, click on the Hide Grid checkbox under Options.

The control panel has been arranged from top to bottom to guide you through the acquisition procedure. There are four basic steps to scanning an image using the FX:

1. Select the application.
2. Select the scan area.
3. Select the resolution.
4. Acquire the image.

G.2. Step 1. Select Application

To set the appropriate filter and other parameters for the type of object you are imaging, click on the Select button under Select Application.

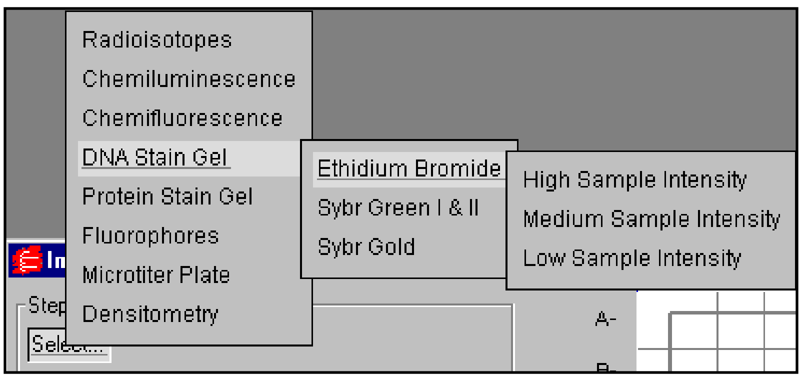


Fig. G-3. Example of an application tree: Ethidium Bromide gel.

The applications and associated settings are listed in a tree that expands from left to right. When you select an application, the software automatically sets the appropriate filter(s) in the FX for that particular application.

First you select the category of your application, next you select your particular application, and finally (if appropriate) you select the intensity of your samples.

To exit the tree without selecting, press the ESC key.

Your selection will be displayed below the Select button.

- Radioisotopes
  - CS- or BI-Screen (Bio-Rad)
  - K-Screen (Kodak)
  - Fuji-Screen
- Chemiluminescence
  - Chemi-Screen (Bio-Rad)
- Chemifluorescence
  - Attophos
- DNA Stain Gel
  - Ethidium Bromide
  - Sybr Green I & II
  - Sybr Gold
- Protein Stain Gel
  - Sypro Orange
  - Sypro Red
  - Nile Red
- Fluorophores
  - FITC
  - R6G
  - TAMRA
  - CY3
  - Texas Red
- Microtiter Plate
  - DNA (Sybr Green I)
  - Protein (Nano Orange)
  - ssDNA (Oligreen)
  - DNA (Picogreen)
  - B-Gal (FDG)
  - GUS (FDG)
- Densitometry
  - Coomassie Blue Gel/Blot
  - Copper Stain Gel/Blot
  - Silver Stain Gel/Blot
  - X-Ray Film (Grey Type)

Fig. G-4. Applications available in the FX acquisition window.

Sample Intensity

When selecting any application except radioisotopes, you will need to specify a sample intensity. This is simply a rough estimate of how much sample is apparent in your gel or other object.

Do not worry if you are not sure about the level of intensity of your sample—if you make a mistake, you can always change the setting and capture another image.

For example, if you specify a low sample intensity and the resulting image has too many saturated pixels, you will receive a warning message. Simply change the setting to medium sample intensity and rescan. If you specify a high sample intensity and the resulting image is too faint, select medium or low sample intensity and rescan.

G.3. Step II. Select Scan Area

To select a scan area, drag your mouse within the scanning window. (In the scanning window, your cursor appearance will change to a cross.) The border of the scan area you are selecting is marked by a frame.

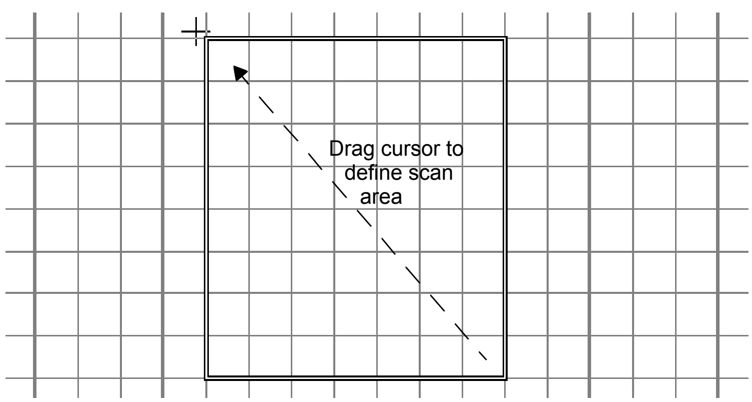


Fig. G-5. Selecting a scan area.

If you are in quadrant mode, note that the frame “locks” onto the next quadrant as you drag. When you release the mouse button, the border changes to a dashed blue line, indicating a selected area.

- To *reposition* the scanning box you have selected, position your cursor inside the box and drag. The entire box will move.

- To *resize* the box, position your cursor on a box side and drag. The side you have selected will move.
- To *redo* the box entirely, position your cursor outside the box and drag. The old box will disappear and a new box will be created.

You can also select the scanning area by entering coordinates in the appropriate fields (Top, Bottom, Left, Right). After you enter a coordinate, the position of the scanning area box will change accordingly.

When selecting, be sure to include the entire area of interest, and be generous with borders. You can always crop the image later.

G.4. Step III. Select Resolution

The FX acquisition window allows you to scan at 50, 100, 200, or 800 micrometers. These resolutions are listed as option buttons in the control panel.

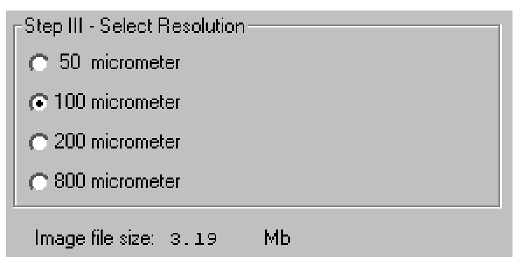


Fig. G-6. Resolution option buttons.

The resolution you select should be based on the size of the objects (e.g., bands, spots) you are interested in. For example:

- 50 micrometer resolution should be reserved for images requiring the highest level of detail, e.g., high density in situ samples, 1,536-well microplates, high density arrays, samples with very closely spaced bands. Files scanned at 50 micrometers can be very large.
- 100 micrometer resolution is useful for typical gels and arrays.

- 200 micrometer resolution is useful for gels with large bands and dot blots.
- 800 micrometer resolution should be reserved for very large objects, such as CAT assays.

File Size of Images

Image File Size (below Select Resolution) shows the size of the scan file you are about to create. If you do not have enough computer memory for the specified file size, an error message will appear when you attempt to scan. If this happens, select a lower resolution or decrease the size of the area to be scanned. (Macintosh users can also increase the application memory partition. See your Macintosh computer documentation for guidance.)

G.5. Acquire the Image

Once you have selected your application, scan area, and resolution, you are ready to acquire an image.

Click on the Acquire button. There may be a short delay while the image laser warms up; then the scanned image will begin to appear in the scanning window, line by line.

To interrupt a scan, click on the Stop button. A message will ask you to confirm the interrupt, and then you will be asked if you want to keep the partial scan. This feature is useful if you overestimated the size of the area you selected.

Note: If the image you are scanning has more than 10 saturated pixels, you will receive a warning message. If this happens, you can go back and select a higher sample intensity in the application tree.

Saving the Image

After the scan is complete, a message will appear asking you if you want to keep the scan. If you select Yes, a separate window will pop up containing the new image. The image will have a default file name that includes the date, time, and user (if known). To save the image, select Save or Save As from the File menu.

You can then analyze the image using Diversity Database's menu and toolbar functions.

G.6. Options

Auto Save After Scan

To automatically save any scan you create, click on the Auto Save After Scan checkbox.

With this checkbox selected, when you click on Acquire, a Save As dialog box will open asking you to specify a file name and location for the image you are about to create. The scan will begin when you click on the Save button.

Make Backup Copy

You can automatically create a backup copy of any scan you create. To do so, first select Auto Save After Scan (see above), then select the Make Backup Copy checkbox.

With this checkbox selected, when you save a scan, a backup copy will be placed in the same directory as the scanned image. Windows backup files will have an ".sbk" extension. Macintosh backup files will have the word "backup" after the file name.

This backup copy will be read-only, which means that you cannot make changes to it. You can open it like a normal file, but you must save it under a different file name before editing the image or performing analysis.

Highlight Saturated Pixels

When this is checked, any saturated pixels in the image will appear marked in red.

Hide Grid

To hide the gridlines in the scanning area window, click on the Hide Grid checkbox.

Appendix H. Keyboard Functions

Key	Normal	Shift	Command (Mac)	Control (PC)
Left	Pan Left			
Up	Pan Up			
Right	Pan Right			
Down	Pan Down			
F1	View Entire Image			
F2	Zoom box	Grab		Zoom Out
F3		Density Tools		
F4	Hide Overlays			
F5	Bars on Bases	Base Ladder		Base Ladder
F6	Sequence Text Window	Calls on Bases		Calls on Bases
F7	Image Tools			
F8		Describe Gel		Describe Gel
F9	Editing Tools	Display Tools		Display Tools
F10	Search for String	Search for Base		Search for Base
F11	Export Sequence			
F12	Save	Save All		
b			Transform	Transform
g			Grab	Grab
h			Help	Help
i			File Info	File Info
o			Open	Open
p				Print Image
q			Exit	Exit
s			Save	Save
t				Plot Cross-Sect.ion

Appendix I. Cross-Platform File Exchange

It is possible to move image data between Discovery Series software applications on different platforms. There are different protocols depending on the platform (PC or Macintosh) you are transferring from and to.

I.1. Macintosh to PC

To transfer a file from a Discovery Series application running on a Macintosh to a Discovery Series application running on a PC, you need to tag the file name with the suffix appropriate to the image file type (i.e., 1-D scan or DNA scan).

For DNA Scans (DNACode) use the suffix: .dsc

For 1-D gels (Diversity Database, Quantity One) use the suffix: .1sc

For database files (Diversity Database) use the suffix: .ddb

I.2. PC to Macintosh

PC versions of Discovery Series files can be read directly by Macintosh applications with no required modifications.

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